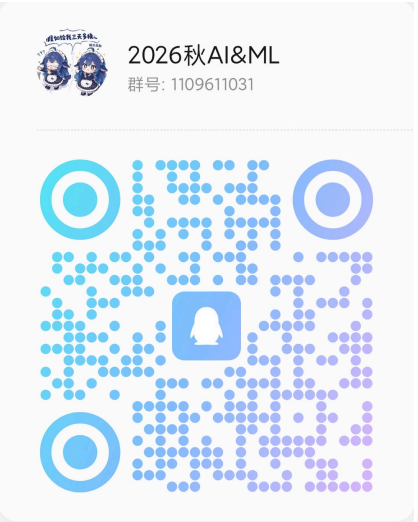
人工智能与机器学习基础

Lecture 06: 神经网络 I

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中国科学技术大学



提纲

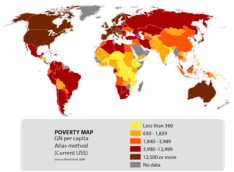
- 多分类问题
- 前馈神经网络
- 神经网络的组成部分
- 深度神经网络

提纲

- **多分类问题**
  - 回归与分类
  - 多分类问题
  - 多分类逻辑回归
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Recap：回归问题

$$x \longrightarrow f \longrightarrow y \in \mathbb{R}$$



贫困地图：卫星地图 ⟶ 资产索引



住房价格：房子信息 ⟶ 价格



到达时间：终点站、天气、时间 ⟶ 到达时间

Recap：线性回归

- 模型：$f_{\mathbf{w}}(x) = \mathbf{w} \cdot \phi(x)$
- 向量表示
  - 参数向量/模型参数：$\mathbf{w} = [w_1, w_2]$
  - 特征提取器/特征向量：$\phi(x) = [1, x]$
- 假设类：$\mathcal{F} = \{f_{\mathbf{w}} : \mathbf{w} \in \mathbb{R}^2\}$ (预测器的f集合)
- 损失函数：$Loss(x, y, \mathbf{w}) = (f_{\mathbf{w}}(x) - y)^2$

$$TrainLoss(\mathbf{w}) = \frac{1}{|\mathcal{D}_{train}|} \sum_{(x,y) \in \mathcal{D}_{train}} Loss(x, y, \mathbf{w})$$

$$\arg\max_{\mathbf{w}} \ln p(\mathbf{y}|\mathbf{x}, \mathbf{w}) = \arg\min_{\mathbf{w}} \sum_{(x,y) \in \mathcal{D}} (y - \mathbf{w} \cdot \phi(x))^2$$

高斯假设下的最大似然估计 = 最小化平方误差

Recap：二分类问题

$$x \longrightarrow f \longrightarrow y \in \{+1, -1\}$$

分类器　　label

二分类任务



欺诈检测: 信用卡交易 → 欺诈或非欺诈



有毒评论检测: 在线评论 → 有害或无害

$y \in \{+1, -1\}$

多分类任务



手写数字识别: 数字图像 → 0，1，2，...

$y \in \{0,1,\cdots,9\}$

$y \in \{1,\cdots,K\}$

Recap：逻辑回归

- 模型：$f_{\mathbf{w}}(x) = \mathbf{w} \cdot \phi(x)$
- 向量表示
  - 类别概率：$\hat{P}(y|x, \mathbf{w})$
  - 逻辑函数：$\sigma(score) = \frac{1}{1+e^{-score}}$
- 目标函数
  - 对于N个样本的数据集 $\{(x_i, y_i)\}_{i=1}^{N}$，其似然估计为

$$\max_{\mathbf{w}} l(\mathbf{w}) = \prod_{i=1}^{N} \hat{P}(y_i|x_i, \mathbf{w})$$

  - 重写为对数似然估计

$$\max_{\mathbf{w}} ll(\mathbf{w}) = \sum_{i=1}^{N} \log\left(\hat{P}(y_i|x_i, \mathbf{w})\right)$$

提纲

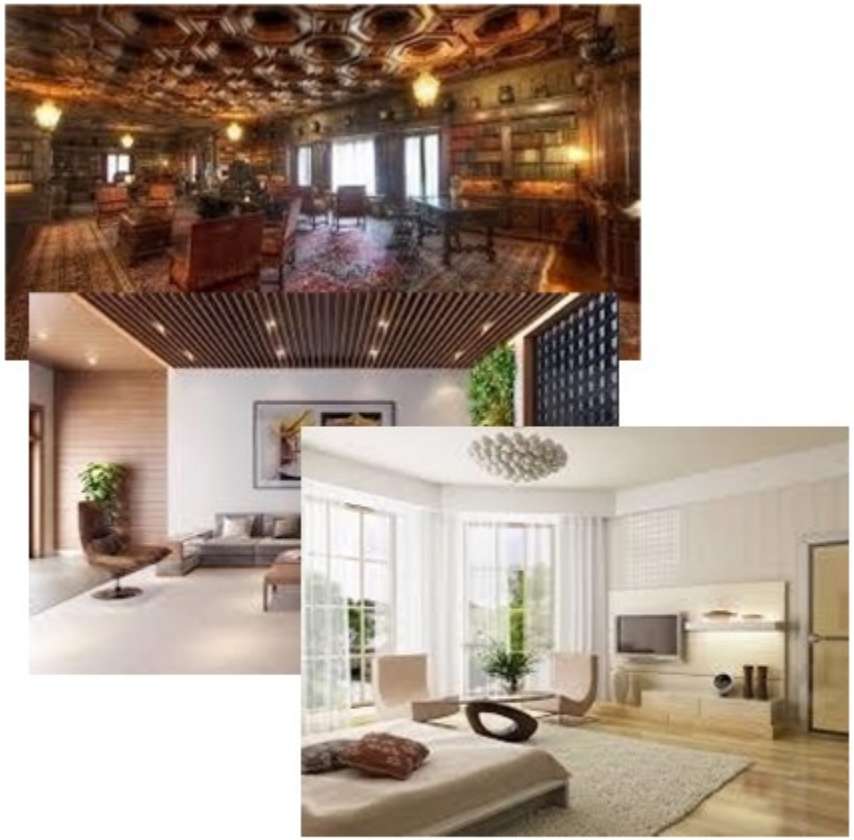
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示例：图像分类

- ImageNet 是计算机视觉中的基础基准数据集
  - 覆盖2万多个类别，其中常用于评测的ILSVRC子集包含1000个类别



示例：图像分类（二分类）

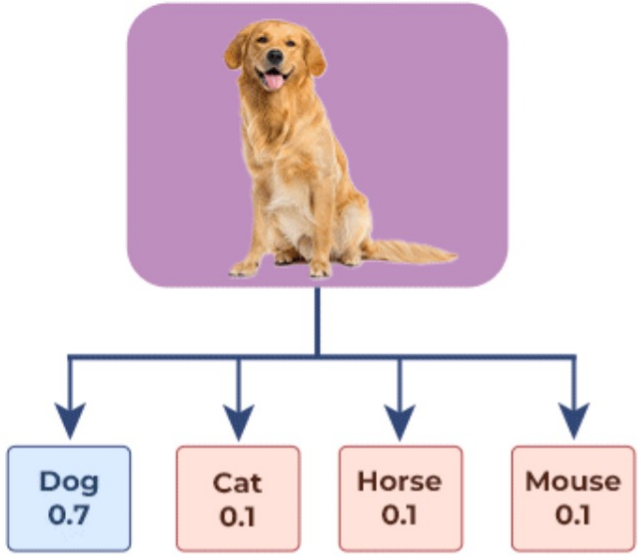


类别1：室内



类别2：室外

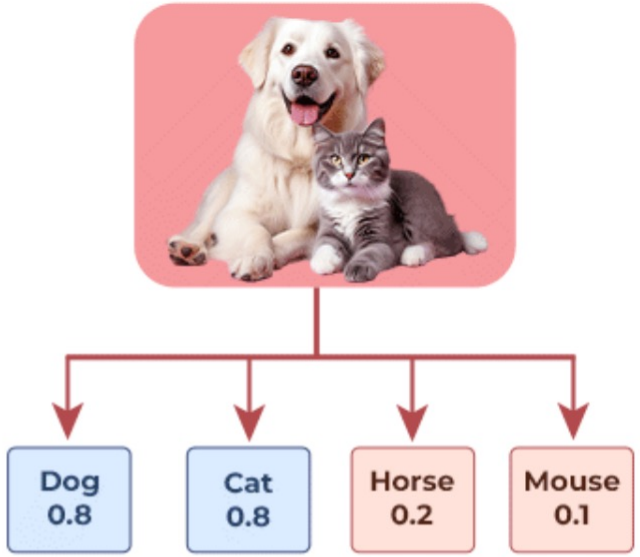
示例：多分类 vs. 多标签



Classes

(pick one class)

- [x] Dog
- [ ] Cat
- [ ] Horse
- [ ] Mouse



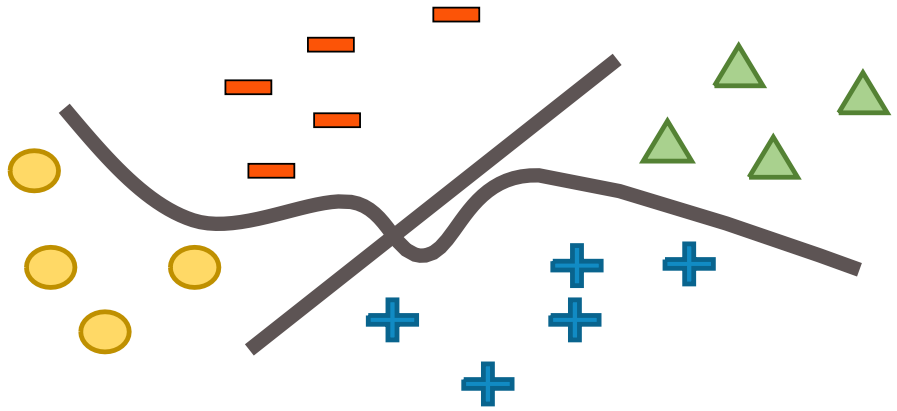
Classes

(pick all the labels present in the image)

- [x] Dog
- [x] Cat
- [ ] Horse
- [ ] Mouse

多分类

- 给定：训练数据$\{(x_i, y_i)\}_{i=1}^{N}$
  - x_i为第i个样本的输入
  - y_i为第i个样本的类别标签



- 目标
  - 学习一个函数能够尽可能输出正确的类别标签

$$f: x \rightarrow \{1,2,\ldots,K\}$$

问： 如何构造f？

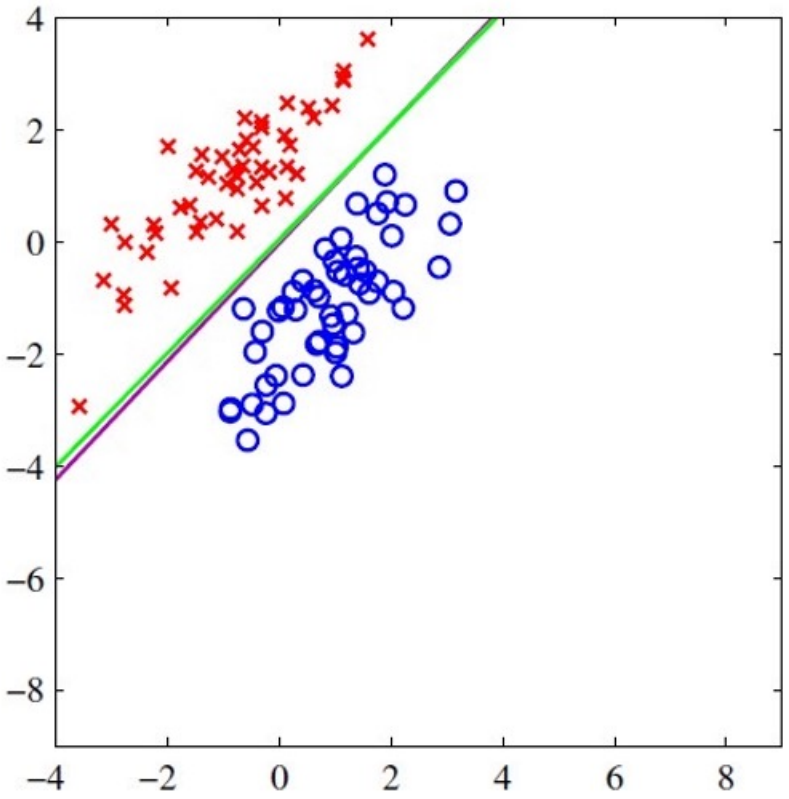
方法1：降维到回归问题

- **给定**：训练数据$\{(x_i, y_i)\}_{i=1}^{N}$
  - x_i为第i个样本的输入
  - y_i为第i个样本的类别标签
- **目标**：学习一个函数$f_{\mathbf{w}}(x) = \mathbf{w}^T\phi(x)$来优化最小化平方损失

$$Loss(x, y, \mathbf{w}) = (f_{\mathbf{w}}(x) - y)^2$$

- **缺点**：即使在二分类中
  - 回归的预测值可能不在[0,1]区间，无法解释为概率
  - RSS(残差平方和）也不是分类问题的合适损失函数
  - 决策边界可能不合理

方法1：降维到回归问题



局限

即使在二分类中，这种方法也不好

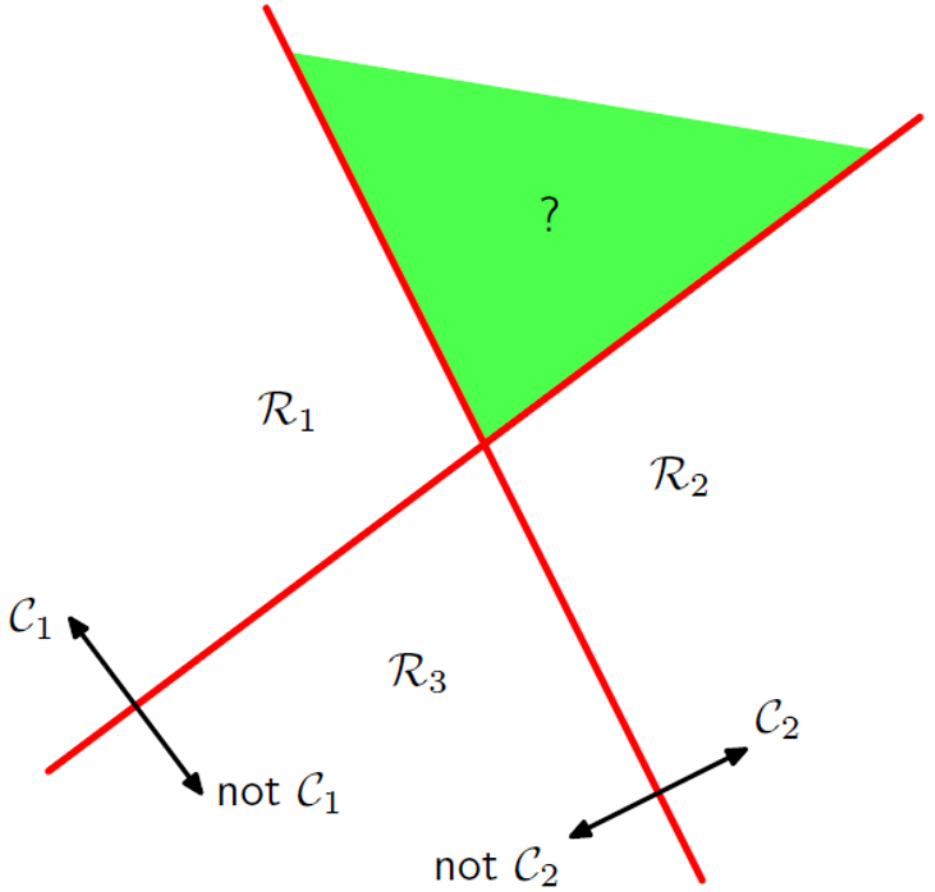
Figure 4.4 The left plot shows data from two classes, denoted by red crosses and blue circles, together with the decision boundary found by least squares (magenta curve) and also by the logistic regression model (green curve), which is discussed later in Section 4.3.2. The right-hand plot shows the corresponding results obtained when extra data points are added at the bottom left of the diagram, showing that least squares is highly sensitive to outliers, unlike logistic regression.

方法2：一对其余

- **模型**：训练 K-1 个分类器 $f_1, f_2, \cdots, f_{K-1}$
  - f_1区分类别1 vs. {2, 3, … , K}
  - f_2区分类别2 vs. {1, 3, … , K}
  - …
  - f_{K-1}区分类别 K-1 vs. {1, 2, … , K-2}
  - 未被分到 {1, 2, 3, … , K-1} 的点，归入类别K

- **缺点**：模糊区域
  - 可能出现 同一个点被多个分类器同时判定为正类；
  - 导致某些样本被归入多个类别，产生歧义

方法2：一对其余

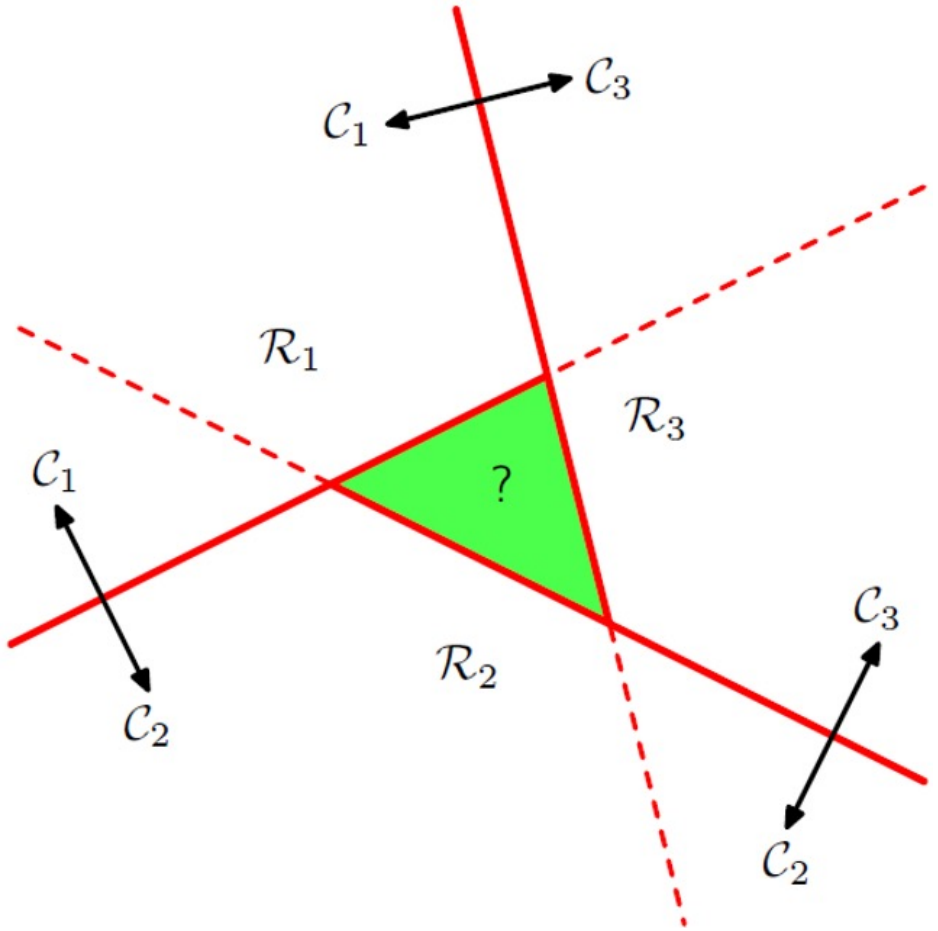


Pattern Recognition and Machine Learning (PRML) 插图

方法3：一对一

- **模型**：训练 $(K-1)K/2$ 个分类器 $f_{(1,2)}, f_{(1,3)}, \cdots, f_{(K-1,K)}$
  - 每个分类器只区分两个类别
  - $f_{(1,2)}$区分类别 1 vs. 2
  - $f_{(1,3)}$区分类别 1 vs. 3
  - …
  - $f_{(K-1,K)}$区分类别 $(K-1)$ vs. K
- **缺点**
  - 计算量大, E.g. ImageNet 中 K=1000
  - 模糊区域问题：一个点可能在不同的 OvO 分类器中被投票到不同类别

方法3：一对一



Pattern Recognition and Machine Learning 插图

方法4：判别函数

- **模型**：训练 K 个打分函数 $s_1, s_2, \ldots, s_K$
  - 分类规则：选择得分最高的类别

$$y = argmax_i s_i(x)$$

  - 第 k 个打分函数

$$s_k(x) = (\mathbf{w_k})^T \phi(x)$$

  - 可解释为类别的条件概率

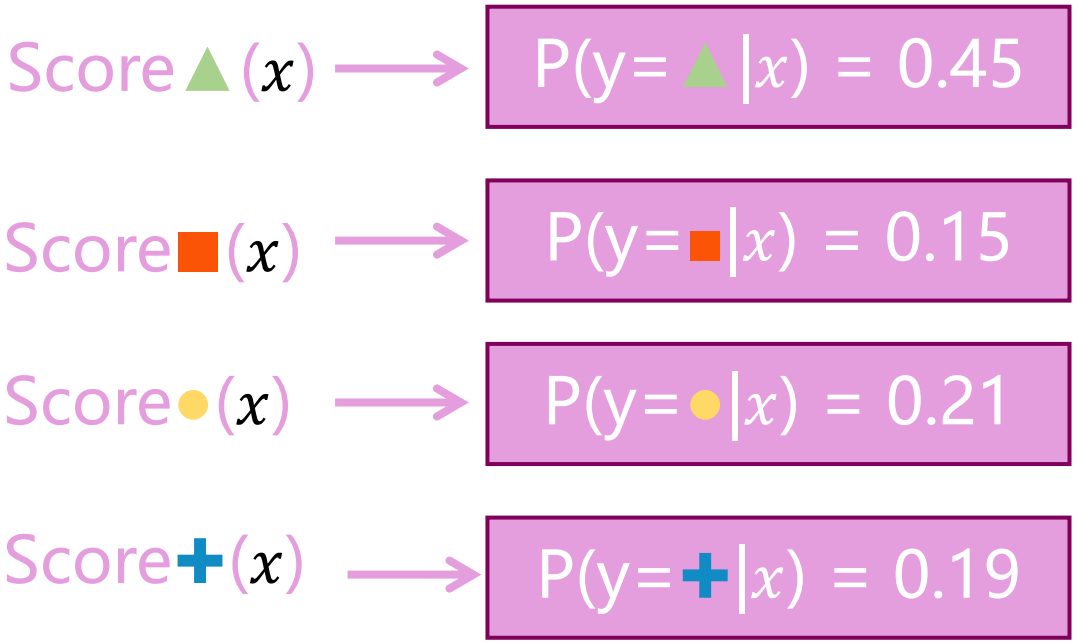
$$s_k(x) \propto p(y = k|x)$$

如何实现呢?

- **优点**
  - 计算开销低：只需计算 K 个得分函数
  - 没有模糊区域：每个样本只会被分到一个类别

条件分布作为判别函数

- 如果判别函数等价于类别条件概率，那么分类器的输出就不仅仅是类别标签，还可以表达预测的置信度 (confidence)



选择其中概率最大的作为最终的多分类预测

概率最大 → $y=$ ▲

如何实现呢?

提纲

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核心：Softmax函数

- Softmax把向量$s = (s_1, s_2, \ldots, s_K)$压缩为概率分布

$$p(s_k) = \frac{exp(s_k)}{\sum_{k'=1}^{K} exp(s_{k'})}$$

$$\text{softmax}(s) = (p(s_1), p(s_2), \ldots, p(s_K))$$

  - 把任意实数向量s映射到一个概率向量 p（所有元素非负且和为 1）

- 类似于“平滑版的 max”
  - 当某个分量 $s_k \gg s_{k'}, \forall k' \neq k$ 时，$p(s_k) \approx 1, p(s_{k'}) \approx 0$
  - 因此, softmax 既能表达相对大小，又能转化为概率分布

交叉熵损失

- 样本预测分数
  - $Score(x) = \mathbf{w}^{\mathsf{T}}\phi(x) \in \mathbb{R}^K, \mathbf{w} \in \mathbb{R}^{D\times K}, \phi(x) \in \mathbb{R}^D$

注意 $\mathbf{w}$ 的维度!

- Softmax函数

$$\hat{p}(y \mid x, \mathbf{w}) = softmax(Score(x))$$

$$\hat{p}(y = k \mid x, \mathbf{w}) = \frac{e^{Score_k(x)}}{\sum_{k'=1}^{K} e^{Score_{k'}(x)}}$$

$$\text{s.t.,} \sum_{k=1}^{K} \hat{p}(y = k \mid x, \mathbf{w}) = 1$$

交叉熵损失

- 最大化对数似然

$$\max_{\mathbf{w}} ll(\mathbf{w}) = \sum_{i=1}^{N} log(\hat{p}(y_i \mid x_i, \mathbf{w}))$$

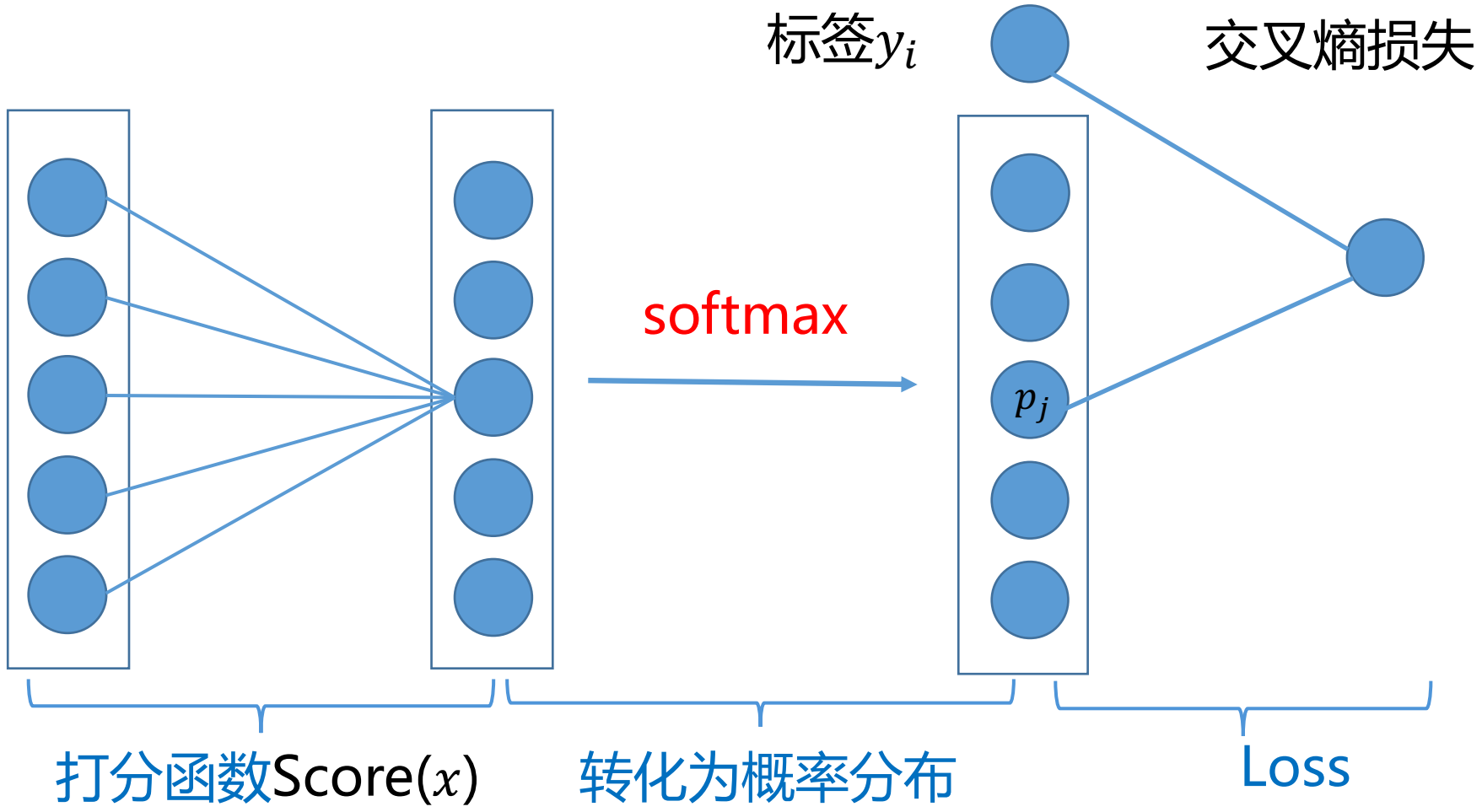
- 最小化交叉熵损失

模型预测的分布

$$\min_{\mathbf{w}} cross_entropy(\mathbf{w}) = -\sum_{i=1}^{N}\sum_{k=1}^{K} 1(y_i = k) \log(\hat{p}(y_i = k \mid x_i, \mathbf{w}))$$

数据的经验分布

多分类逻辑回归



提纲

- 多分类问题
- **前馈神经网络**
  - 表示学习
  - 直观理解
  - 示例
- 神经网络的组成部分
- 深度神经网络

Recap：机器学习流程

- 数据：收集数据集
- 特征：通过特征提取器将原始输入转化为特征向量

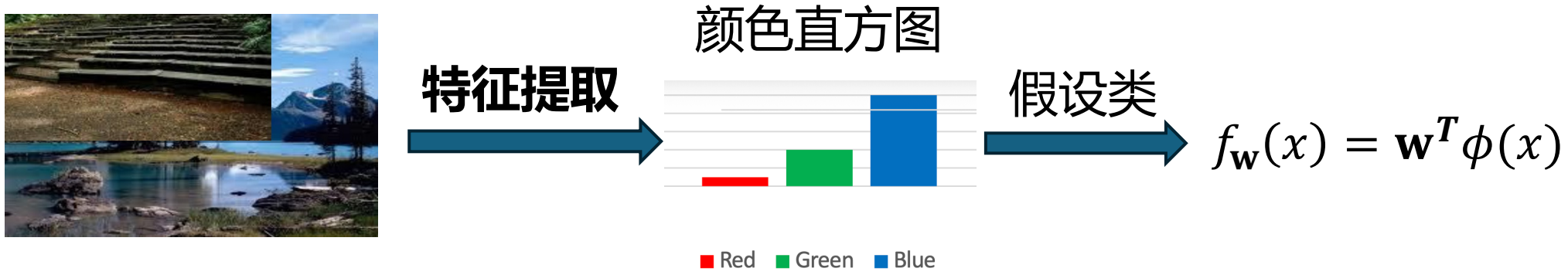
$$\phi(\mathrm{x})$$

- 模型：选择假设空间与损失函数

$$f_{\mathbf{w}}(x) = \mathbf{w} \cdot \phi(x)$$

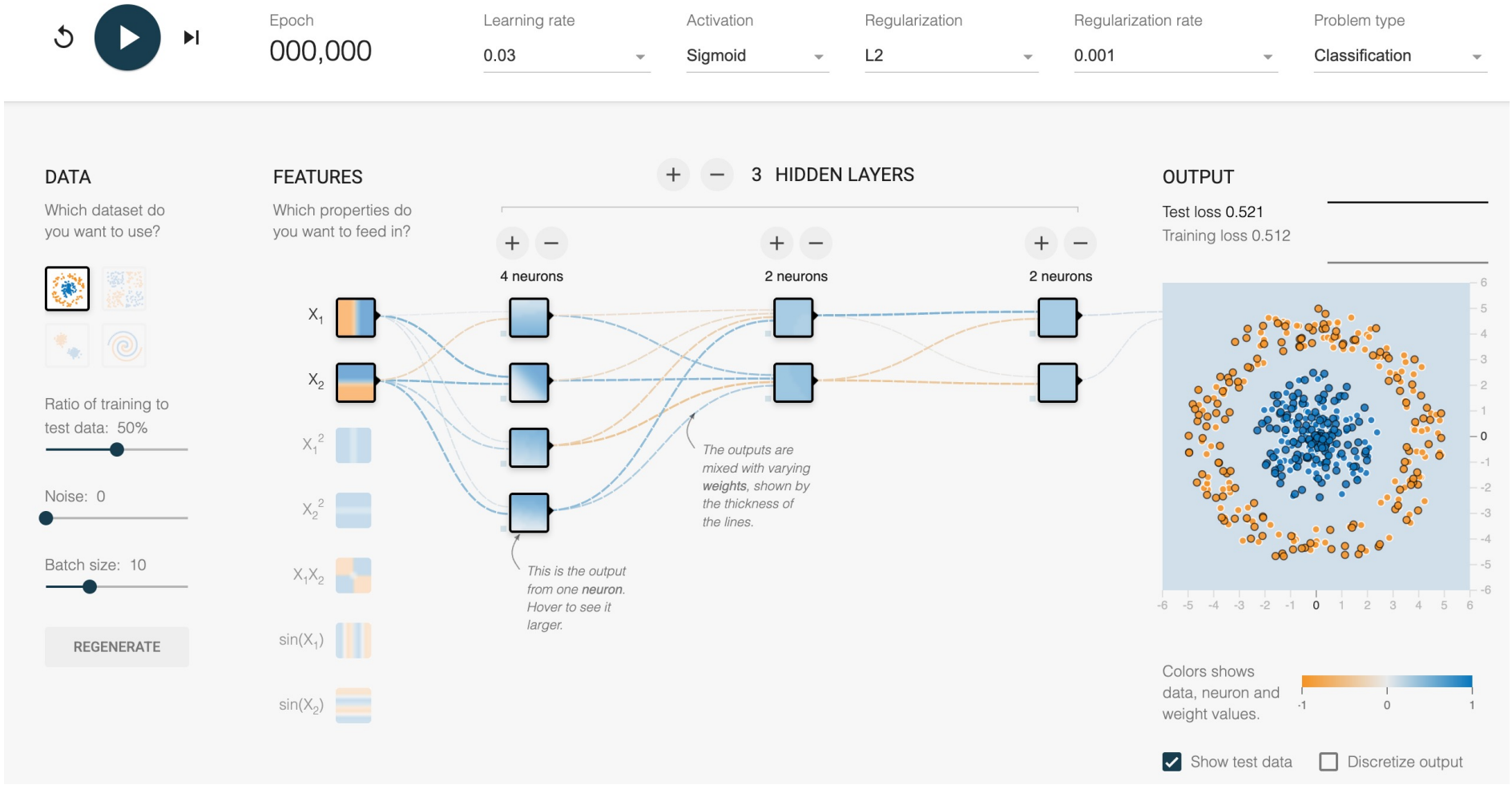
- 优化：最小化经验损失/最大化似然估计

样例：在图片分类中的特征



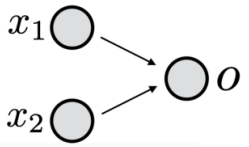
- 描述图像中不同颜色出现的分布情况；
- 对光照变化和旋转具有一定的鲁棒性；
- 常用于早期图像分类和检索任务

样例：在图片分类中的特征



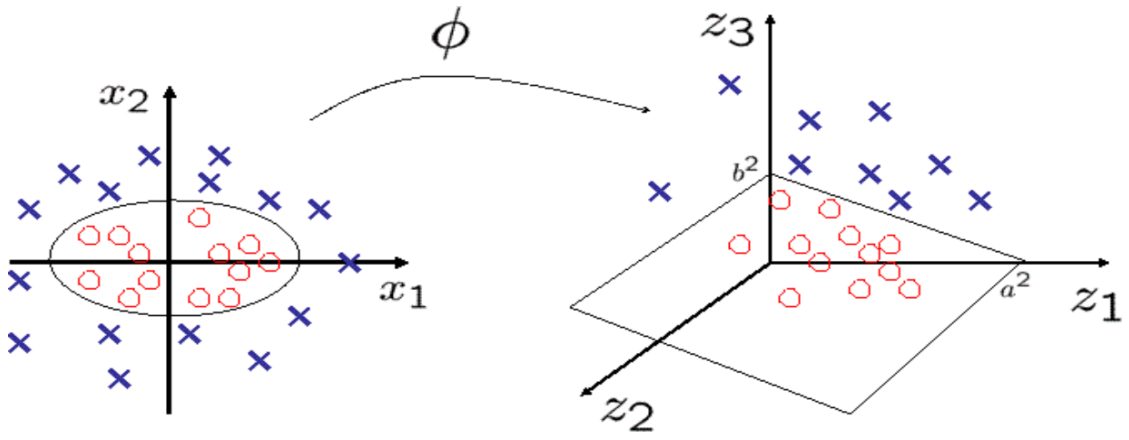
https://playground.tensorflow.org/

question



特征是模型的一部分

- **特征不是独立存在的，而是模型的重要组成部分**
  - 特征决定了模型能够捕捉什么信息；
  - 特征与模型参数一起决定了预测函数的形式；
  - 不同的特征选择会直接影响模型的拟合能力与泛化性能



假设类

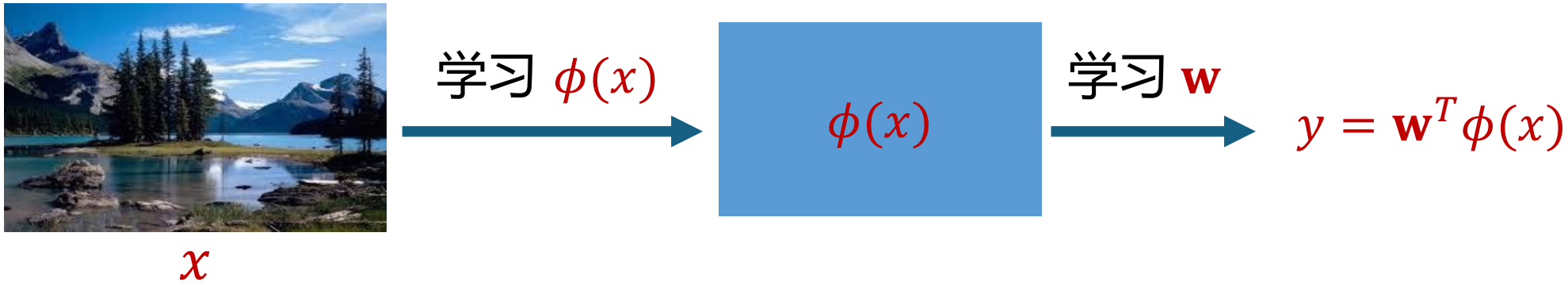
$$f_{\mathbf{w}}(x) = \mathbf{w}^T \phi(x)$$

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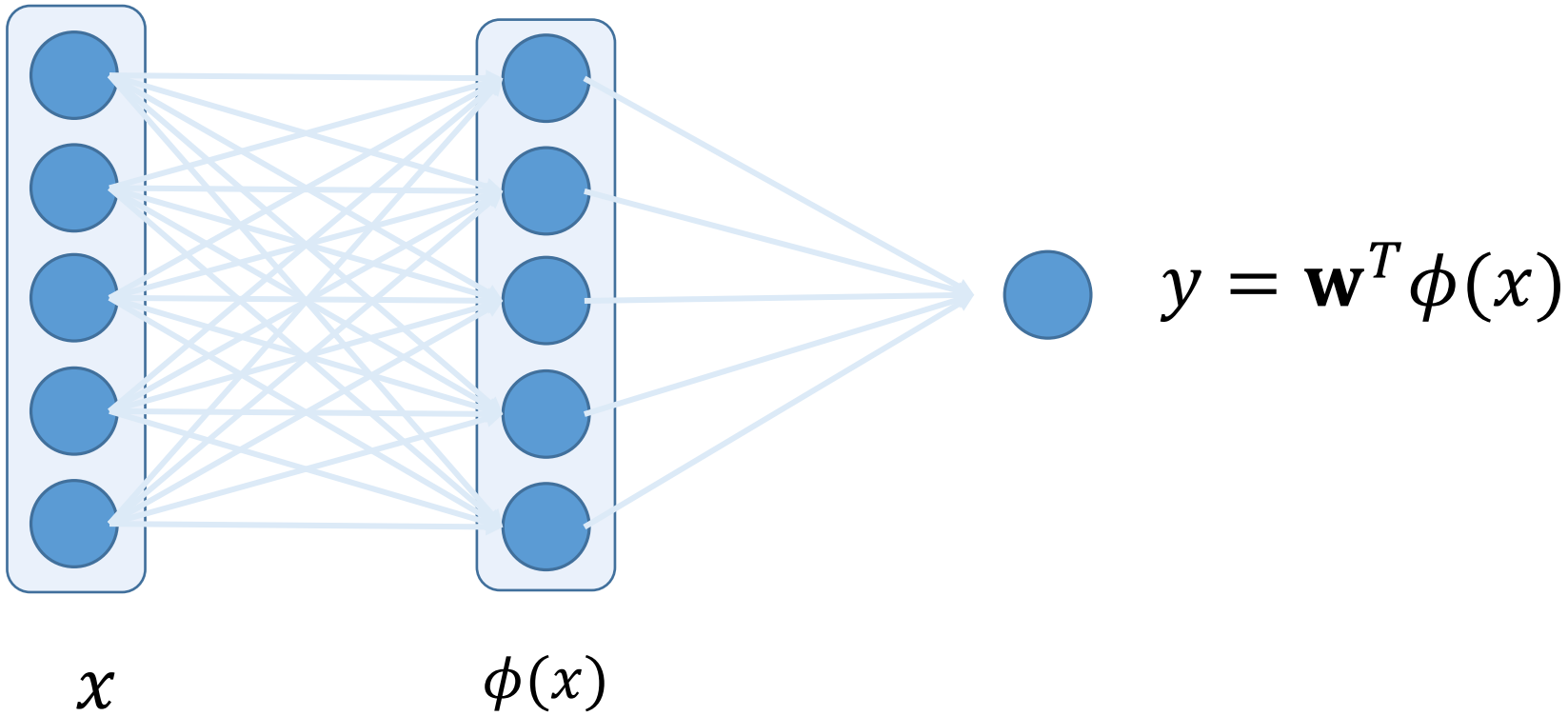
动机：表示学习

- **表示学习的思想**
  - **不仅学习模型参数，还学习特征提取器 $\phi(x)$；**
  - 通过多层网络自动从原始数据中提取更抽象、更有判别力的特征



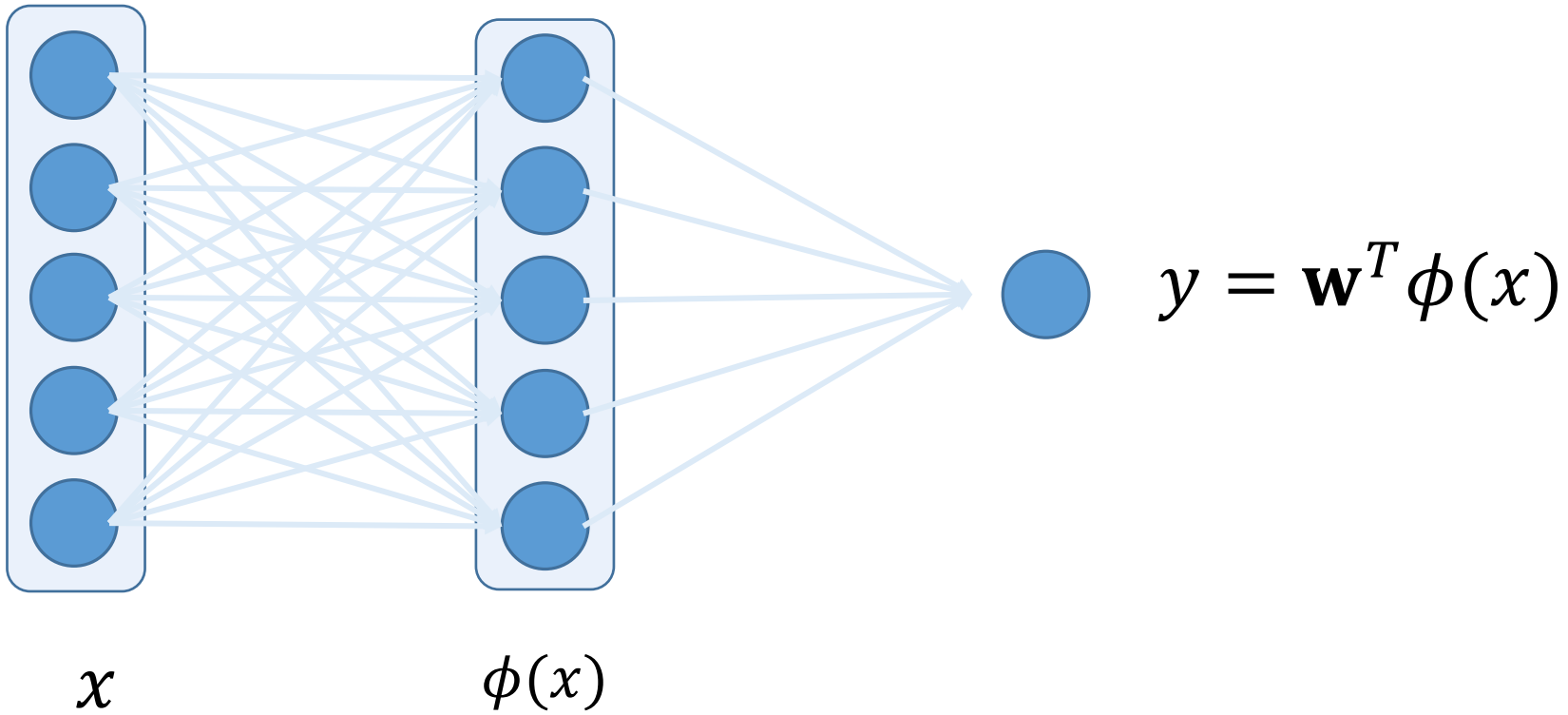
前馈神经网络

- 将特征提取器$\phi(x)$的每一维都看作是需要学习的表示



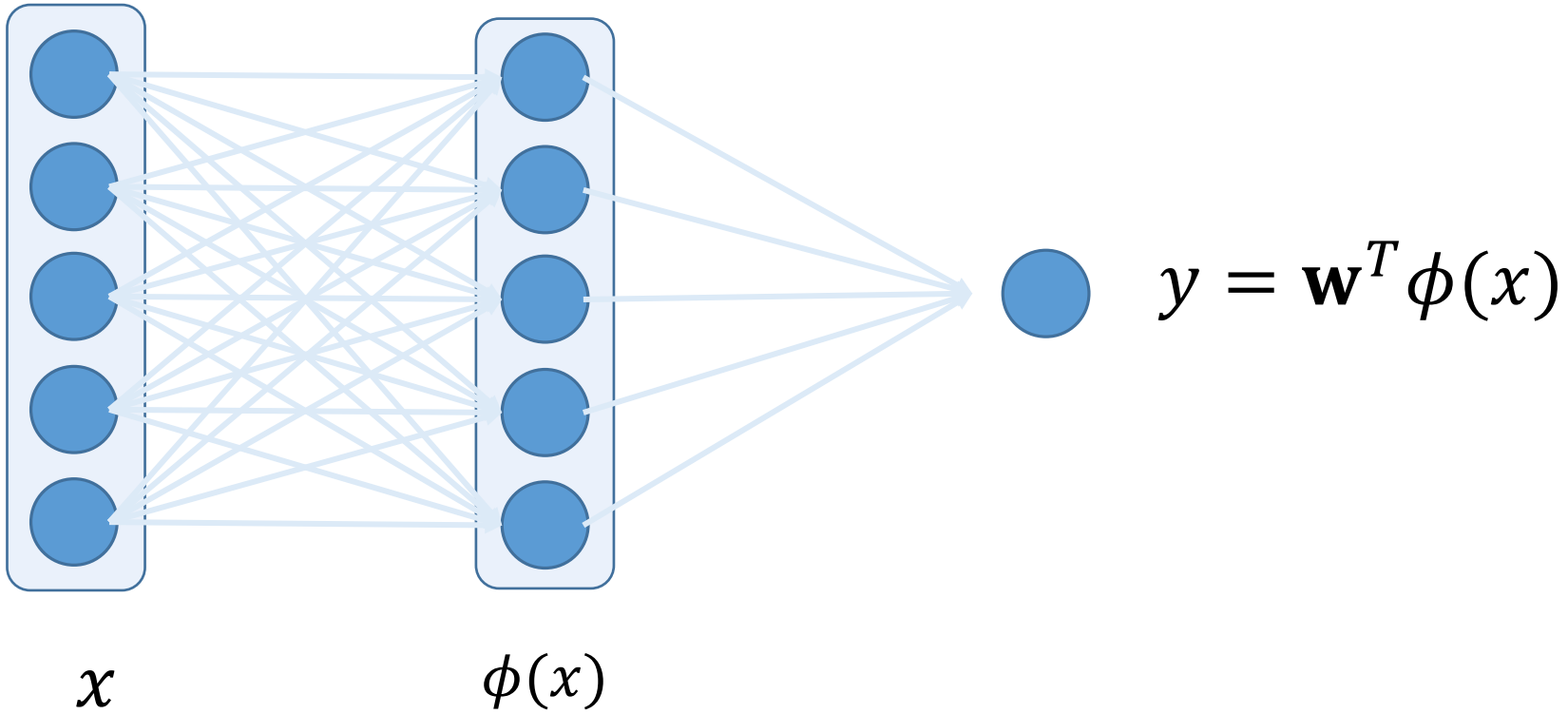
前馈神经网络

- 如果仅用线性变换 $\phi_i(x) = \theta_i^T x$，不同线性组合之间仍是线性关系，模型无法表达复杂的非线性模式



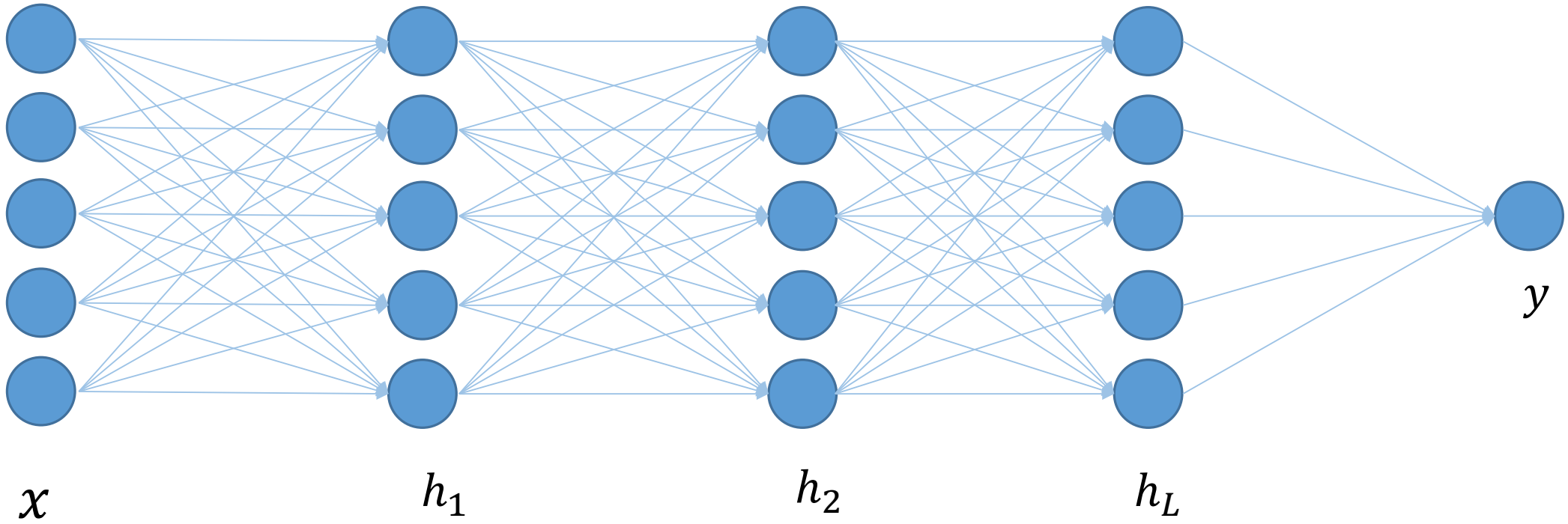
前馈神经网络

- 引入非线性：$\phi_i(x) = \sigma(\theta_i^T x)$，其中, σ(·) 是非线性激活函数
  - 常见非线性函数：Sigmoid、Tanh、ReLU, …



前馈神经网络

- 多层网络的每一层都在前一层的表示上进一步抽象
  - 表达能力更强：能够逼近任意复杂函数（**通用逼近定理**）
  - 自动学习多层次的表示，无需手工设计特征

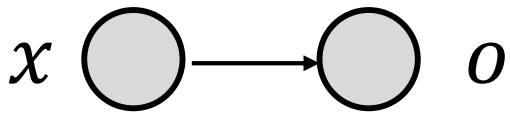


提纲

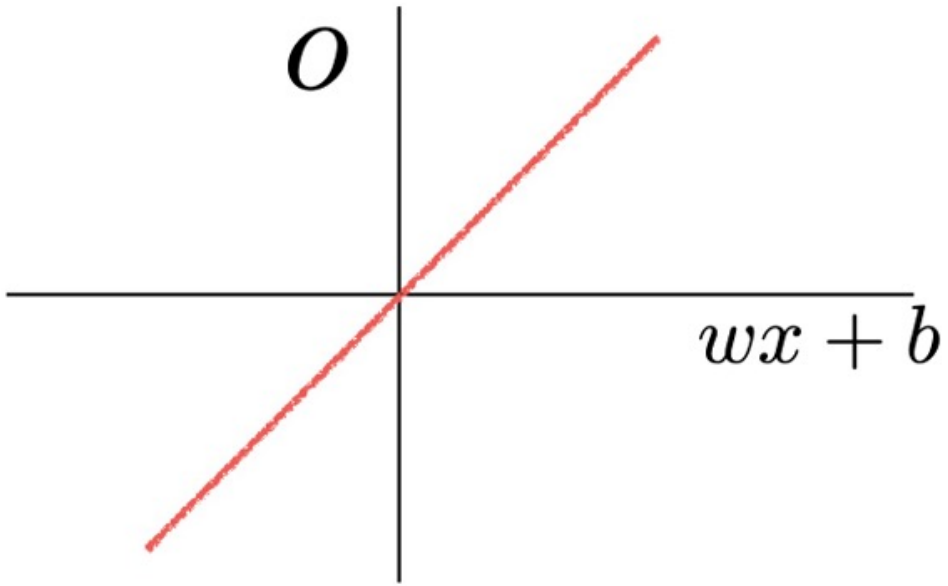
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前馈神经网络直观理解

- 线性激活函数

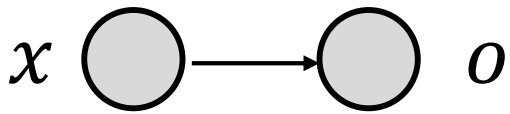


$$o = wx + b$$

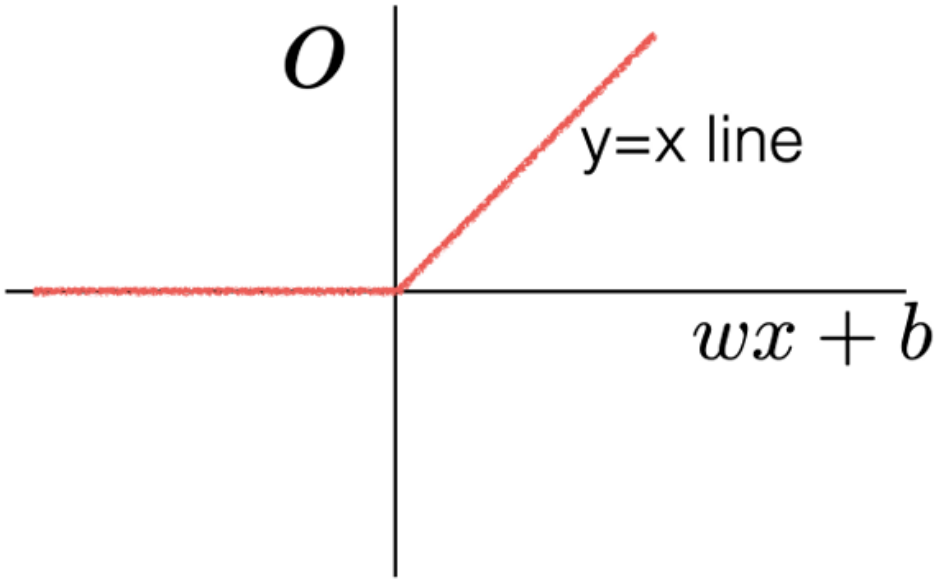


前馈神经网络直观理解

- 分段线性激活函数

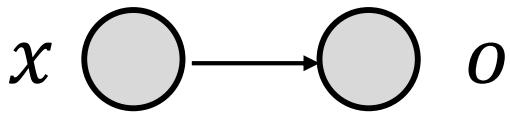


$$o = ReLu(wx + b)$$

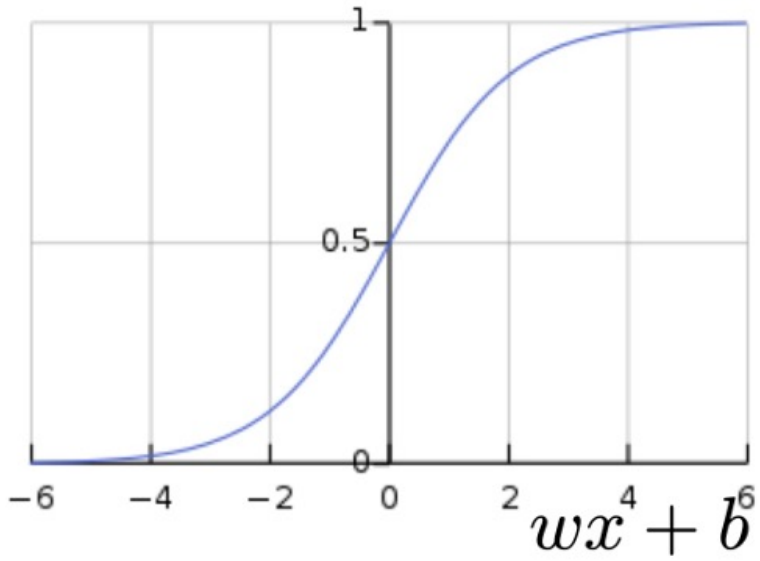


前馈神经网络直观理解

- **Sigmoid 激活函数**

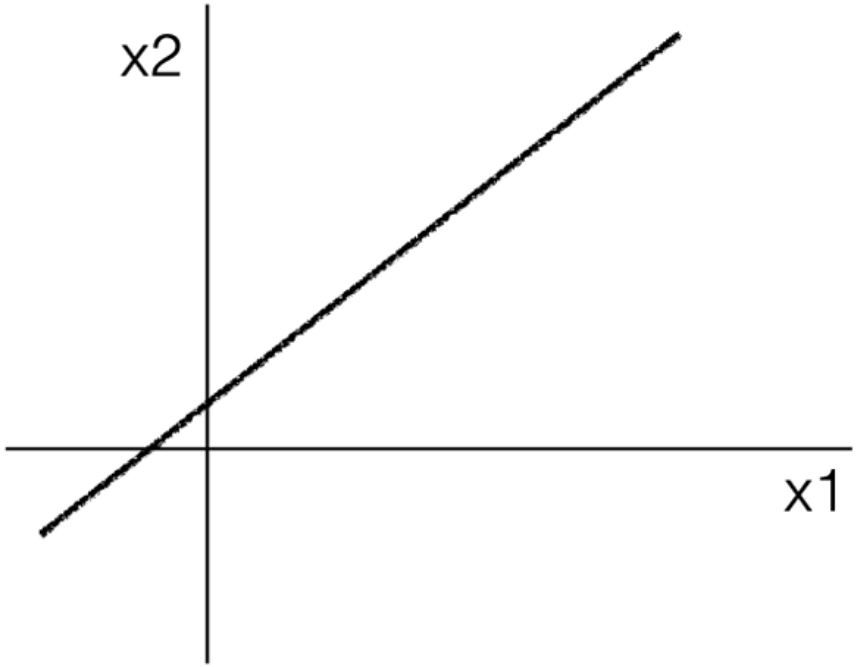


$$o = \frac{1}{1 + \exp(-wx - b)}$$



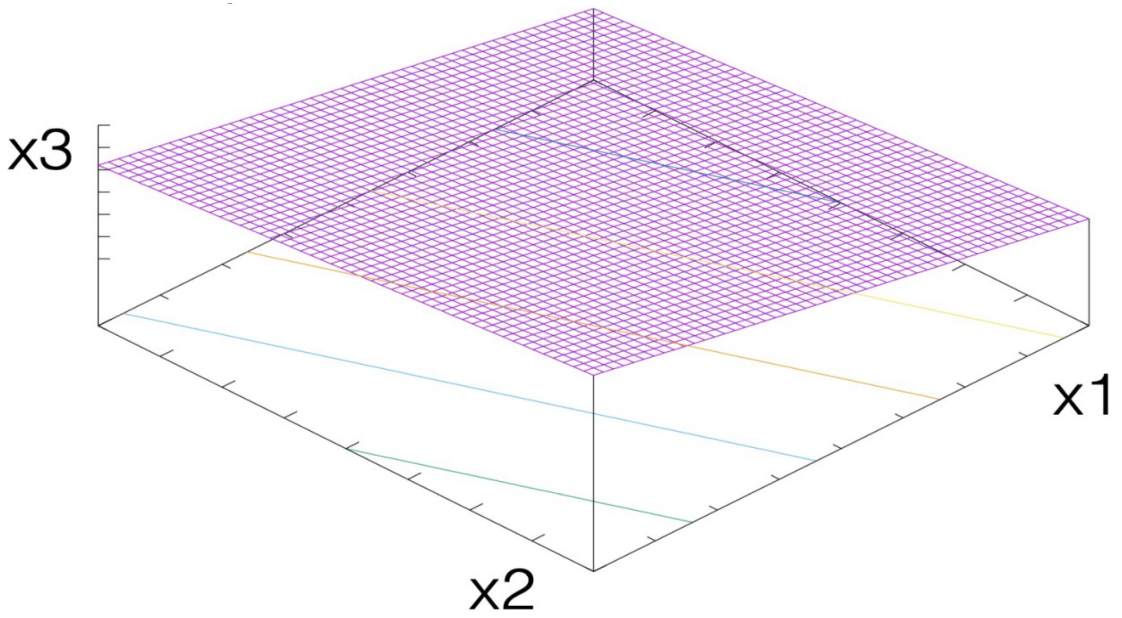
前馈神经网络直观理解

- 直线方程：$x_1 = m * x_2 + c$



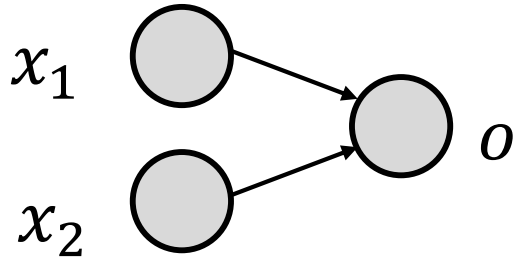
前馈神经网络直观理解

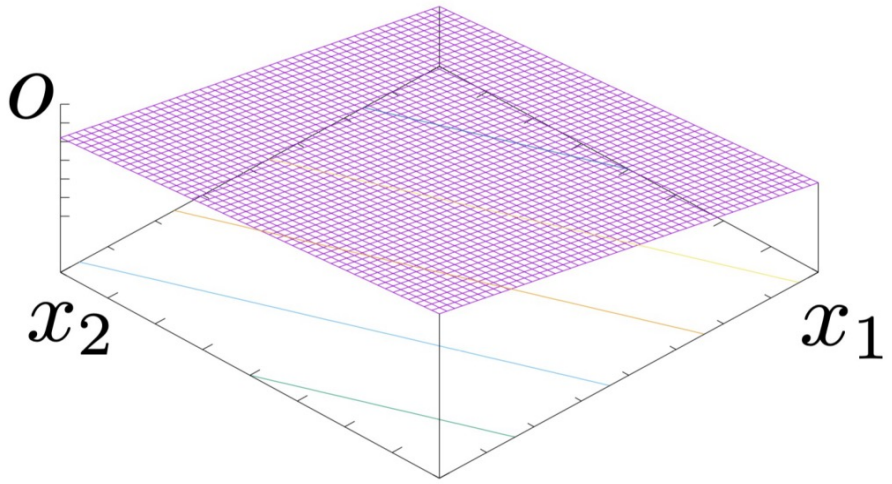
- 平面方程: $0 = m_1 * x_1 + m_2 * x_2 + m_3 * x_3 + c$
- 超平面方程：$0 = m_1 * x_1 + m_2 * x_2 + \cdots + m_d * x_d + c$



前馈神经网络直观理解

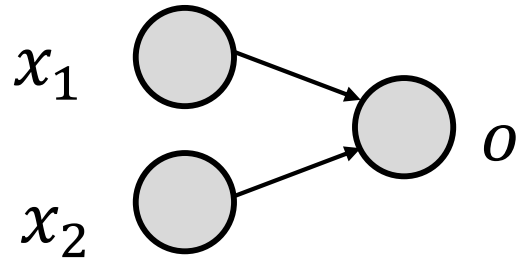
- $x = (x_1, x_2) \in \mathbb{R}^2$
- $0 = w_1 x_1 + w_2 x_2 + b$

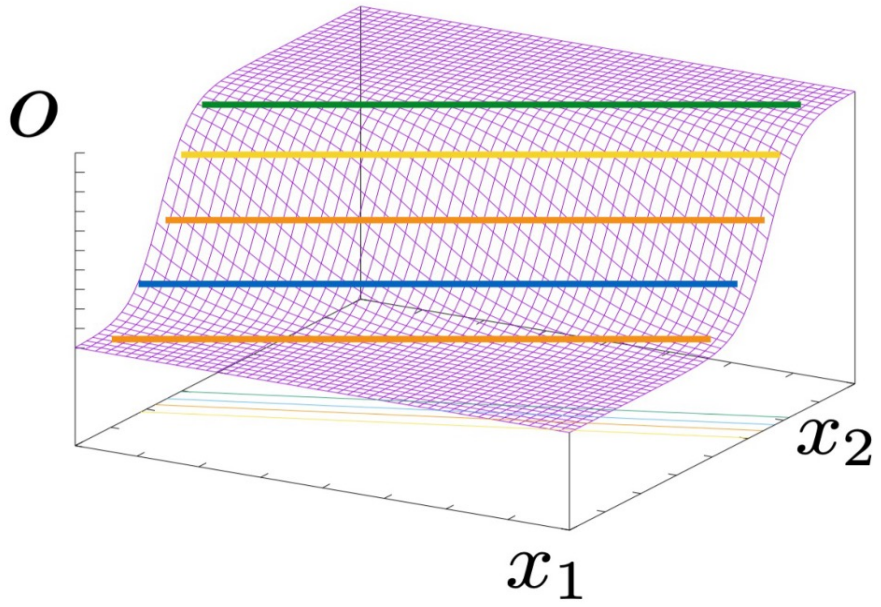


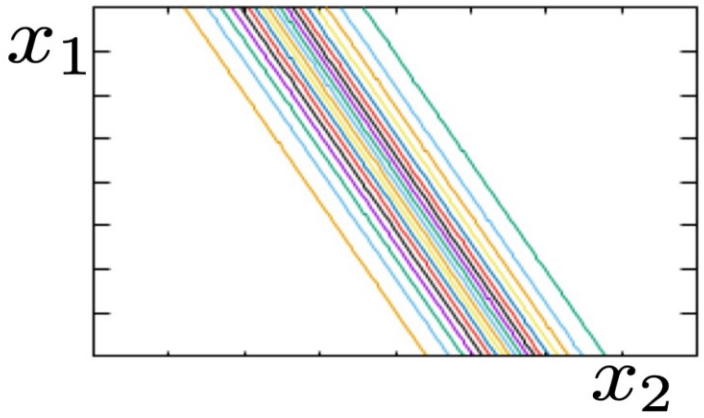


前馈神经网络直观理解

- $x = (x_1, x_2) \in \mathbb{R}^2$
- $0 = \sigma(w_1 x_1 + w_2 x_2 + b)$

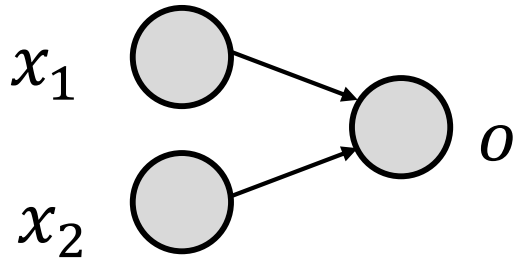


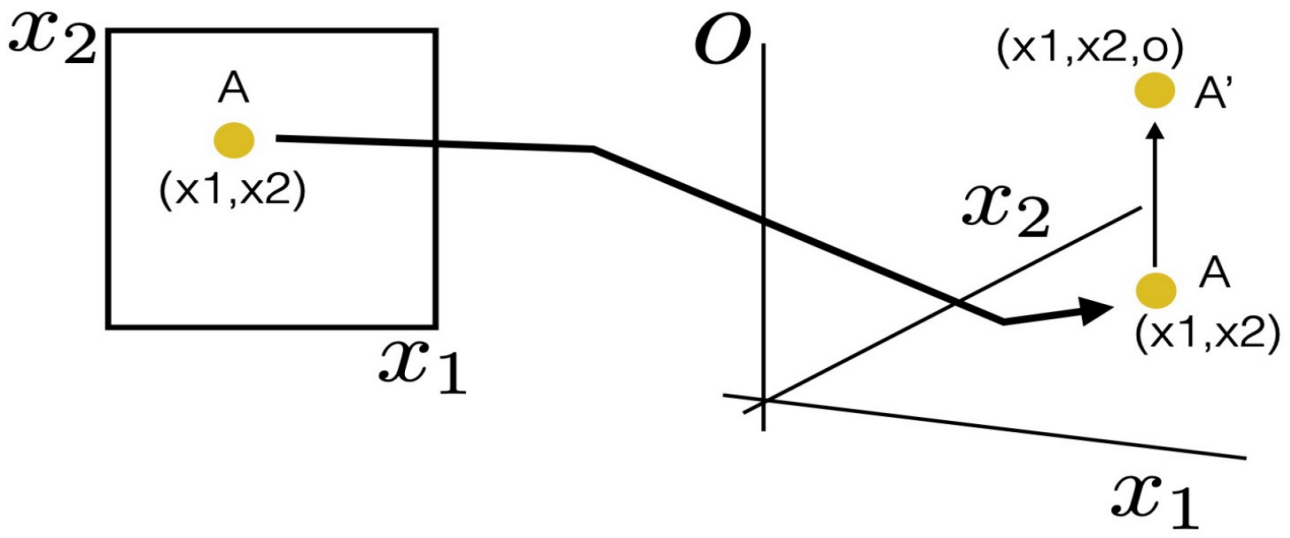




前馈神经网络直观理解

- $x = (x_1, x_2) \in \mathbb{R}^2$
- $0 = \sigma(w_1 x_1 + w_2 x_2 + b)$

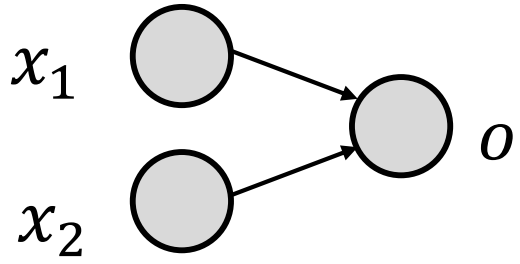


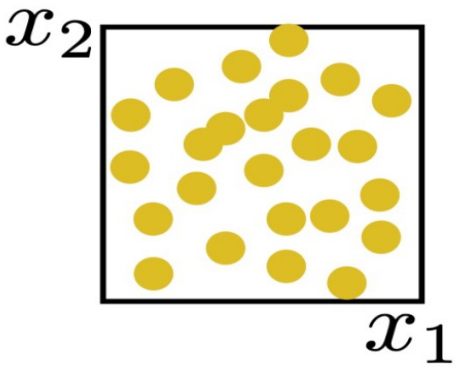


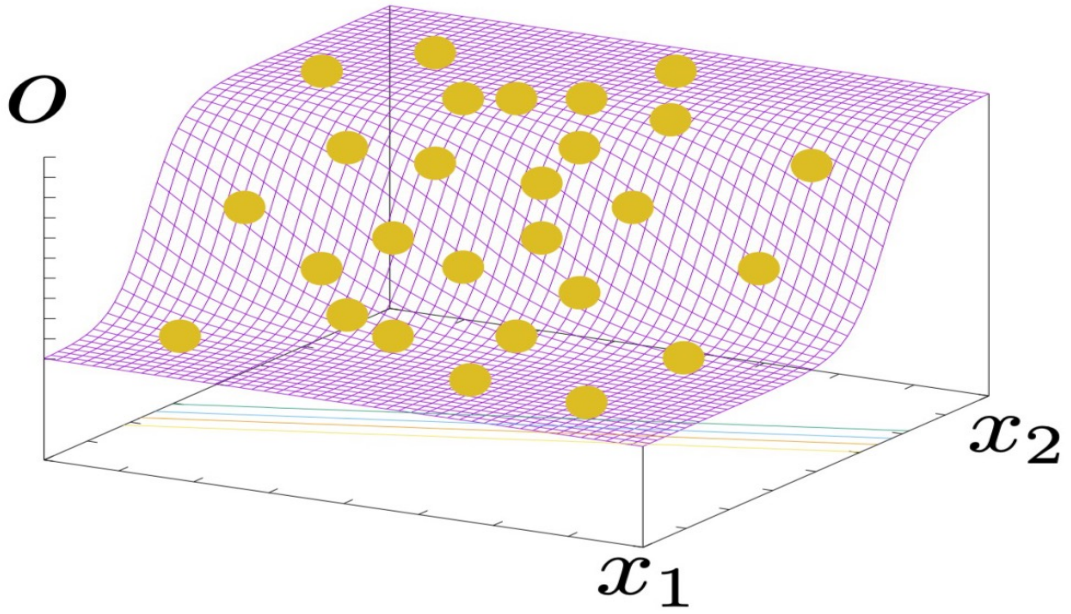
$A' = (x_1, x_2, o)$

前馈神经网络直观理解

- $x = (x_1, x_2) \in \mathbb{R}^2$
- $0 = \sigma(w_1 x_1 + w_2 x_2 + b)$

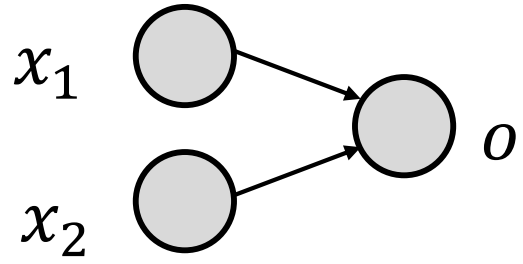


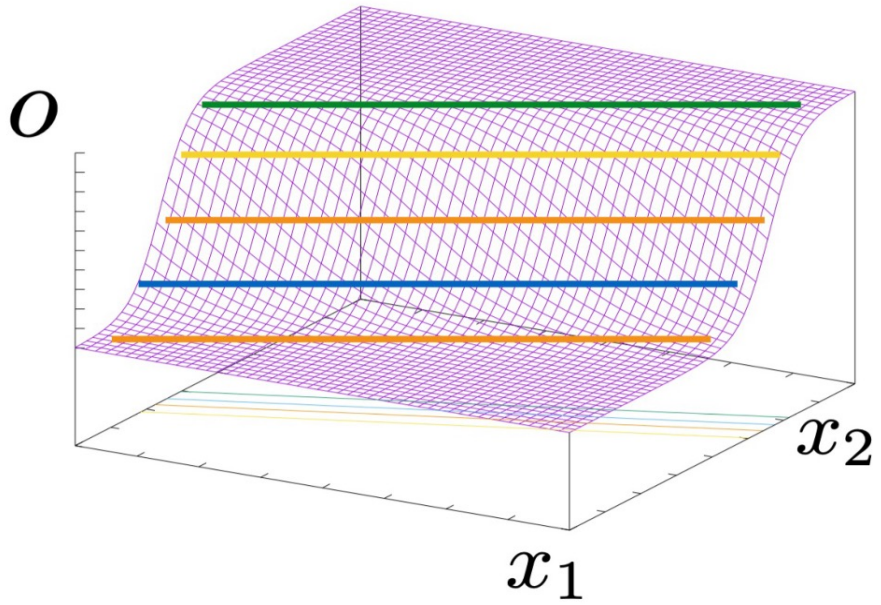


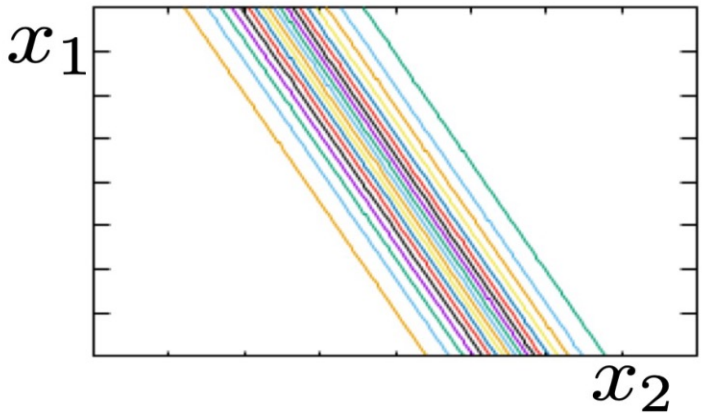


前馈神经网络直观理解

- $x = (x_1, x_2) \in \mathbb{R}^2$
- $0 = \sigma(w_1 x_1 + w_2 x_2 + b)$

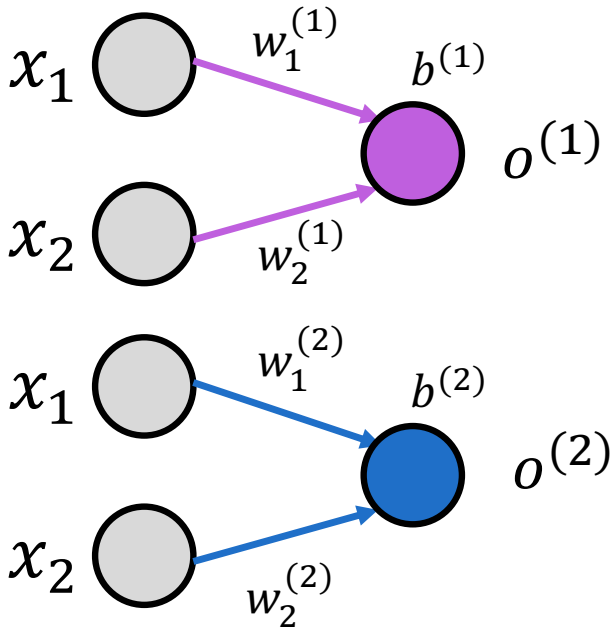






前馈神经网络直观理解

- 叠加起来

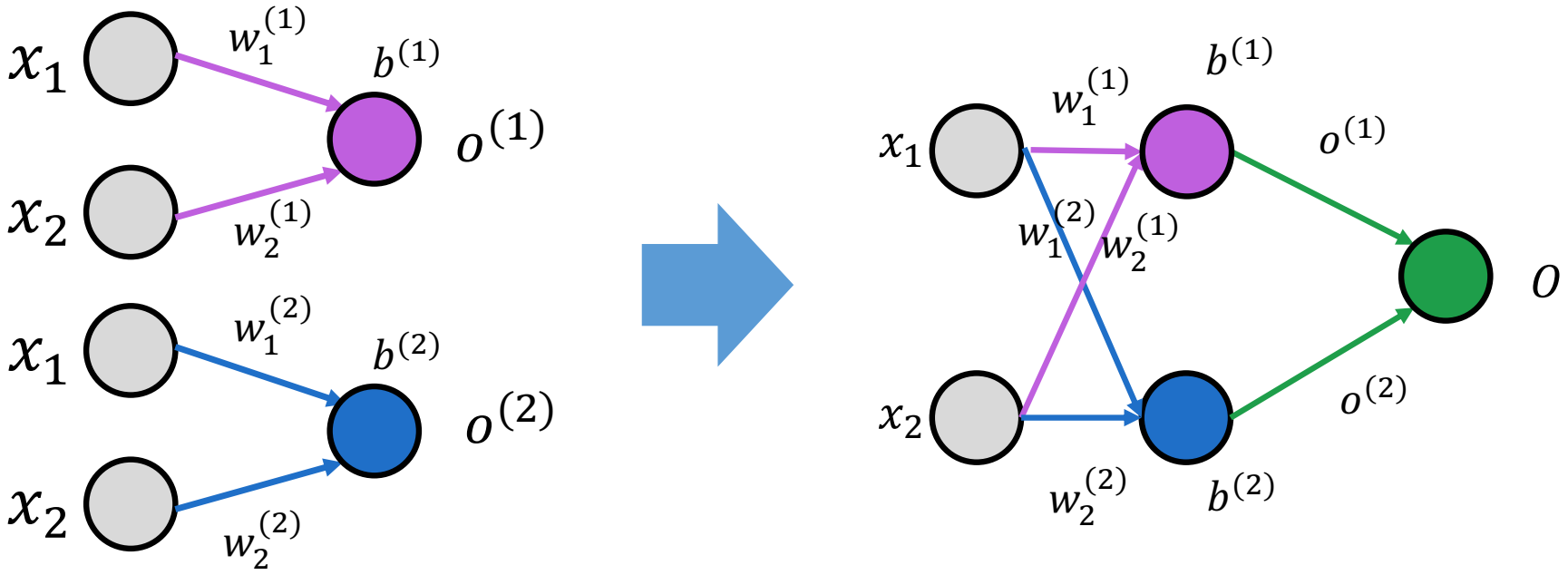


$$o^{(1)} = w_1^{(1)}x_1 + w_2^{(1)}x_2 + b^{(1)}$$

$$o^{(2)} = w_1^{(2)}x_1 + w_2^{(2)}x_2 + b^{(2)}$$

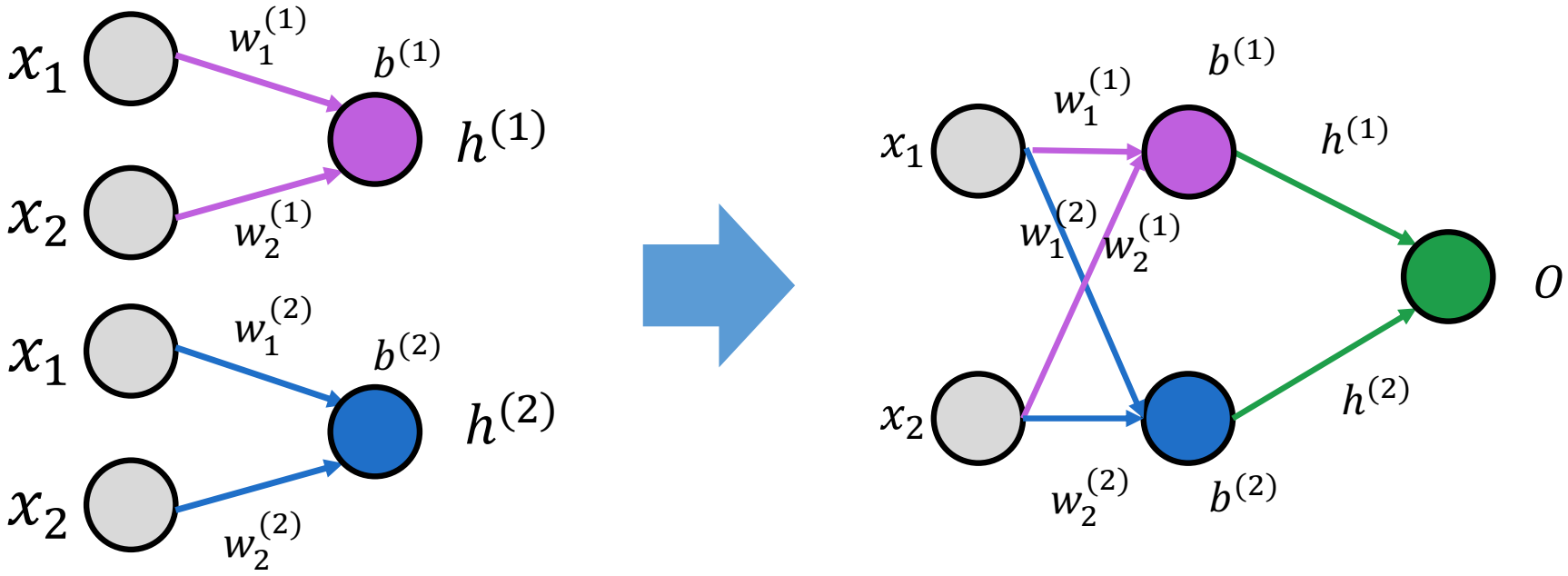
前馈神经网络直观理解

- 叠加起来

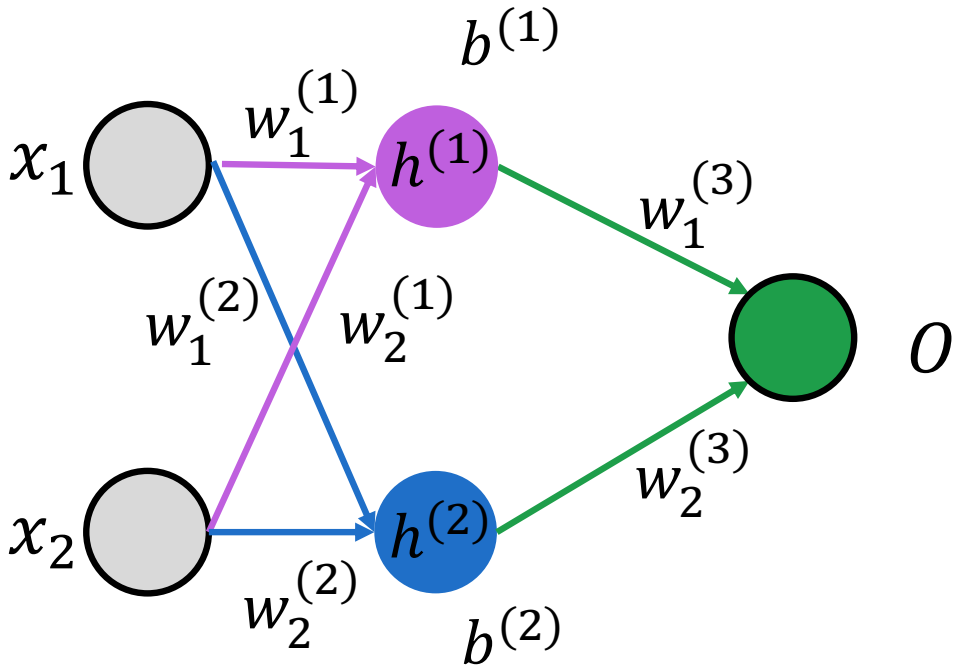


前馈神经网络直观理解

- 叠加起来



前馈神经网络直观理解

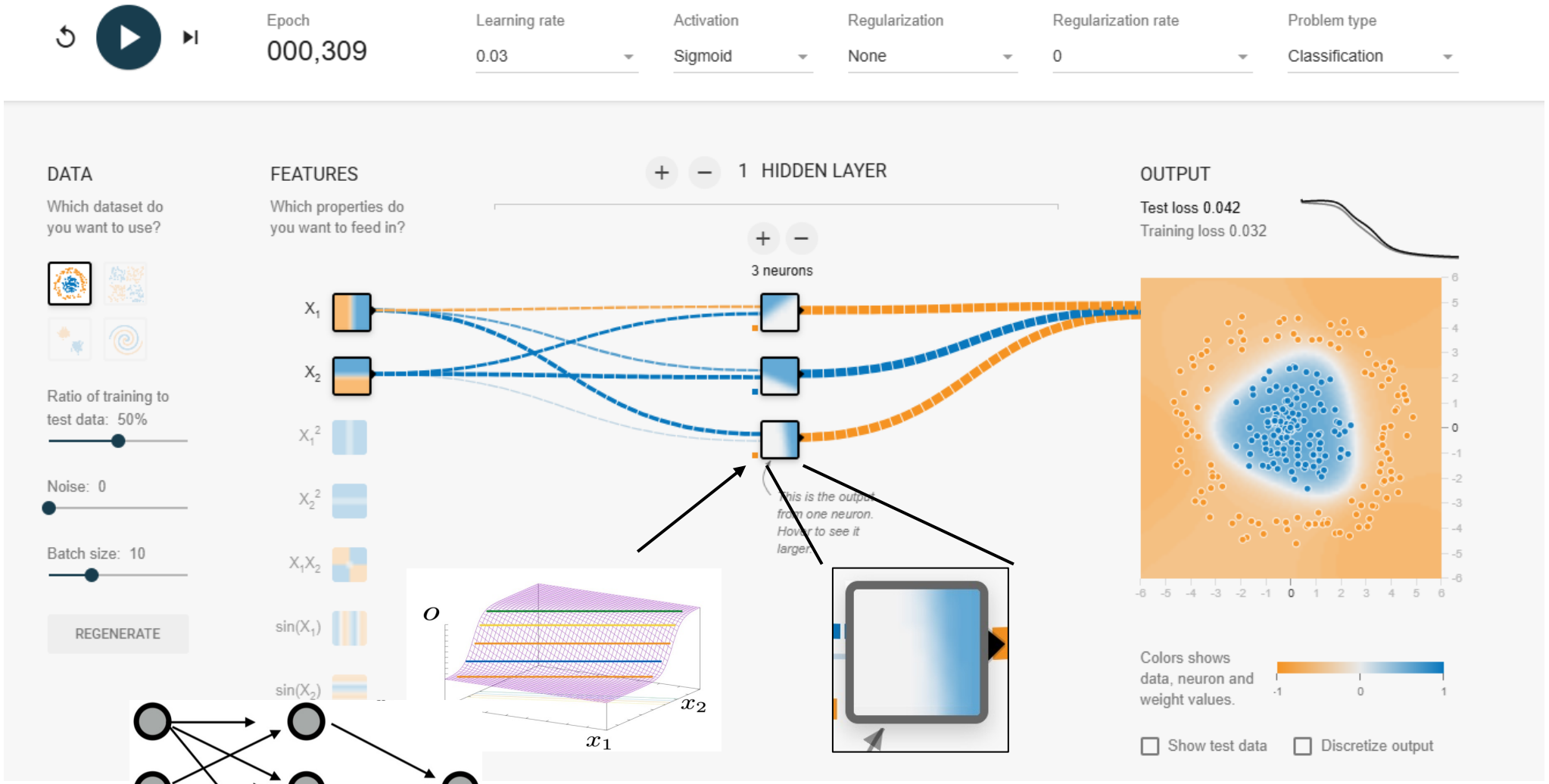


$$h^{(1)} = \sigma(w_1^{(1)}x_1 + w_2^{(1)}x_2 + b^{(1)})$$

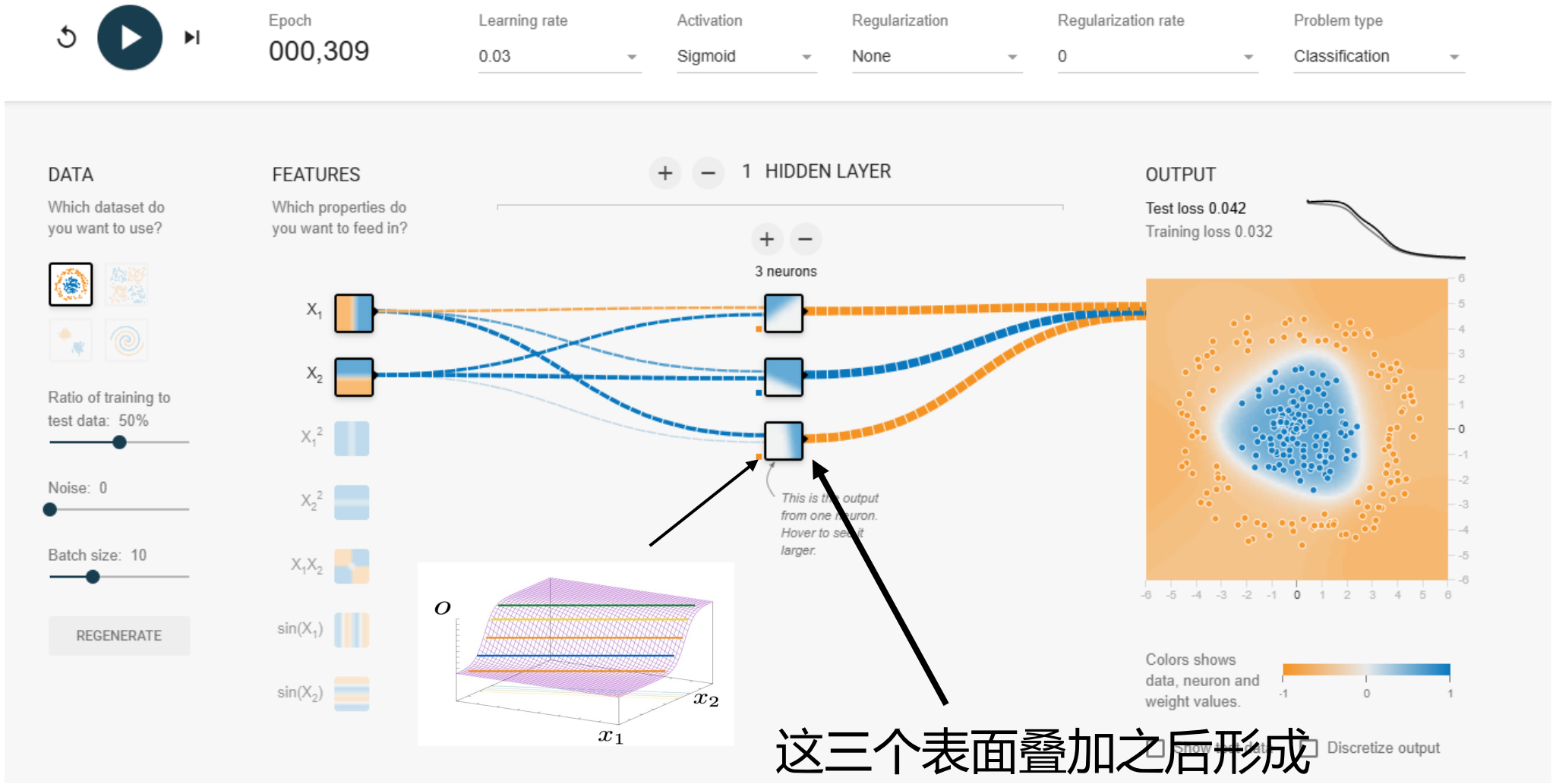
$$h^{(2)} = \sigma(w_1^{(2)}x_1 + w_2^{(2)}x_2 + b^{(2)})$$

$$o = \sigma(w_1^{(3)}h^{(1)} + w_2^{(3)}h^{(2)} + b^{(3)})$$

前馈神经网络直观理解


Epoch
000,309
Learning rate
0.03
Activation
Sigmoid
Regularization
None
Regularization rate
0
Problem type
Classification
DATA
Which dataset do you want to use?
Ratio of training to test data: 50%
Noise: 0
Batch size: 10
REGENERATE
FEATURES
Which properties do you want to feed in?
1 HIDDEN LAYER
3 neurons
OUTPUT
Test loss 0.042
Training loss 0.032
Colors shows data, neuron and weight values.
Show test data
Discretize output

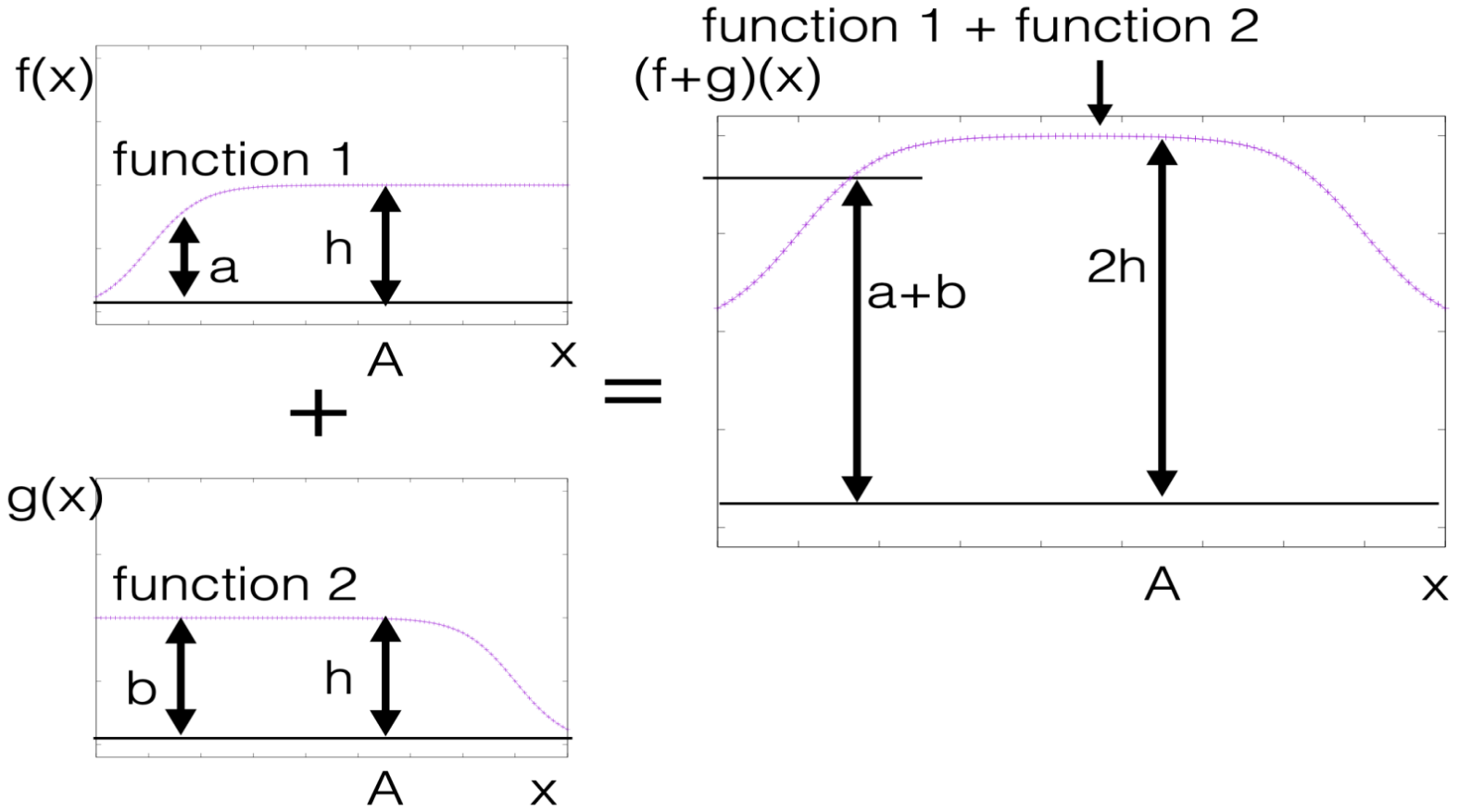
前馈神经网络直观理解



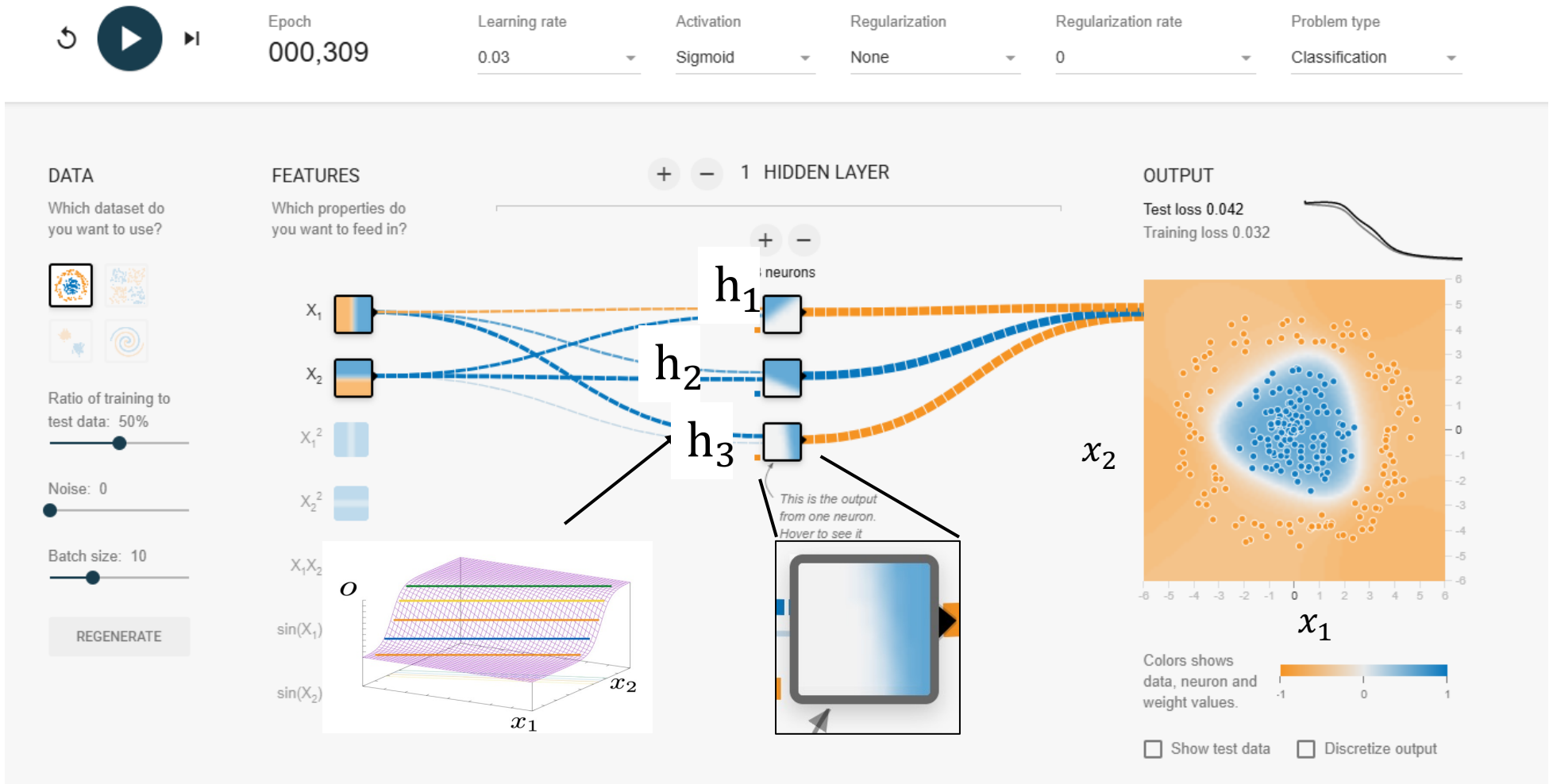
这三个表面叠加之后形成
一个三角形的决策边界

前馈神经网络直观理解

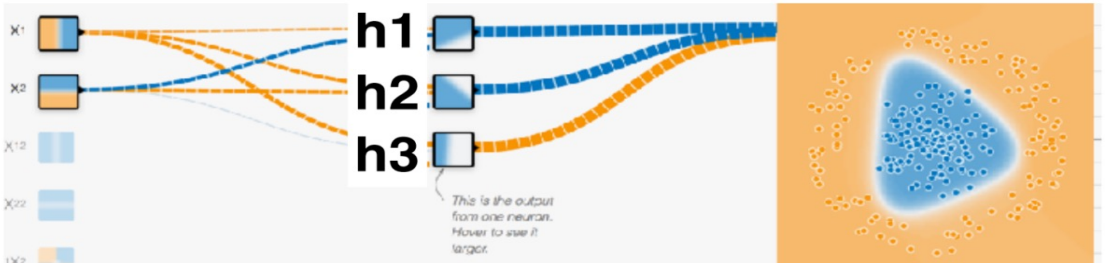
- 函数叠加



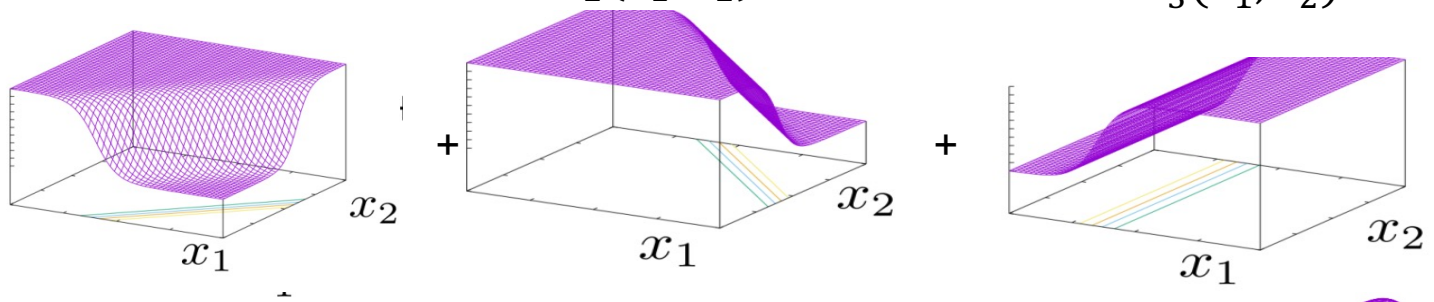
前馈神经网络直观理解


Epoch
000,309
Learning rate
0.03
Activation
Sigmoid
Regularization
None
Regularization rate
0
Problem type
Classification
DATA
Which dataset do you want to use?
Ratio of training to test data: 50%
Noise: 0
Batch size: 10
REGENERATE
FEATURES
Which properties do you want to feed in?
1 HIDDEN LAYER
neurons
X_1
X_2
X_1^2
X_2^2
X_1X_2
$\sin(X_1)$
$\sin(X_2)$
h_1
h_2
h_3
This is the output from one neuron. Hover to see it
o
x_1
x_2
OUTPUT
Test loss 0.042
Training loss 0.032
x_2
x_1
Colors shows data, neuron and weight values.
Show test data
Discretize output

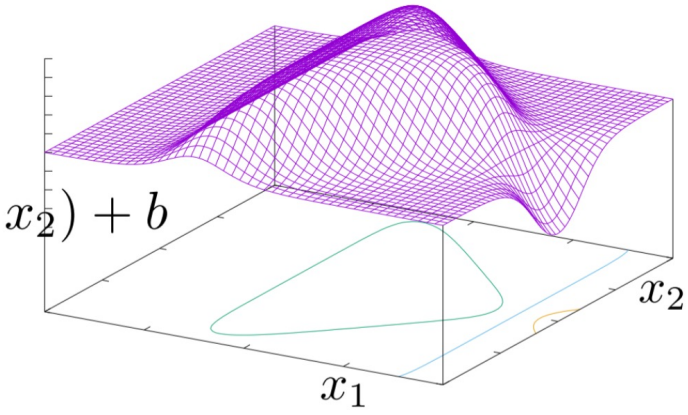
前馈神经网络直观理解



$h_1(x_1, x_2)$ $h_2(x_1, x_2)$ $h_3(x_1, x_2)$

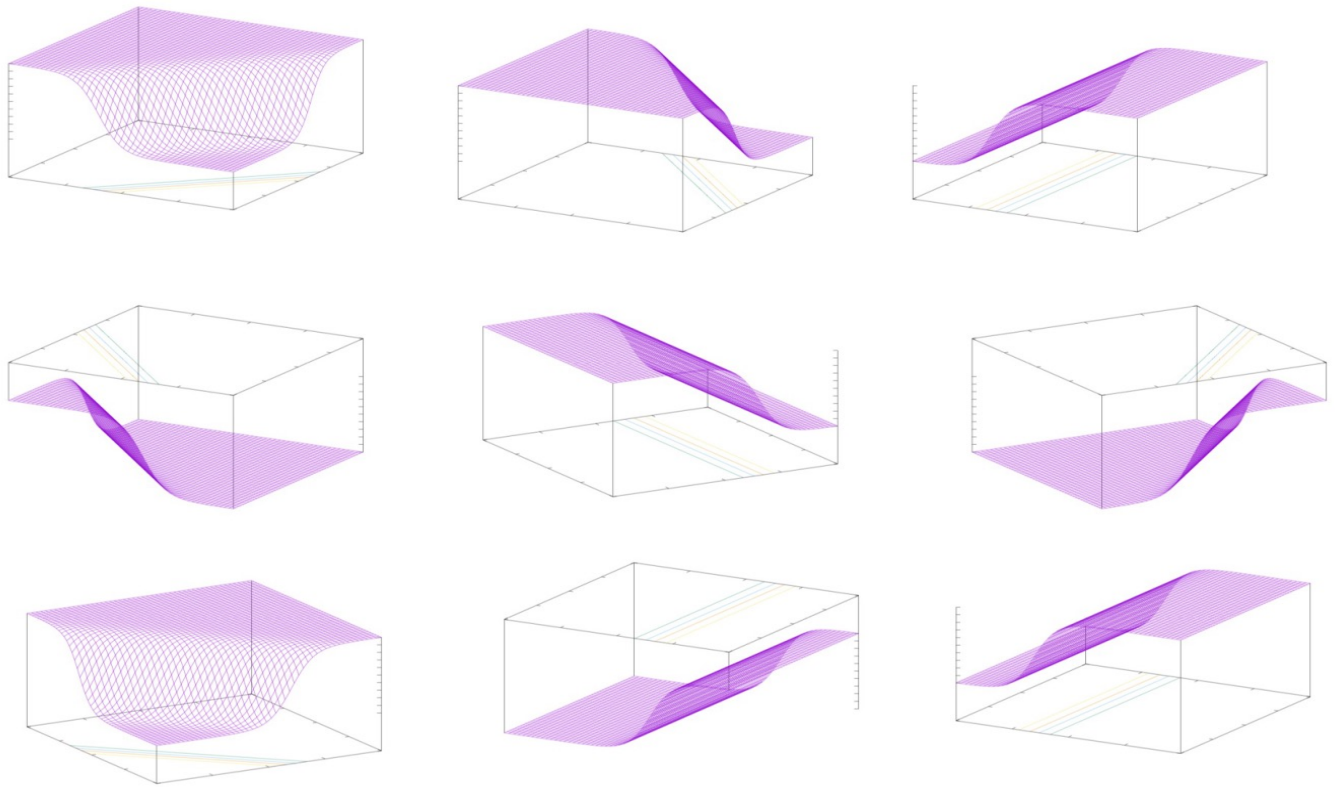


$$w_1 h_1(x_1, x_2) + w_2 h_2(x_1, x_2) + w_3 h_3(x_1, x_2) + b$$



前馈神经网络直观理解

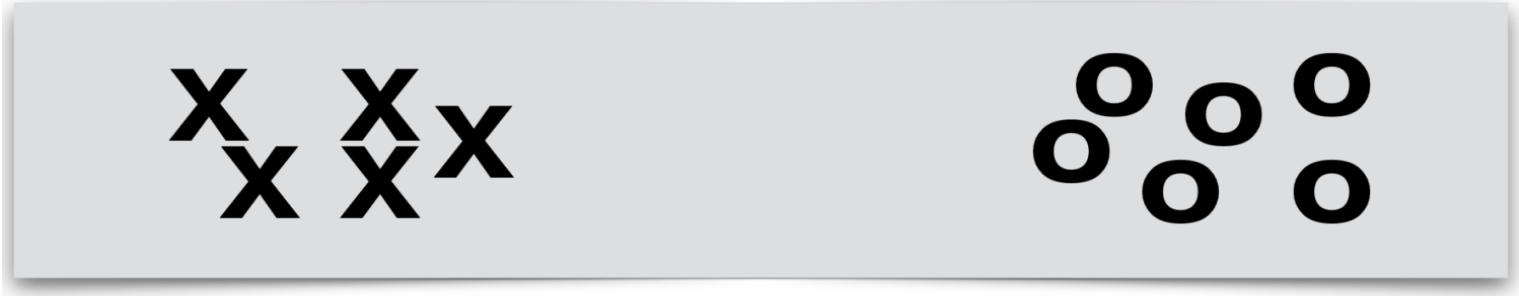
- 如果把足够多的“台阶”曲面叠加起来，能否构造出任意函数?



通用逼近定理

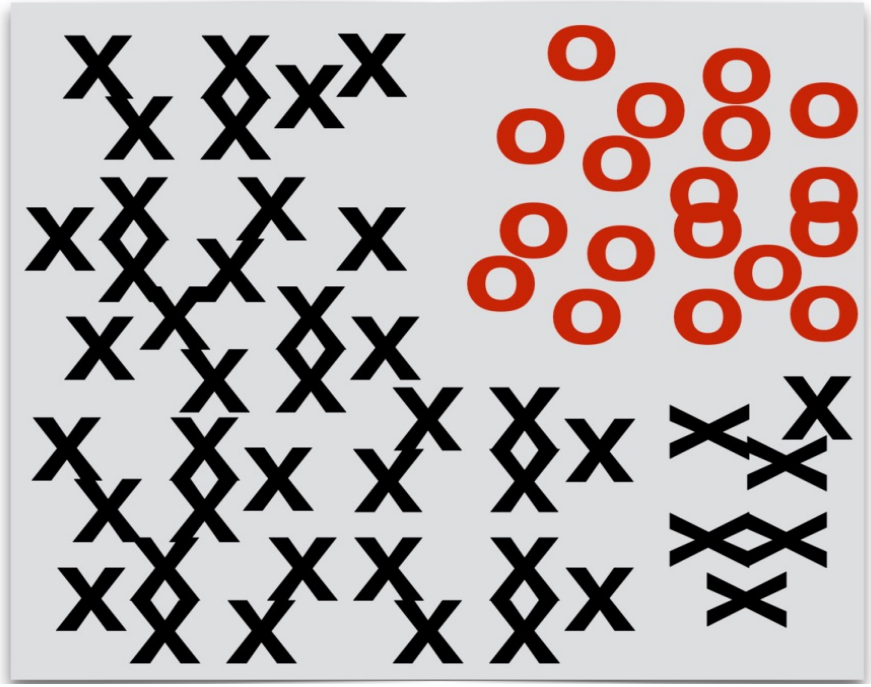
前馈神经网络直观理解

- 游戏示例
  - 拿出一张纸，画出如图所示的图案
  - 任务：**用一条直线**剪开，把“X”和“O”完全分开



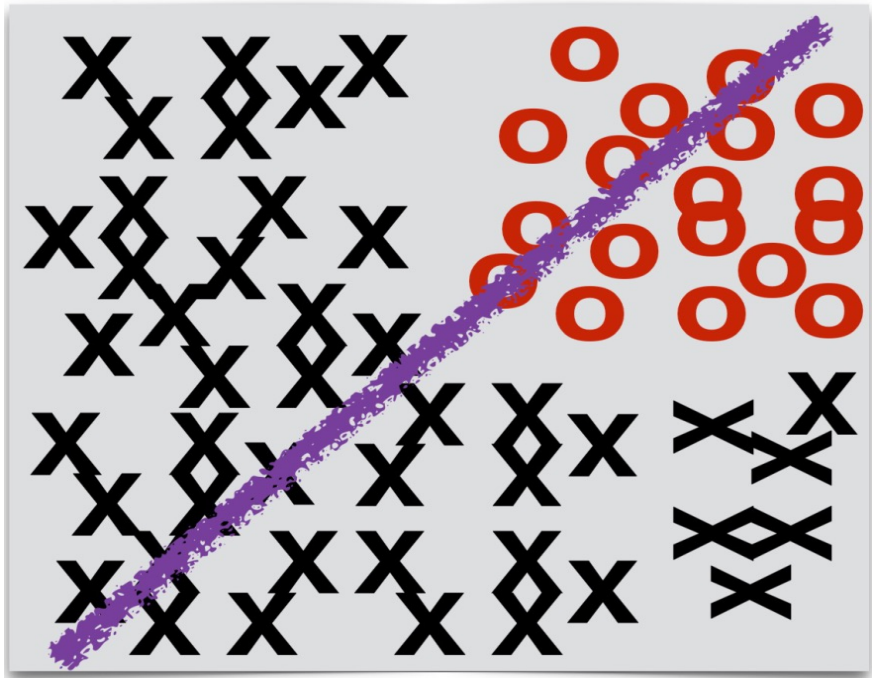
前馈神经网络直观理解

- 用**一条**直线剪开！



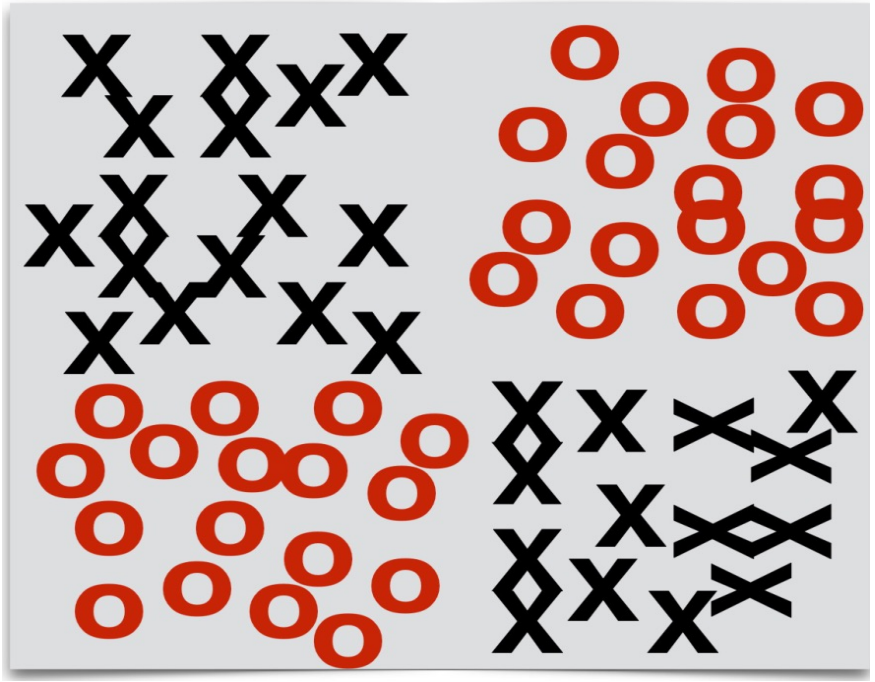
前馈神经网络直观理解

- 用**一条**直线剪开！



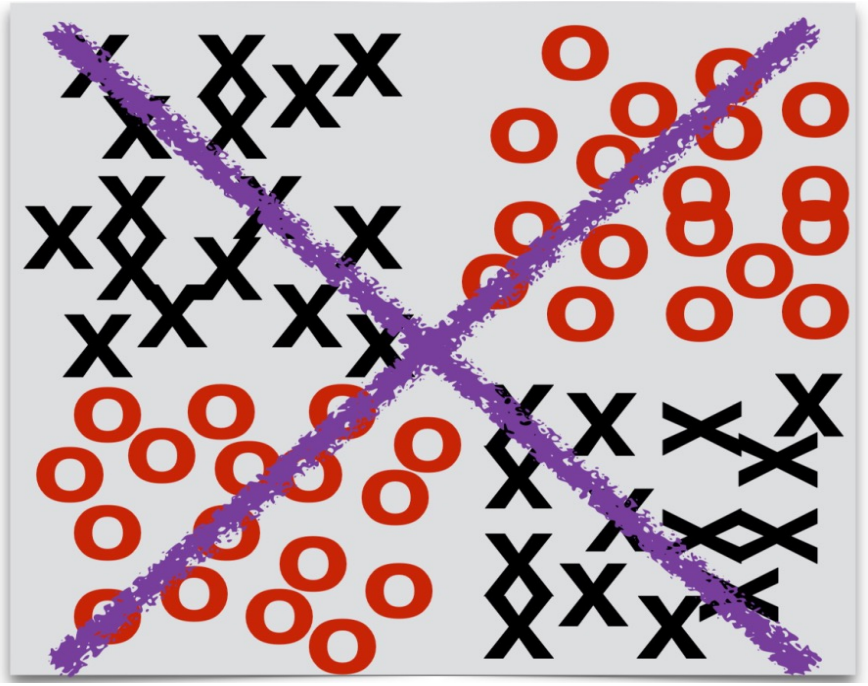
前馈神经网络直观理解

- 用**一条**直线剪开！



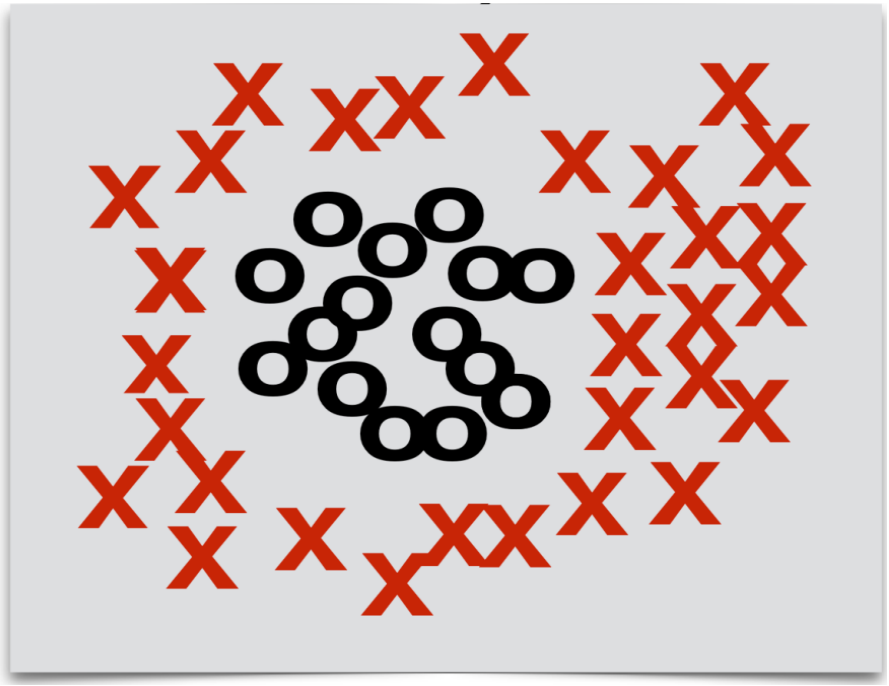
前馈神经网络直观理解

- 用**一条**直线剪开！



前馈神经网络直观理解

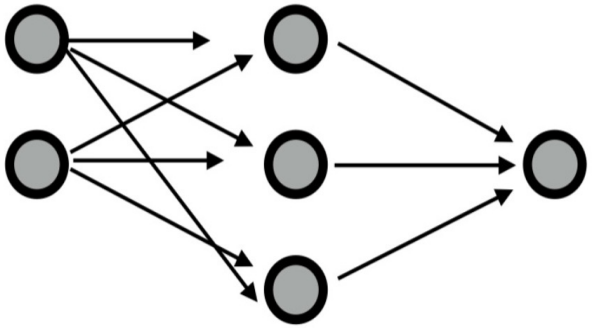
- 用**一条**直线剪开！



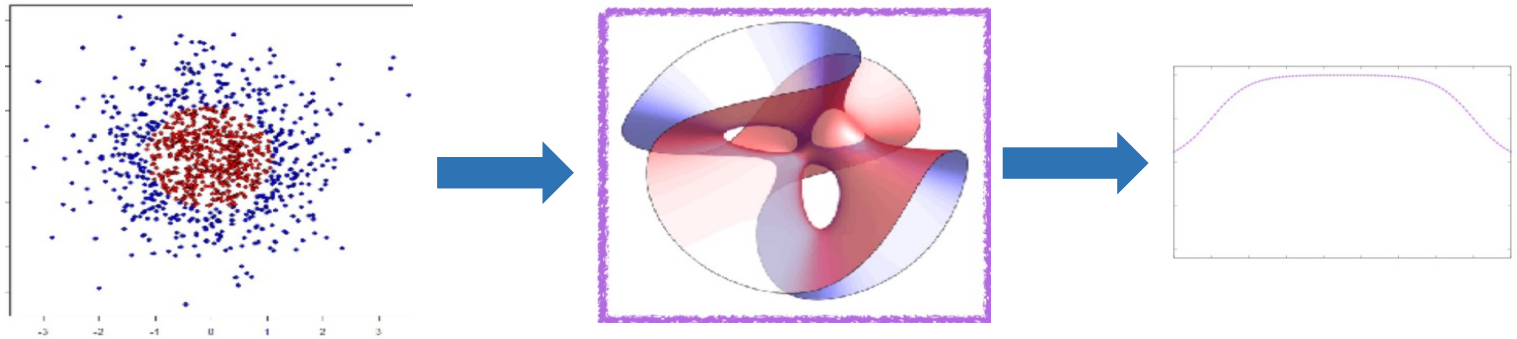
前馈神经网络直观理解

- 神经网络的**数据流形视角**

数据输入
第一层



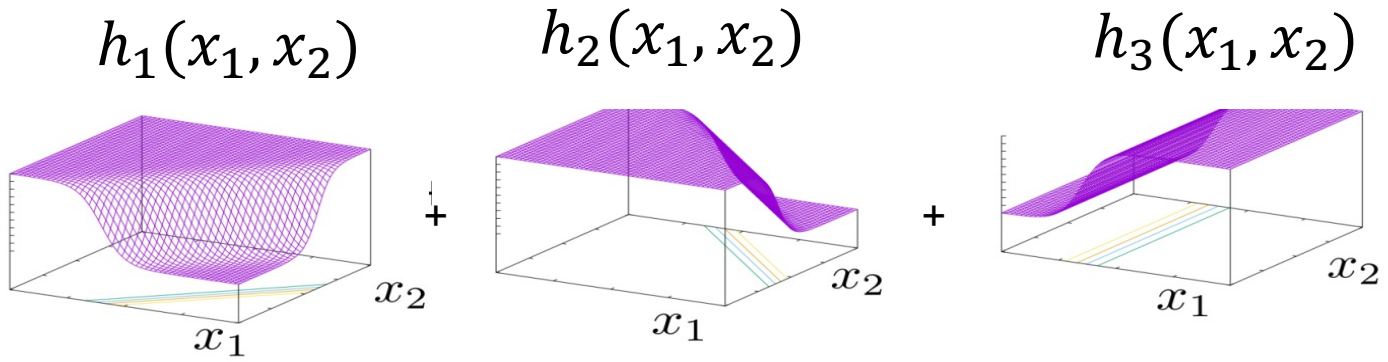
最后一层给出
网络的输出

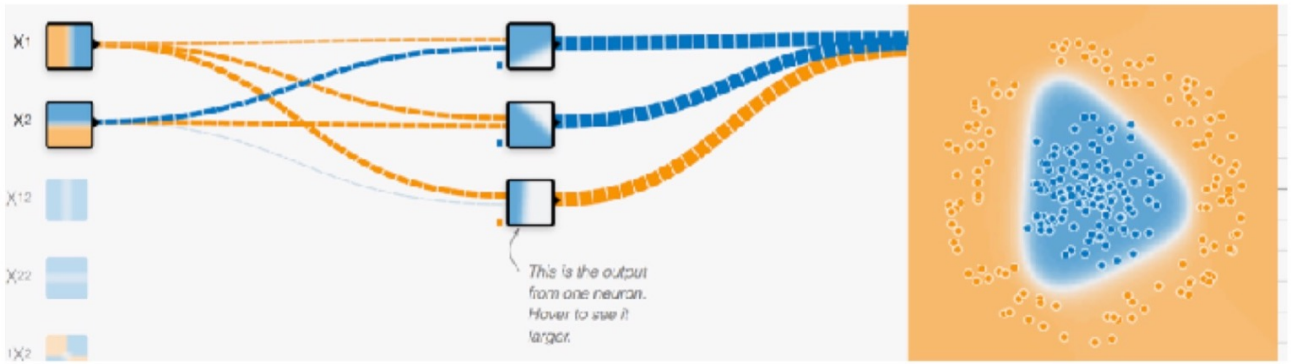


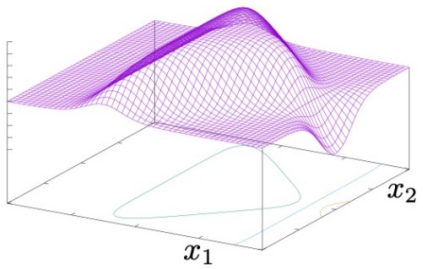
数据　　隐藏层　　输出

前馈神经网络直观理解

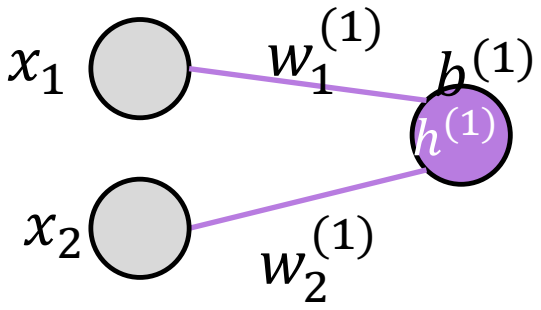
- 神经网络的**函数视角**

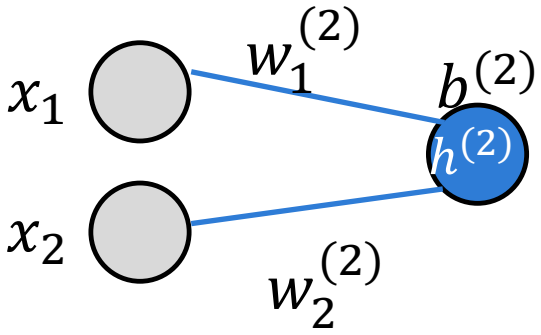




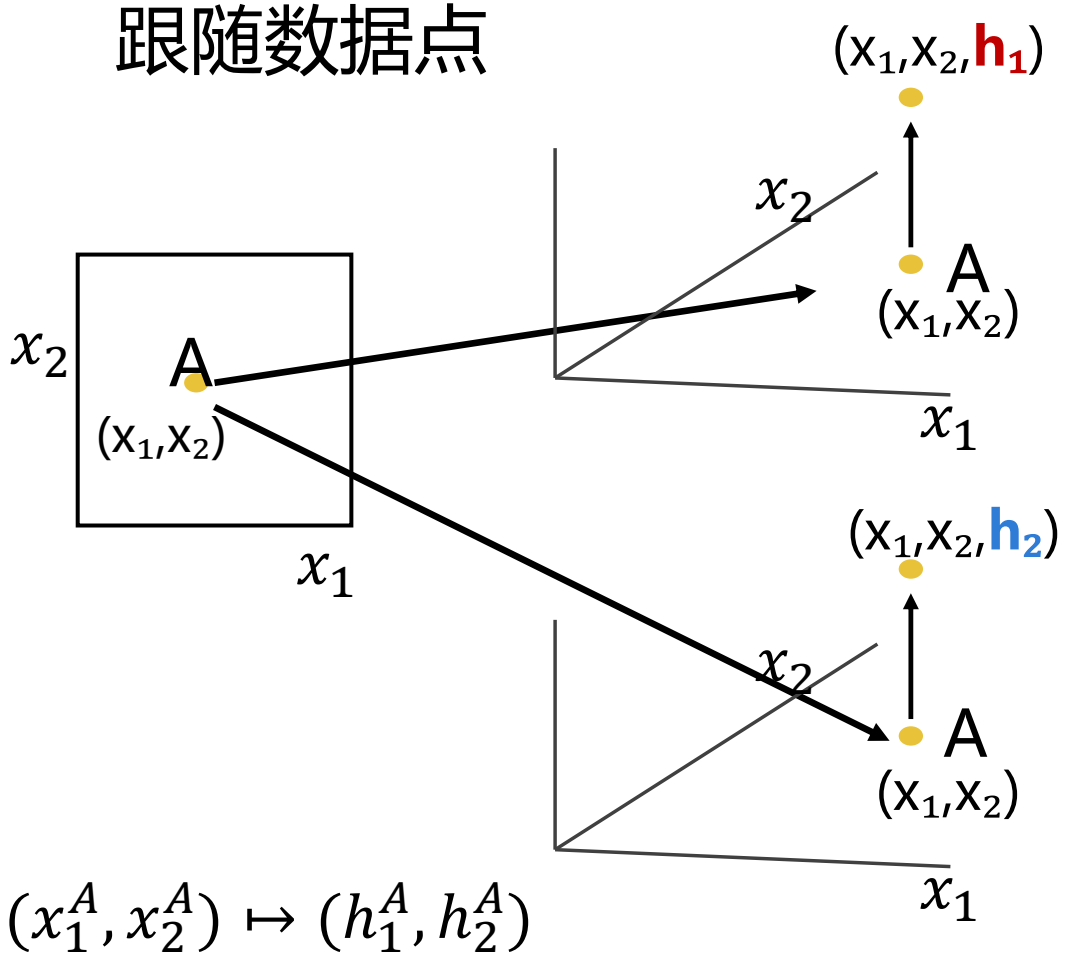


前馈神经网络直观理解



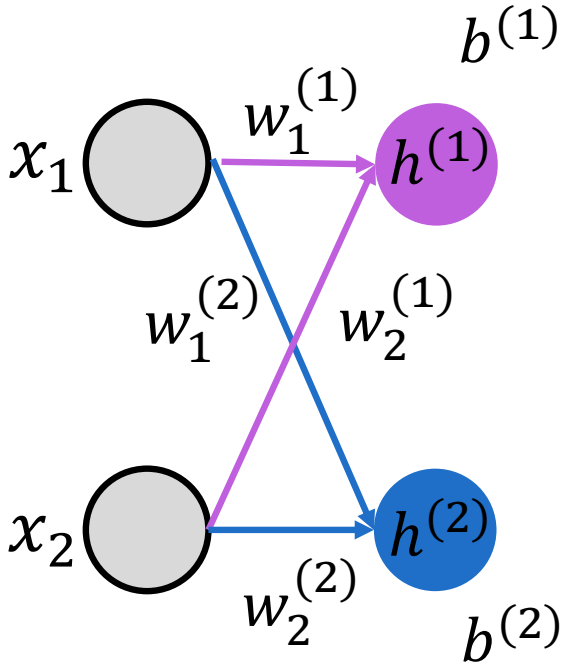


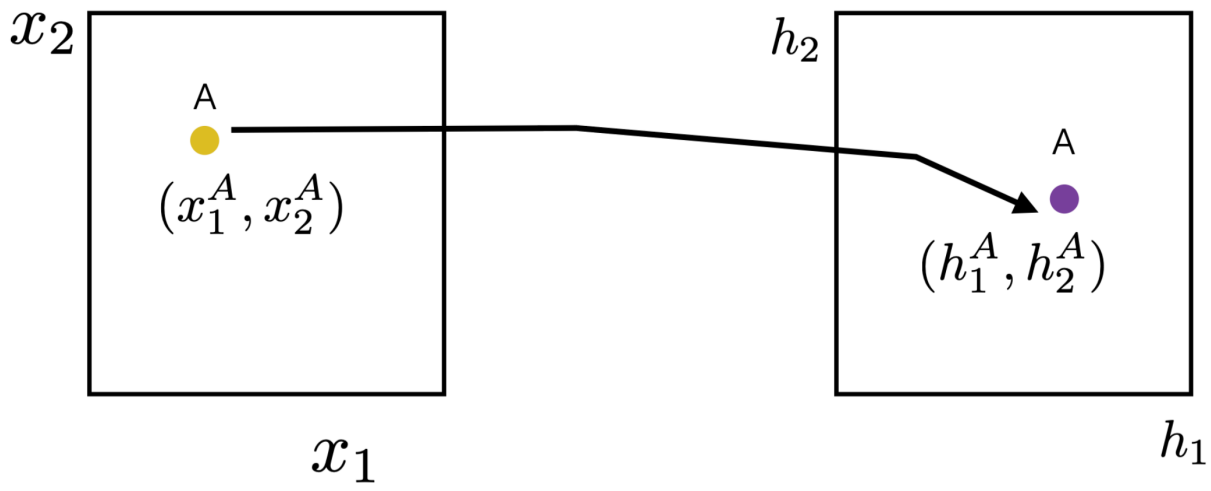
跟随数据点



$$(x_1^A, x_2^A) \mapsto (h_1^A, h_2^A)$$

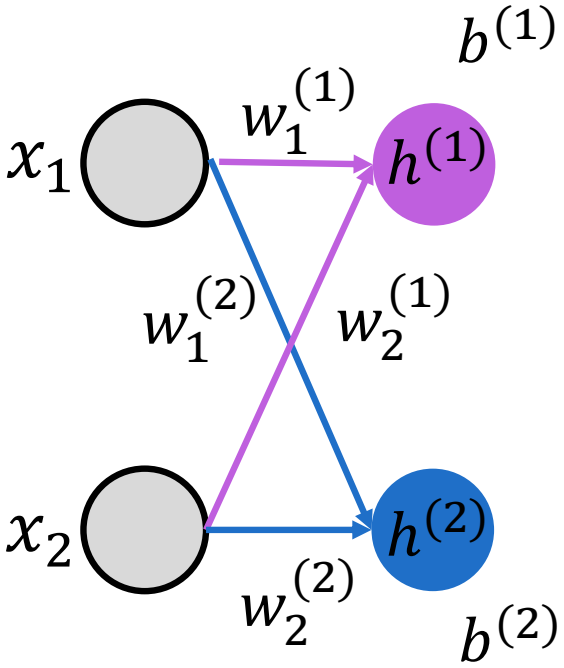
前馈神经网络直观理解

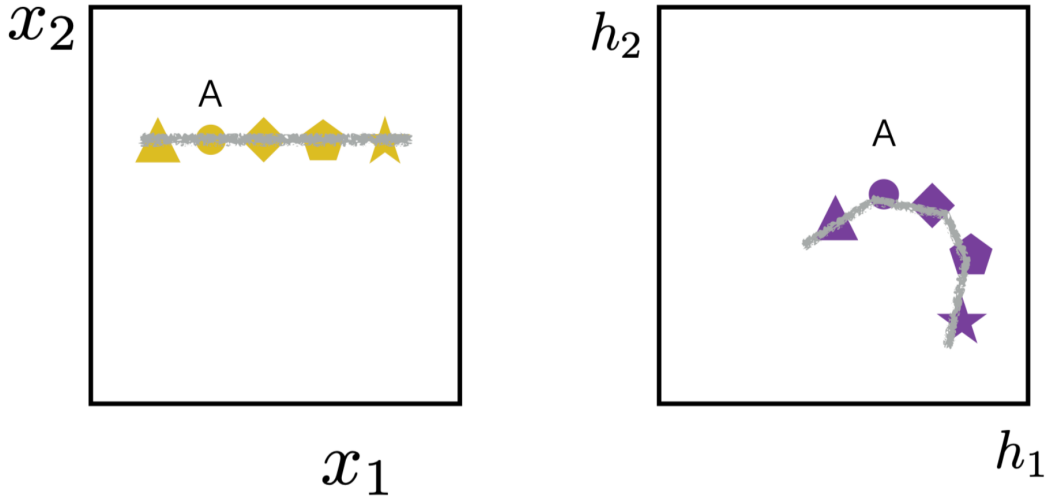




$$(x_1^A, x_2^A) \mapsto (h_1^A, h_2^A)$$

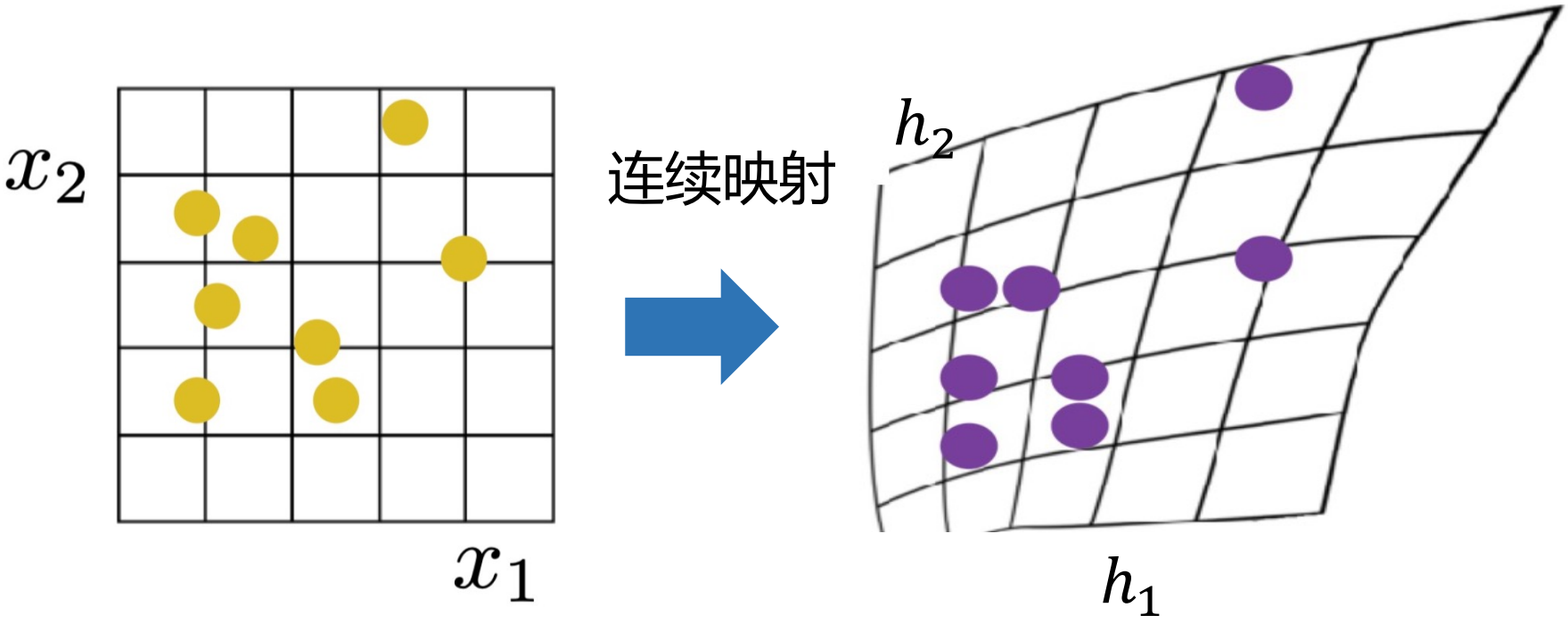
前馈神经网络直观理解





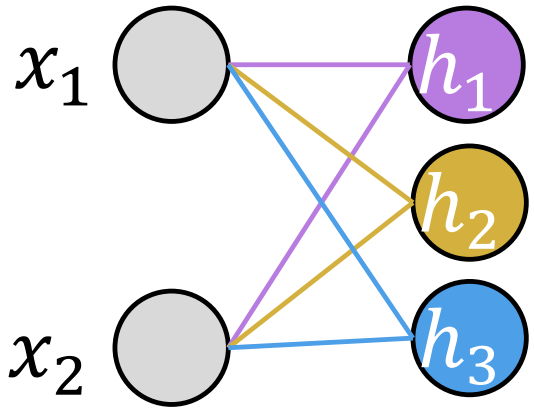
邻域关系保持不变

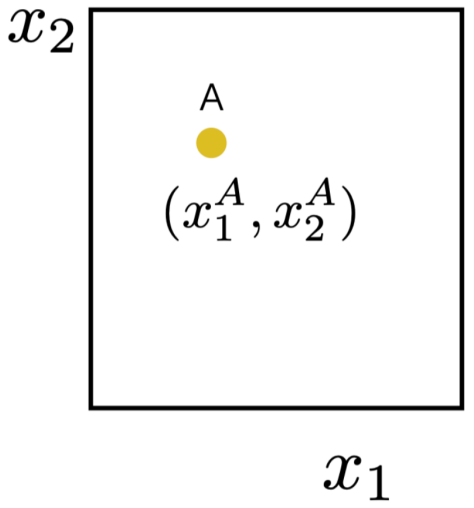
前馈神经网络直观理解

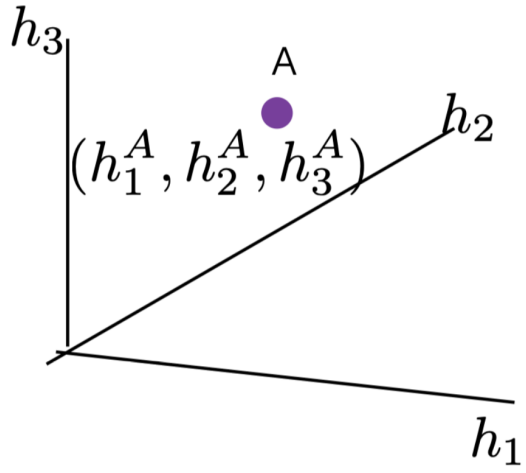


$$(x_1^A, x_2^A) \mapsto (h_1^A, h_2^A)$$

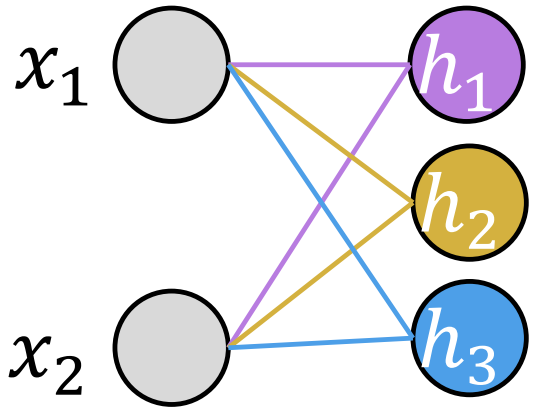
前馈神经网络直观理解

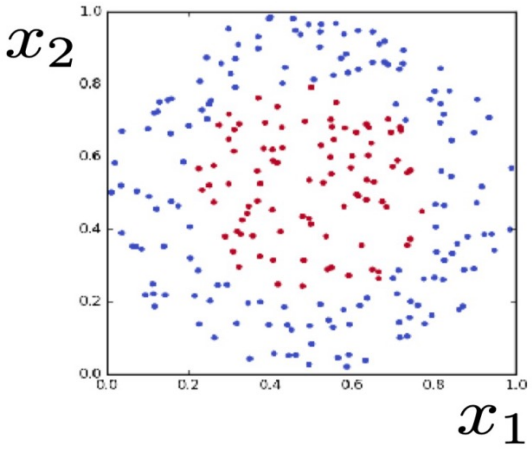


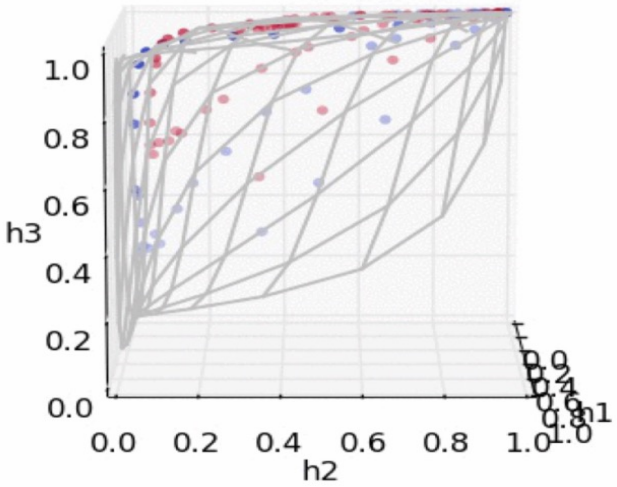




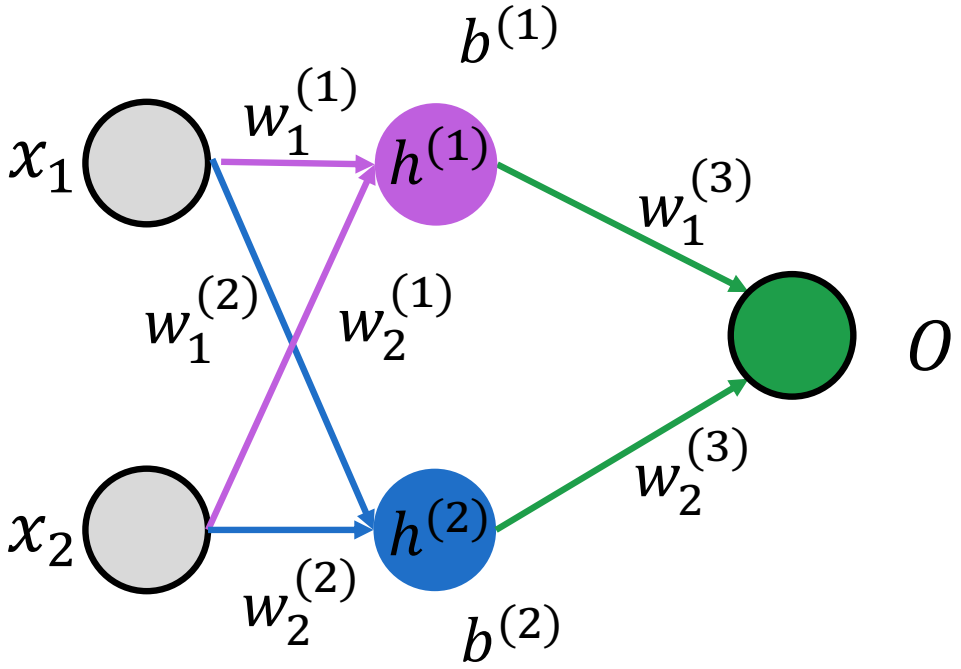
前馈神经网络直观理解

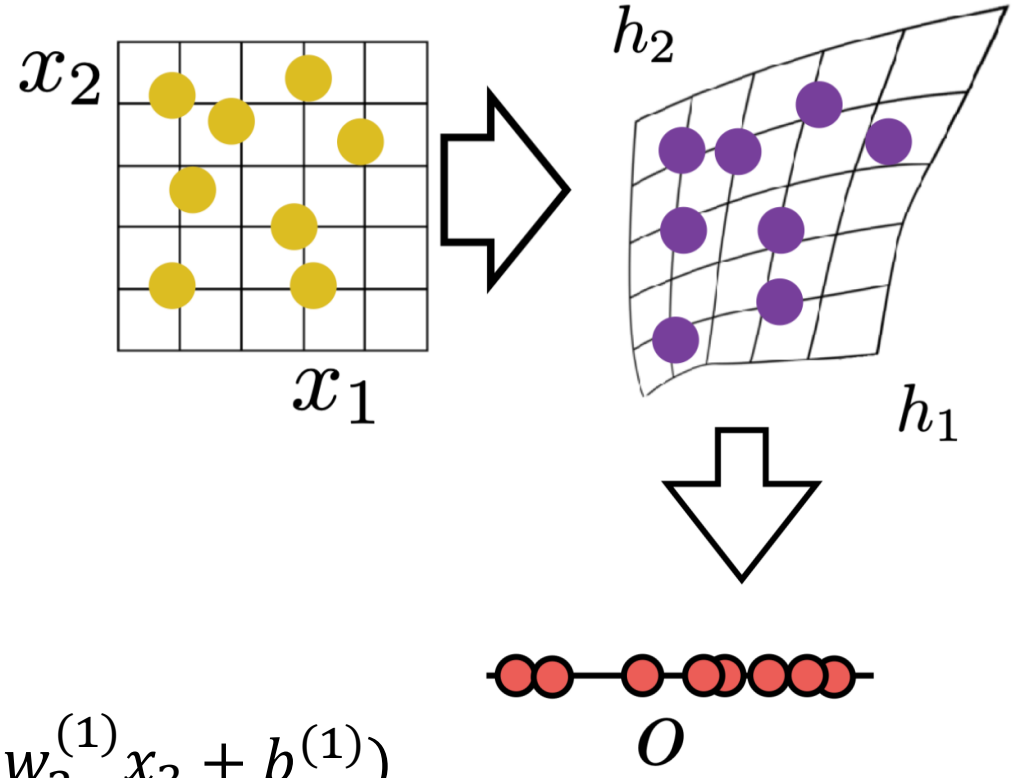






前馈神经网络直观理解





$$h^{(1)} = \sigma(w_1^{(1)}x_1 + w_2^{(1)}x_2 + b^{(1)})$$

$$h^{(2)} = \sigma(w_1^{(2)}x_1 + w_2^{(2)}x_2 + b^{(2)})$$

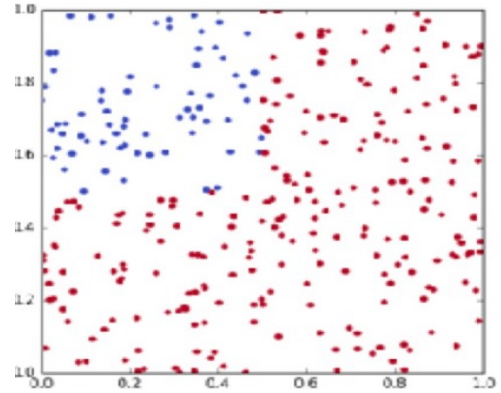
$$o = \sigma(w_1^{(3)}h^{(1)} + w_2^{(3)}h^{(2)} + b^{(3)})$$

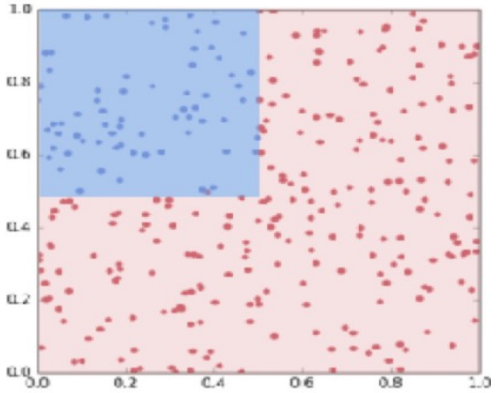
提纲

- 多分类问题
- **前馈神经网络**
  - 表示学习
  - 直观理解
  - 示例
- 神经网络的组成部分
- 深度神经网络

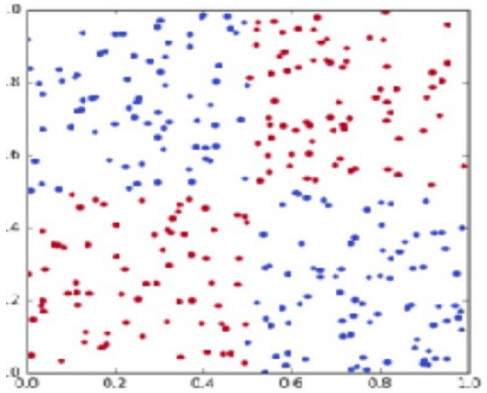
Playground 上的真实例子

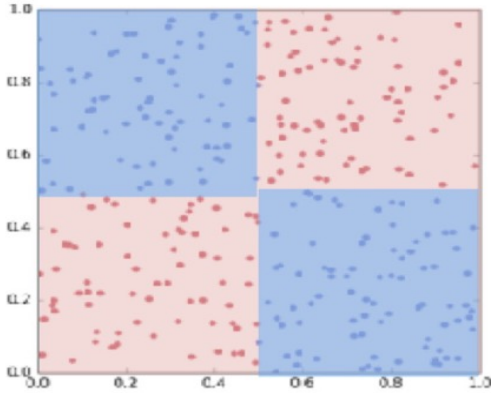
角度数据



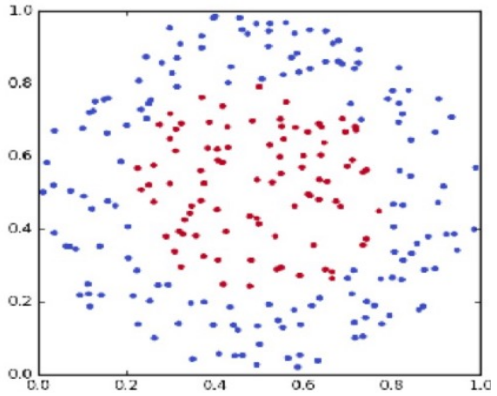


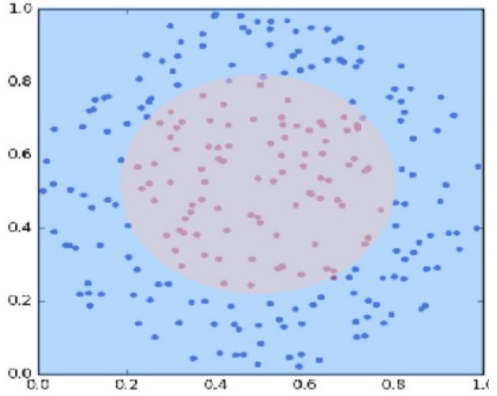
异或数据





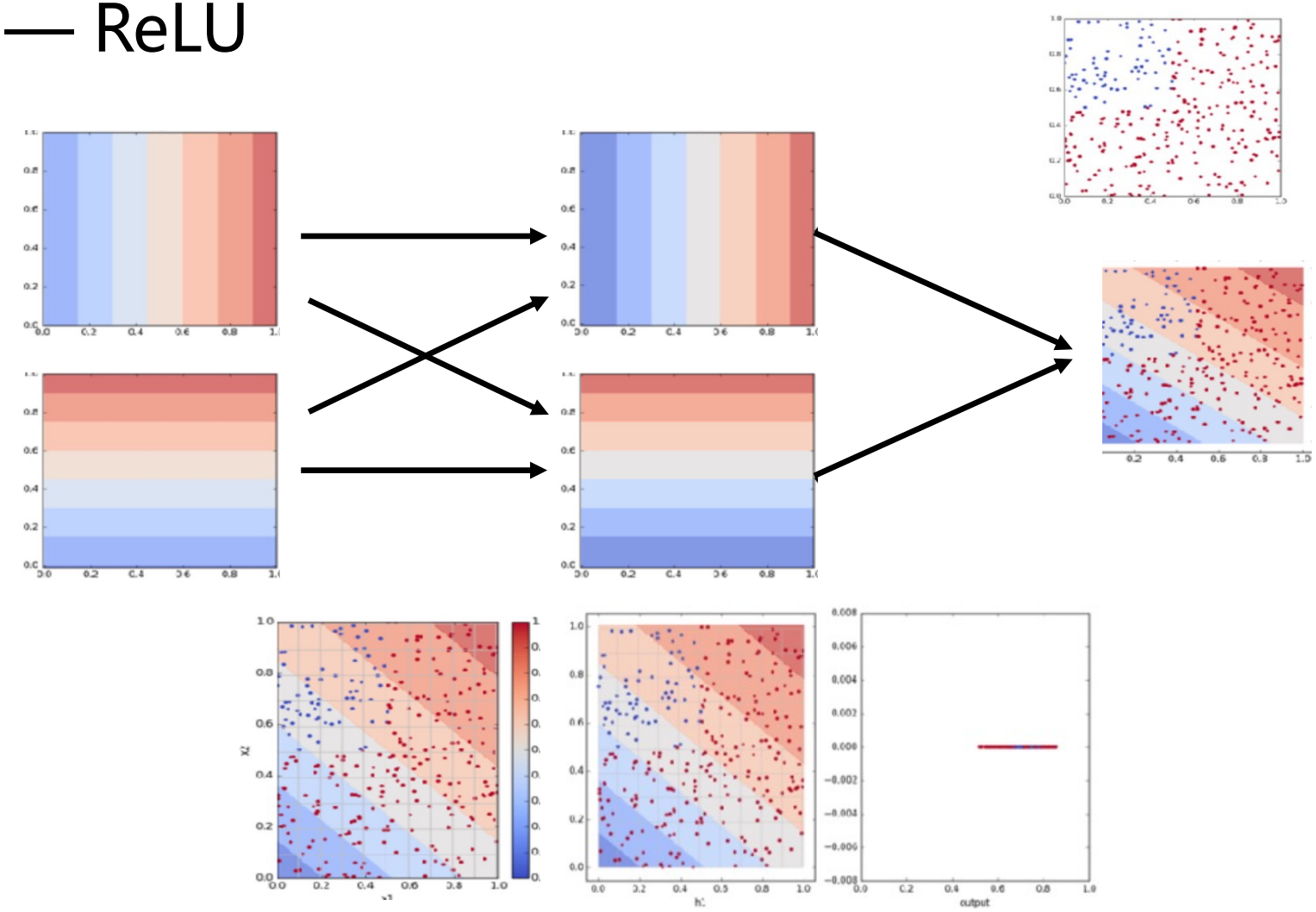
环形数据





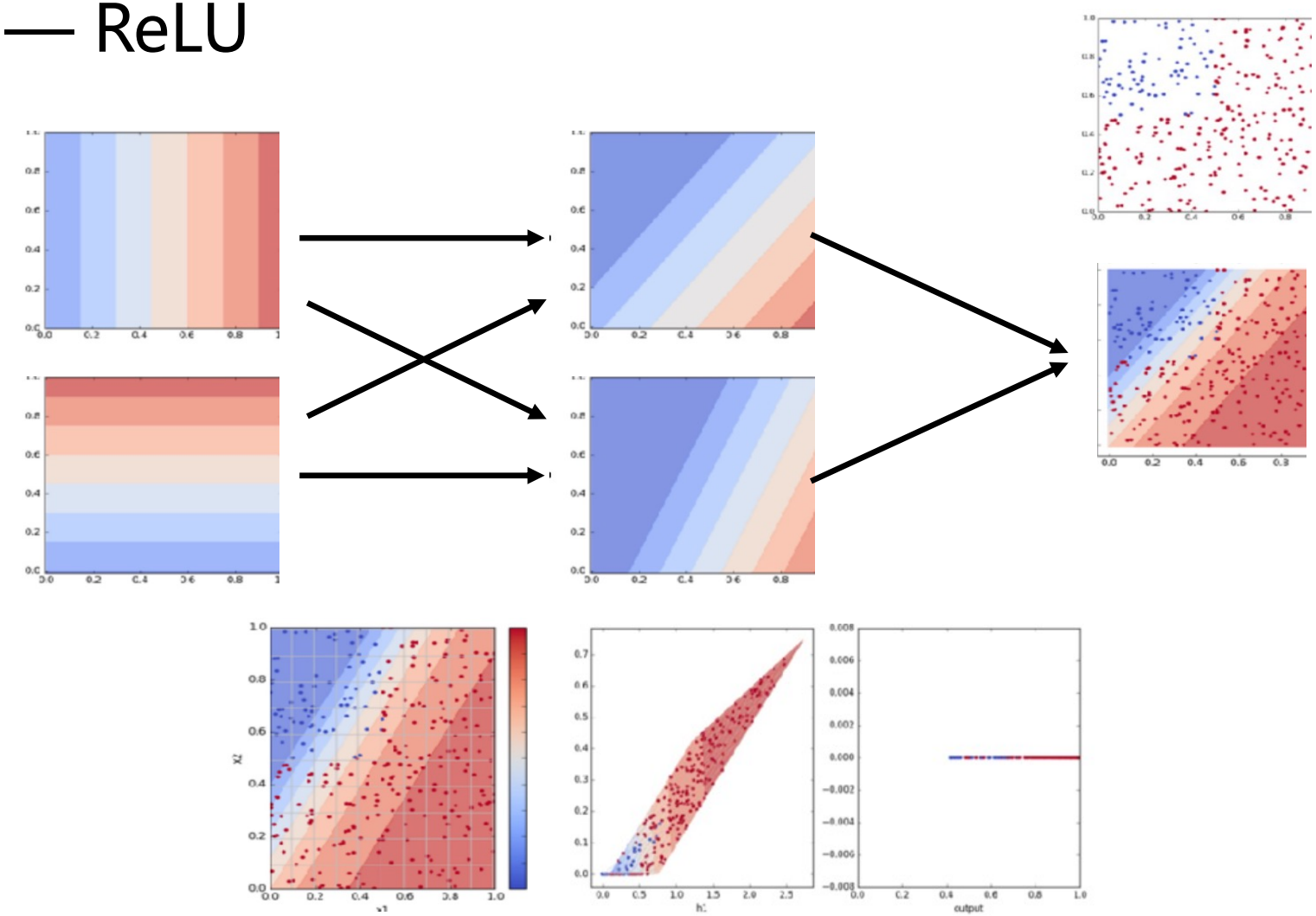
Playground 上的真实例子

- 角度数据 — ReLU



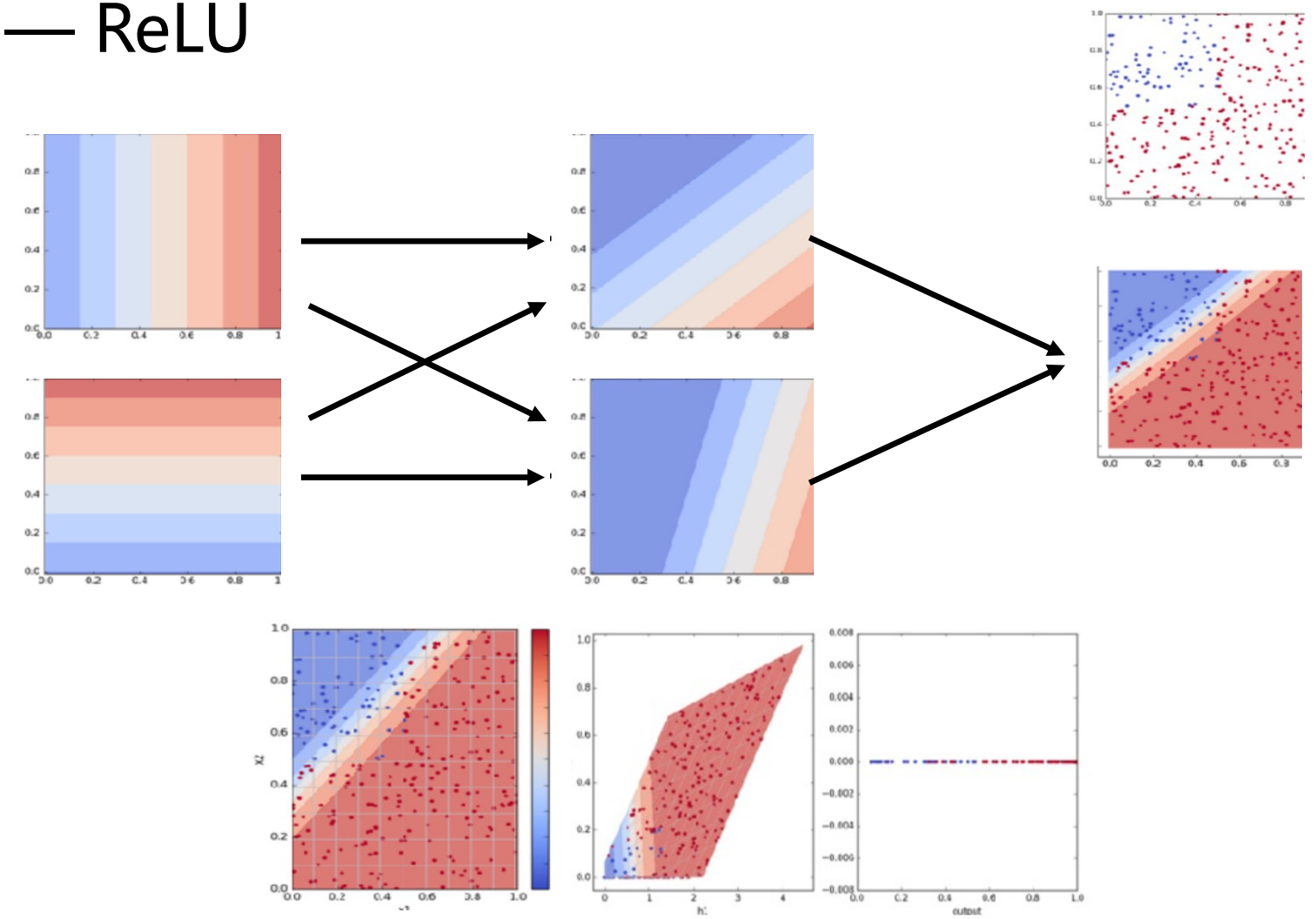
Playground 上的真实例子

- 角度数据 — ReLU



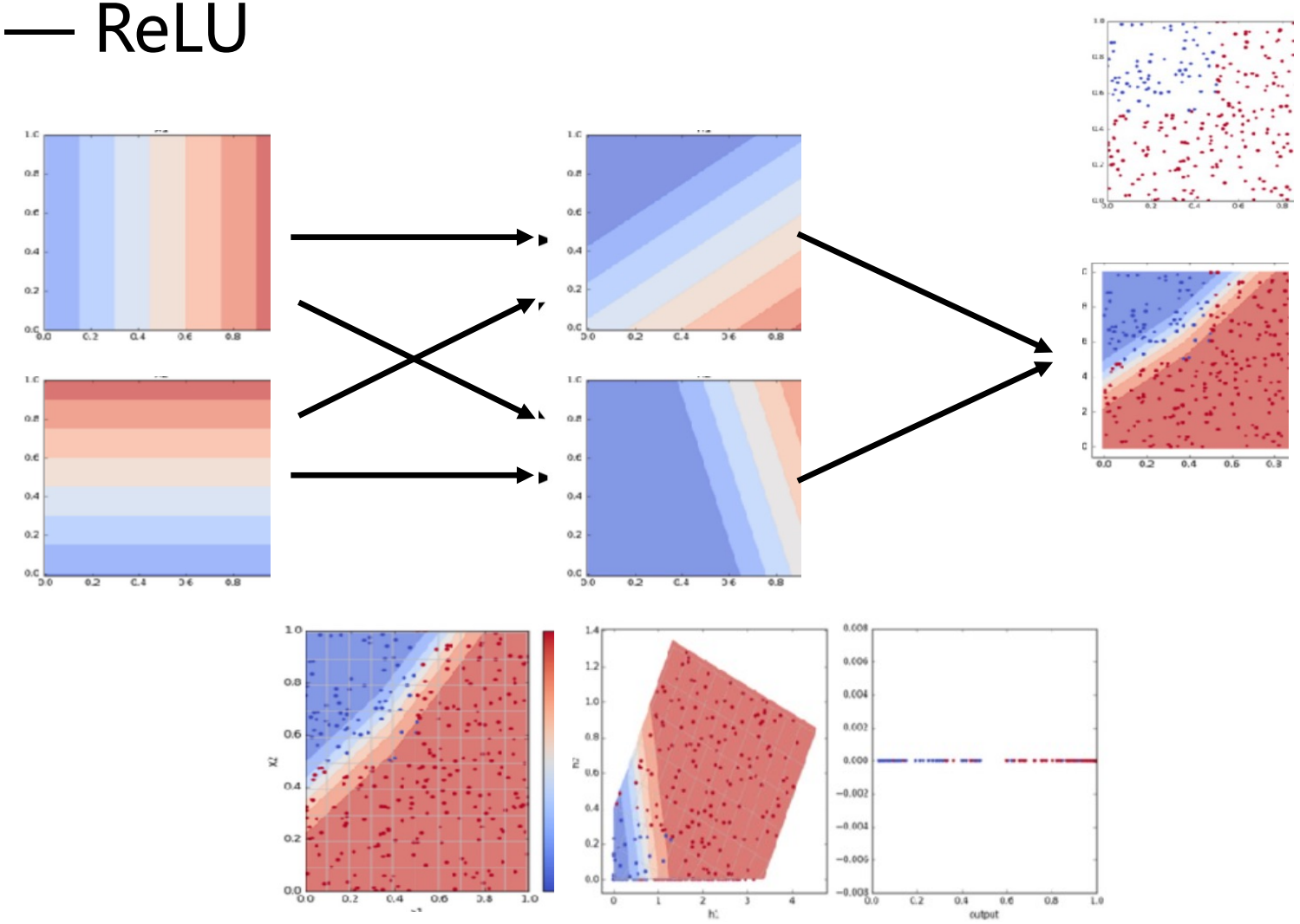
Playground 上的真实例子

- 角度数据 — ReLU



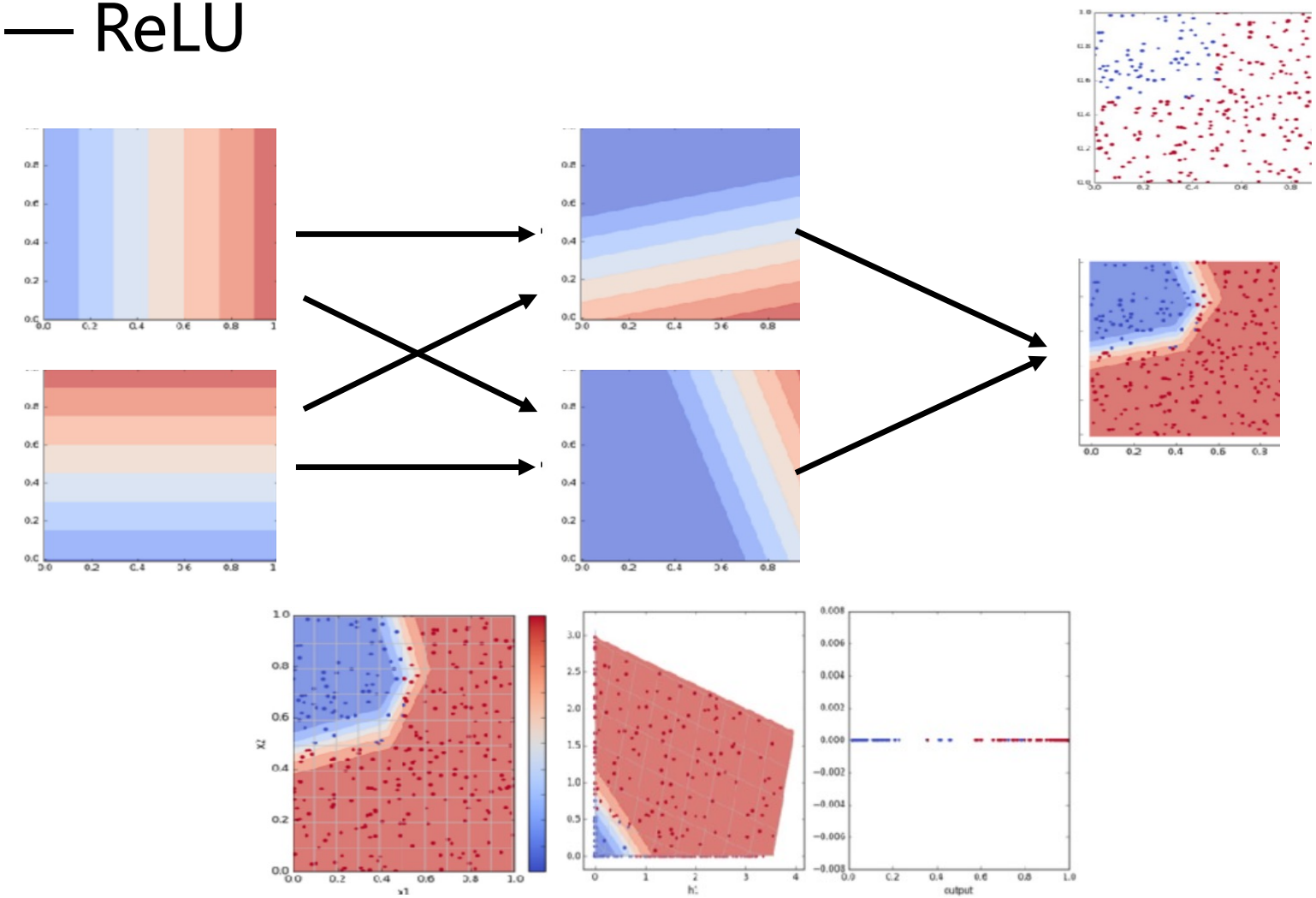
Playground 上的真实例子

- 角度数据 — ReLU



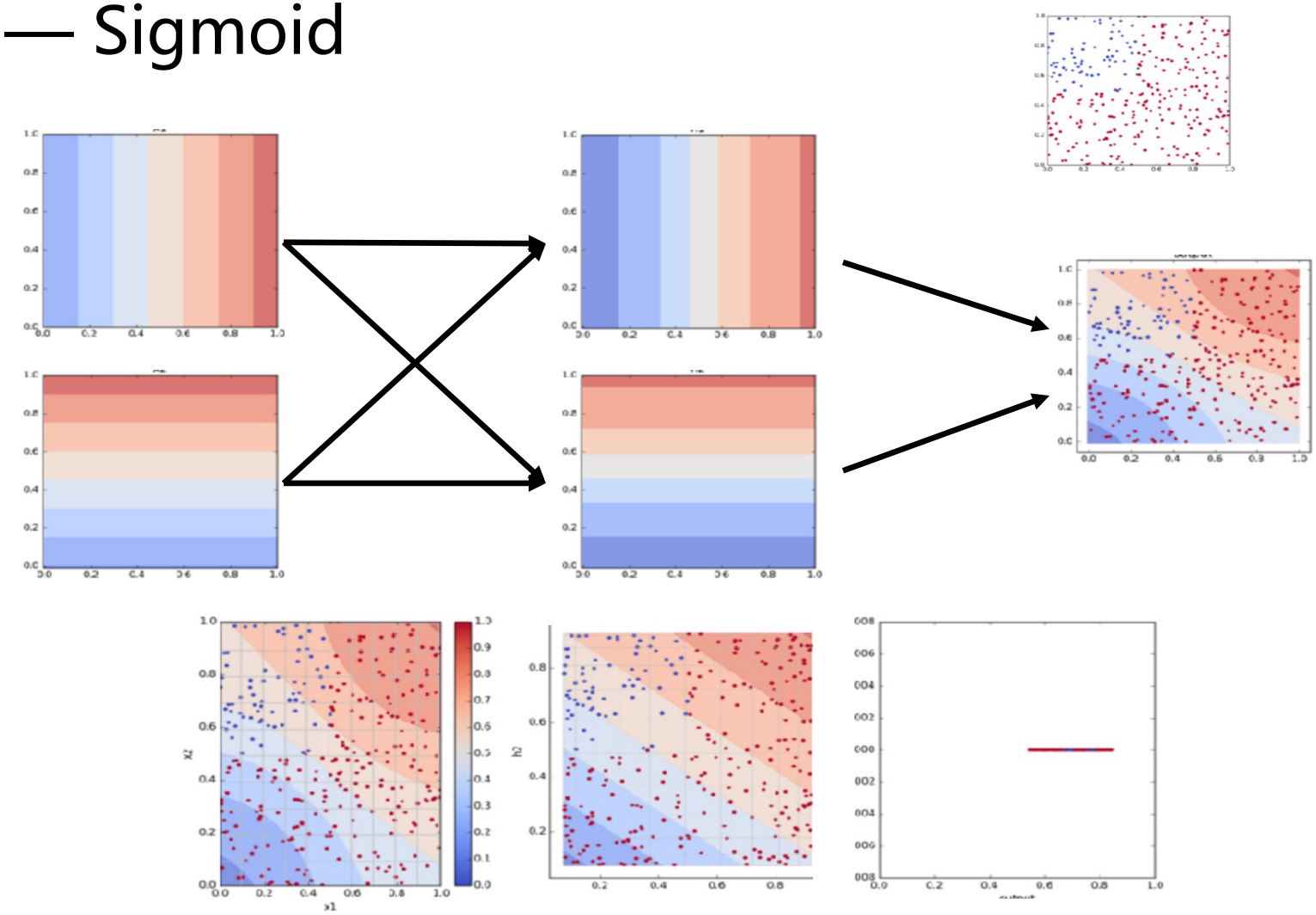
Playground 上的真实例子

- 角度数据 — ReLU



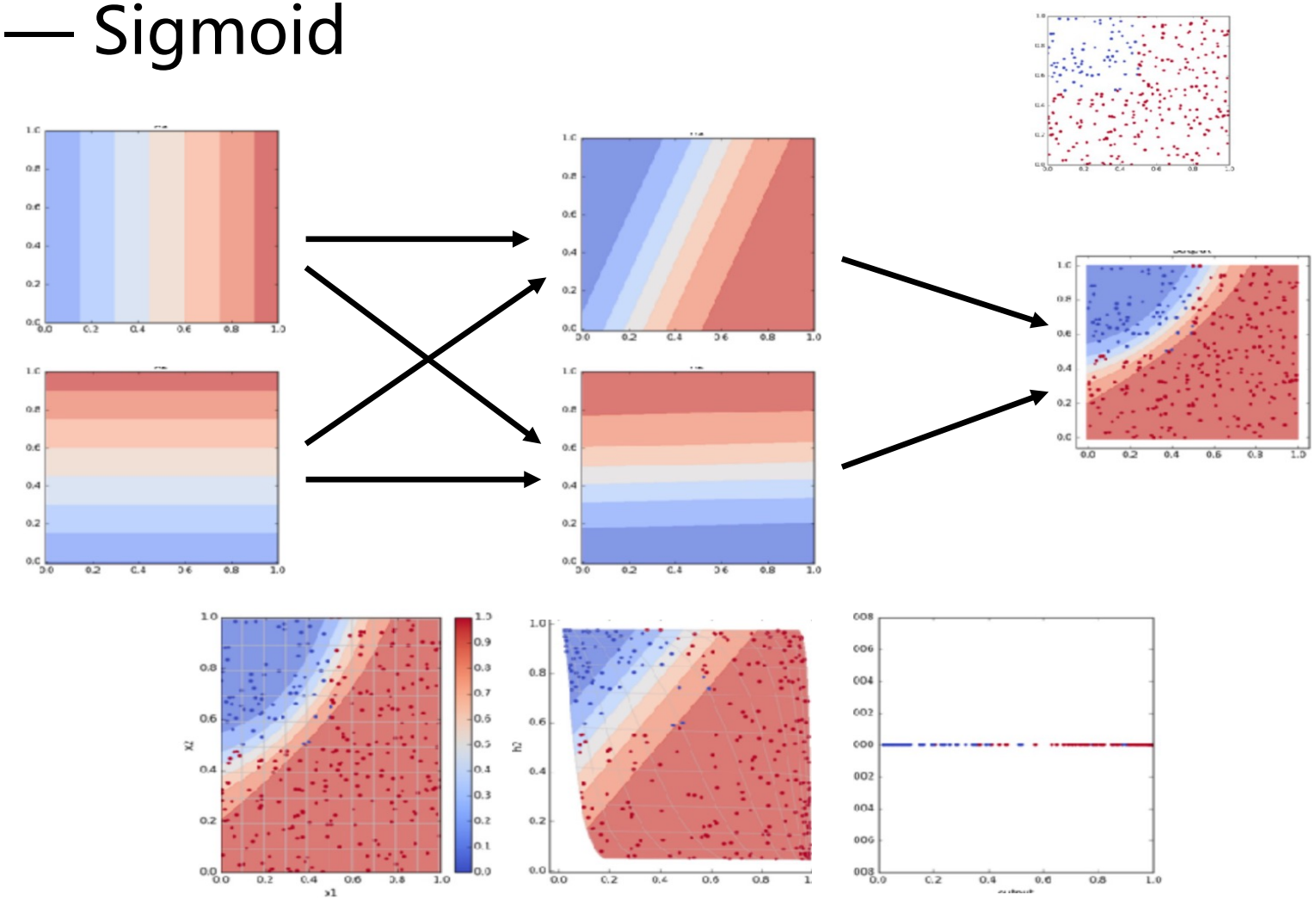
Playground 上的真实例子

- 角度数据 — Sigmoid



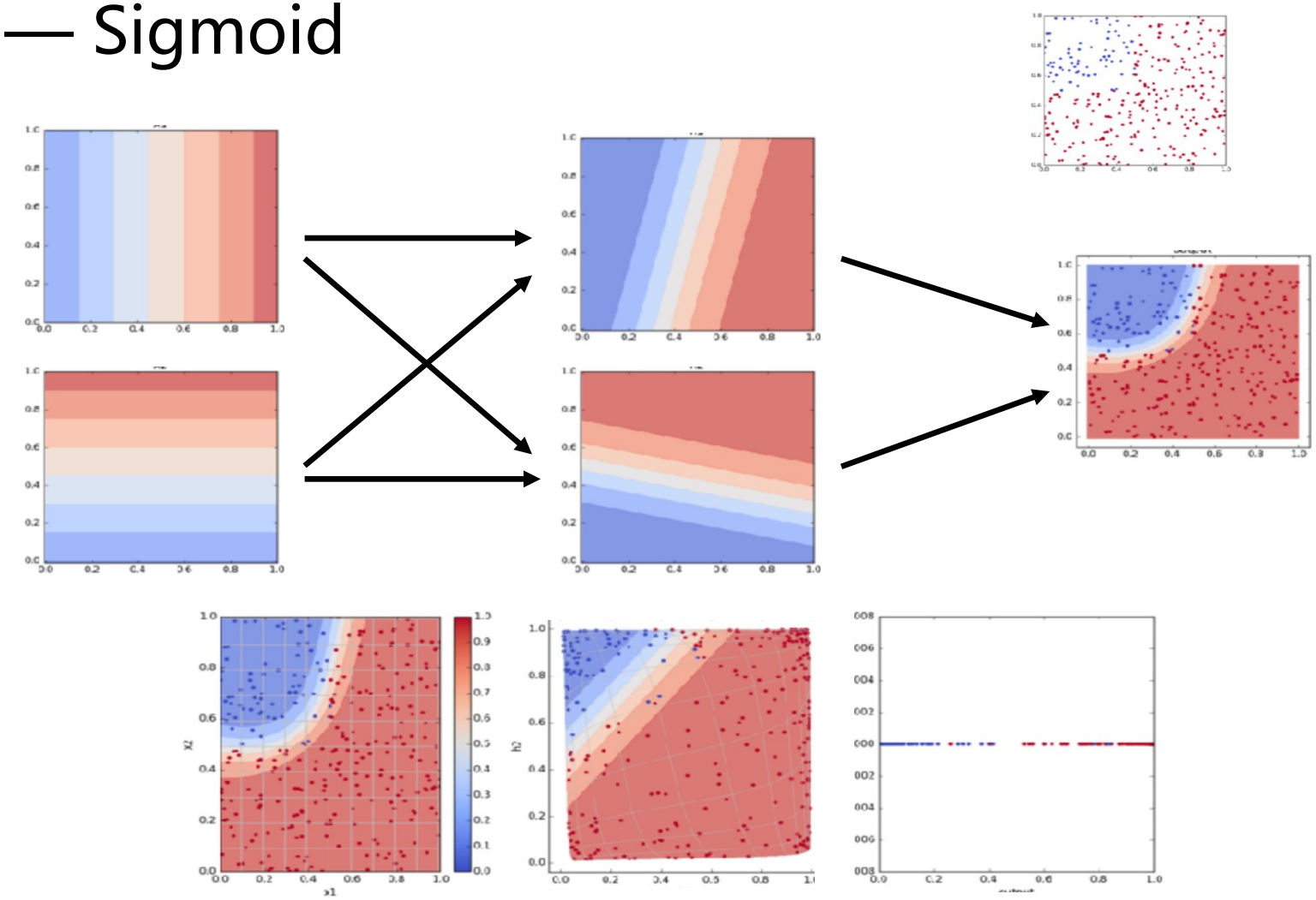
Playground 上的真实例子

- 角度数据 — Sigmoid



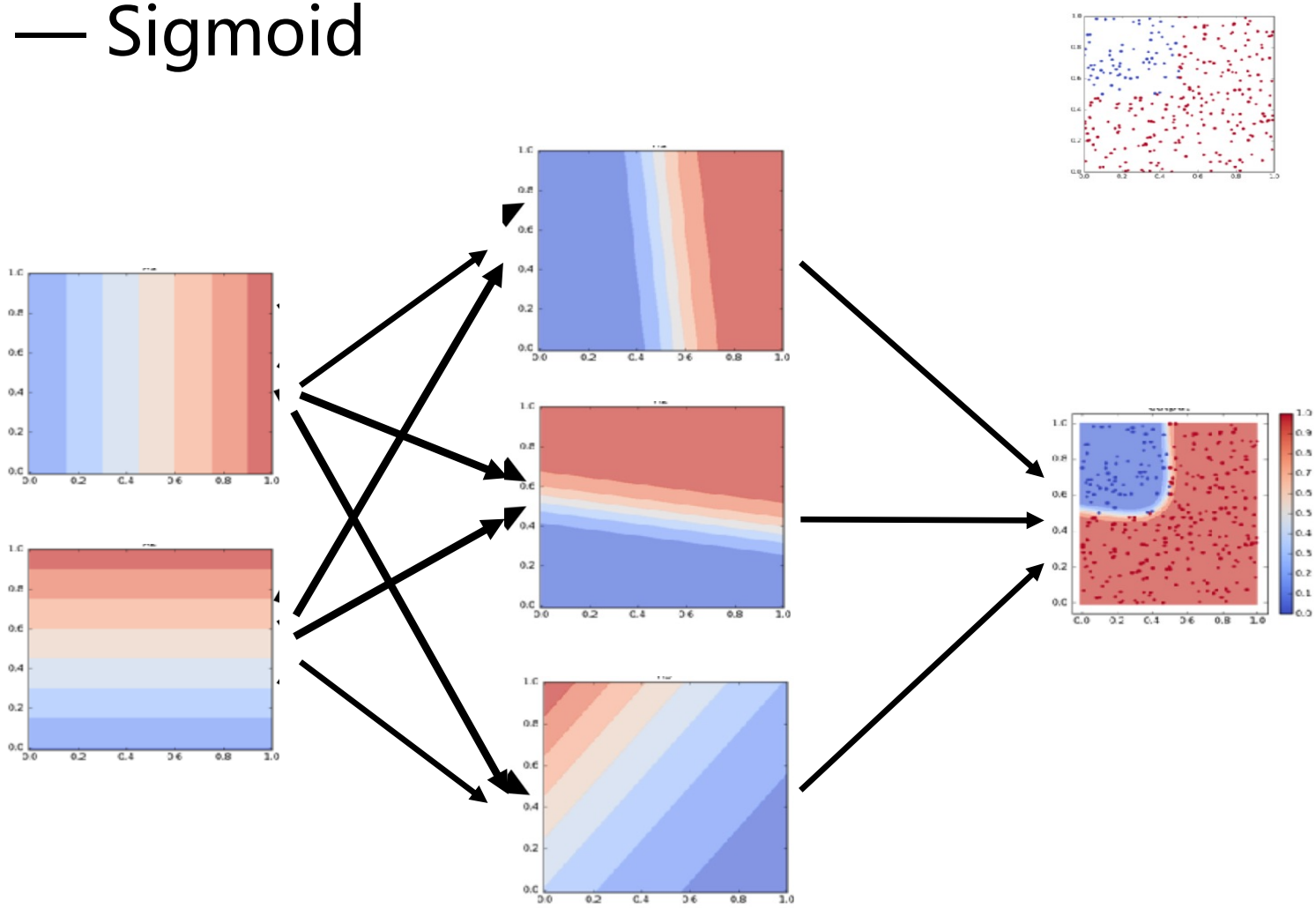
Playground 上的真实例子

- 角度数据 — Sigmoid



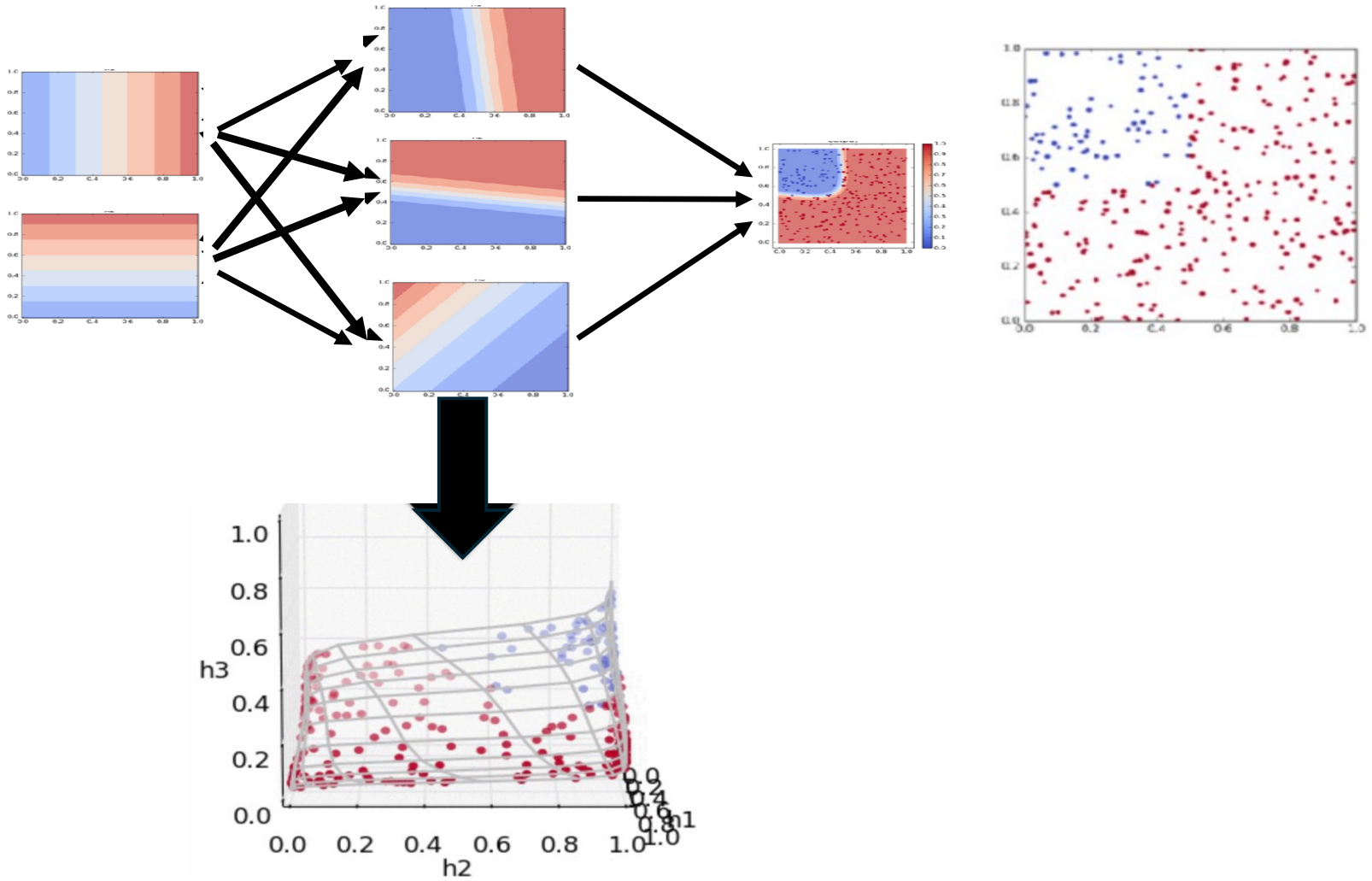
Playground 上的真实例子

- 角度数据 — Sigmoid



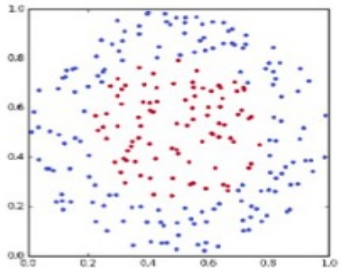
Playground 上的真实例子

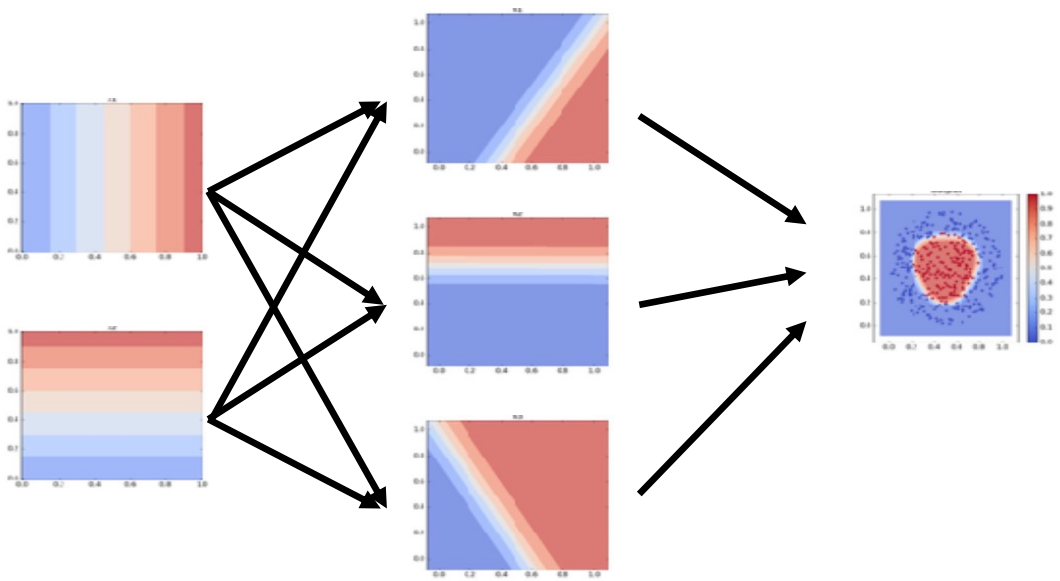
- 角度数据 — Sigmoid



Playground 上的真实例子

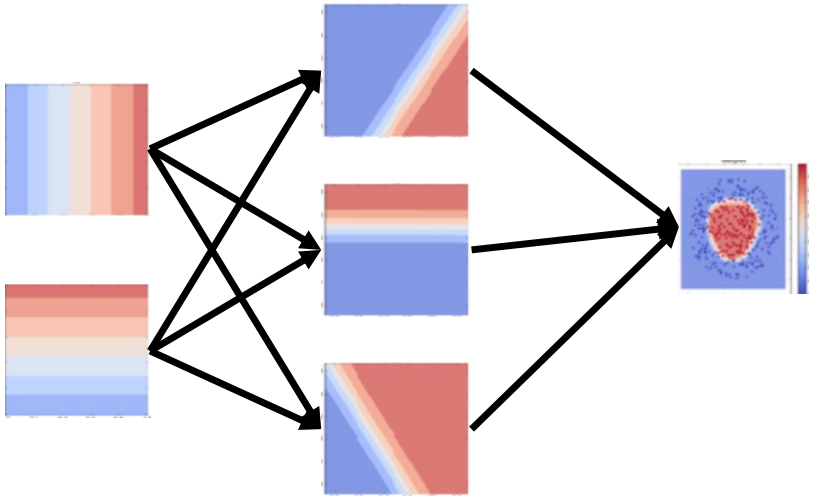
- 让我们来看看空间的变化：环状数据

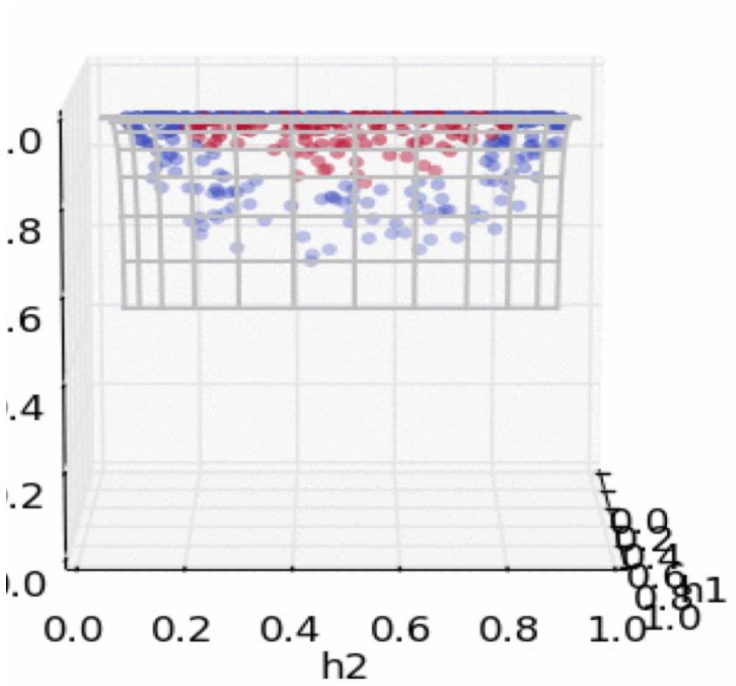




Playground 上的真实例子

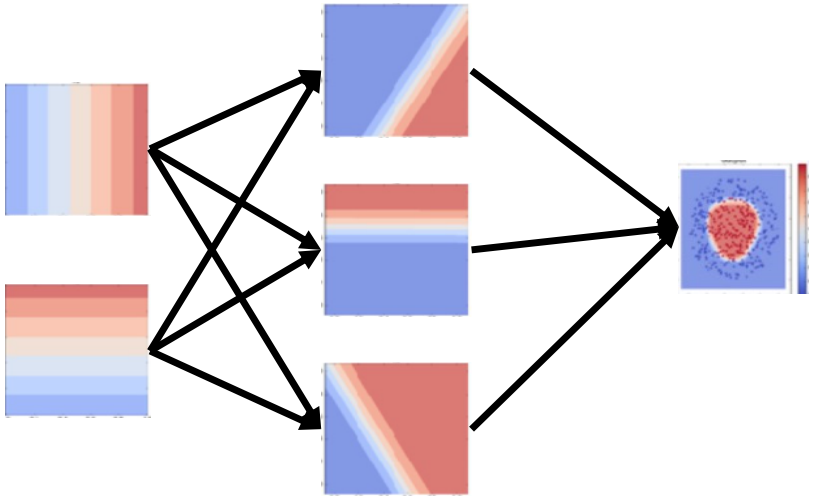
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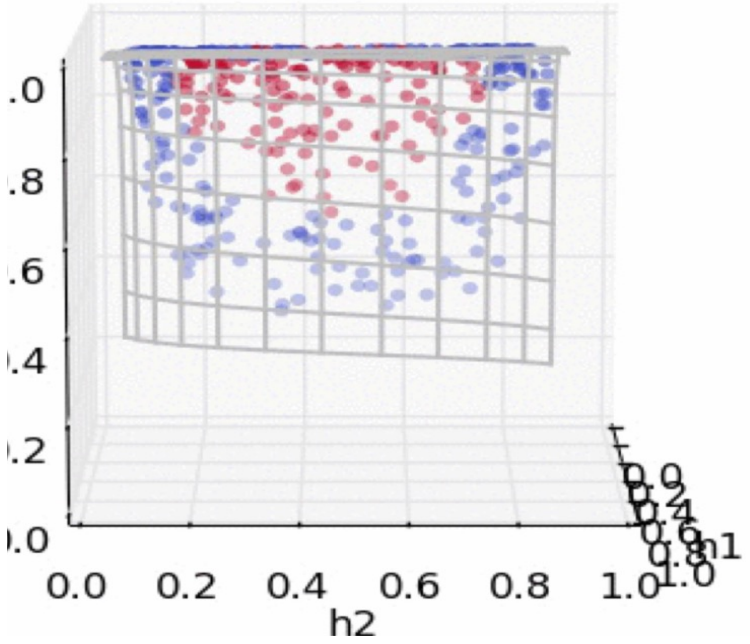




Playground 上的真实例子

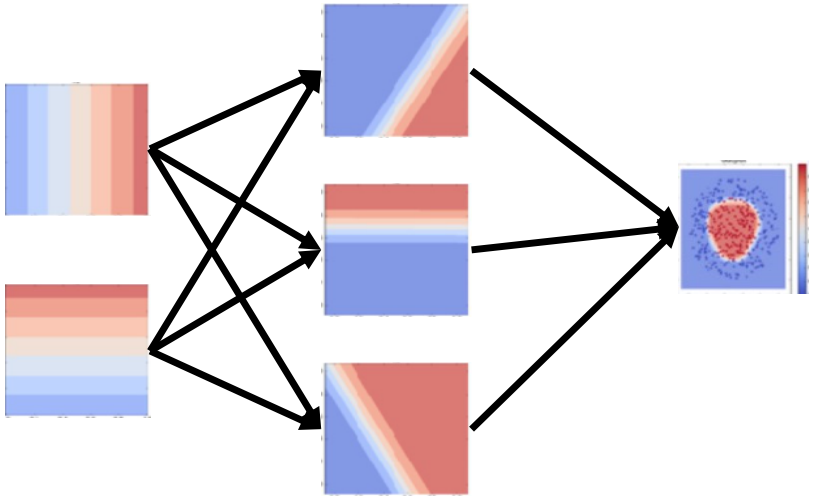
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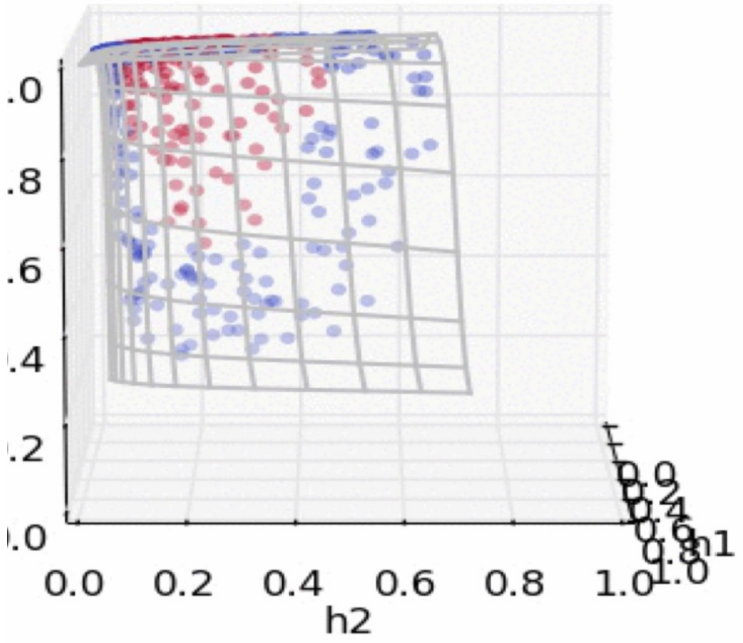




Playground 上的真实例子

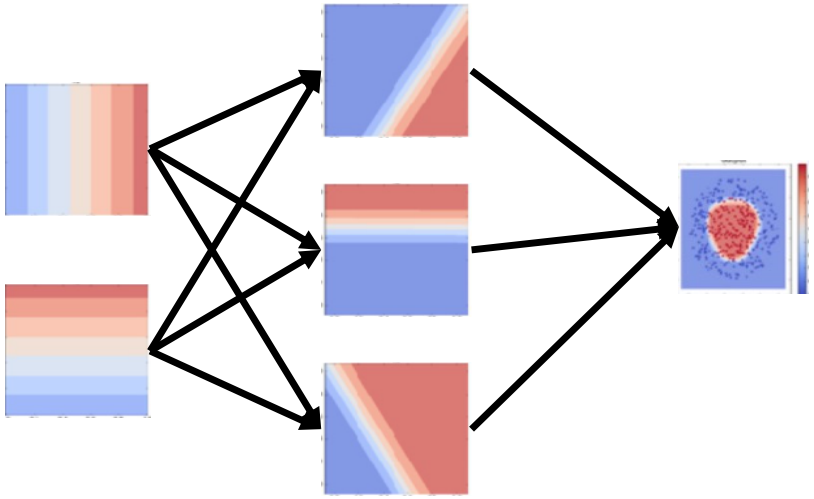
- 让我们来看看空间的变化：环状数据

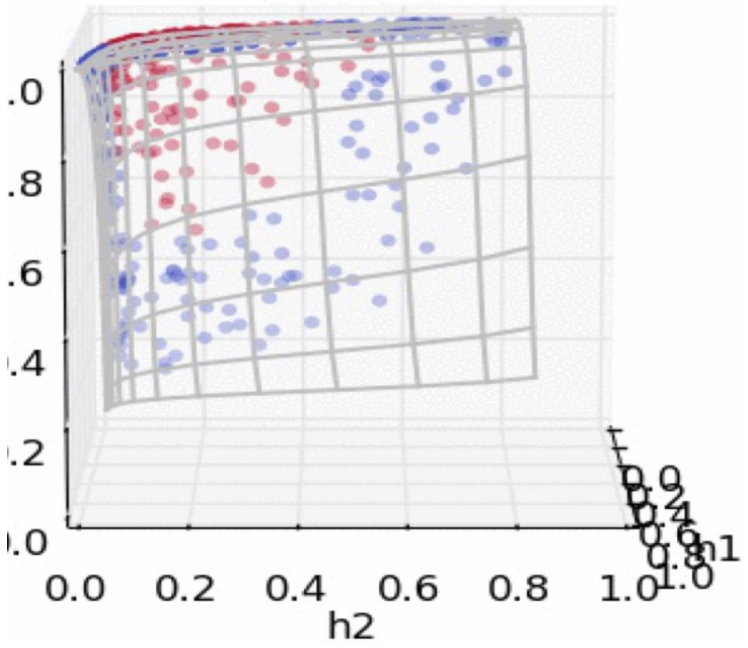




Playground 上的真实例子

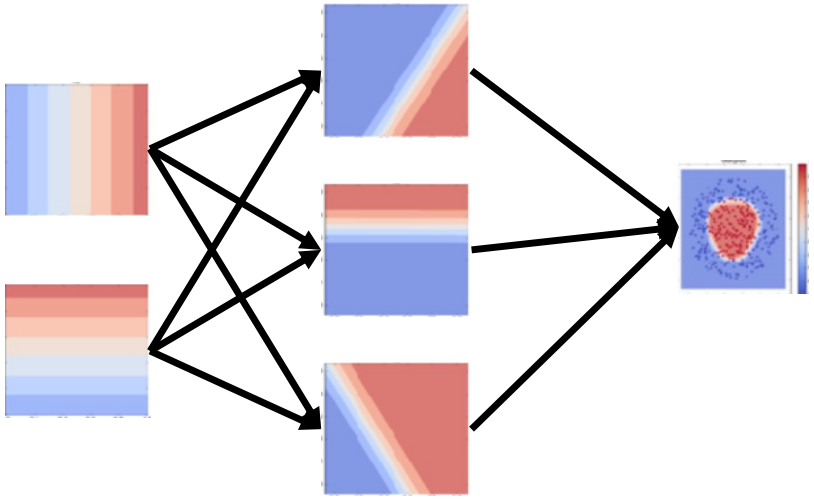
- 让我们来看看空间的变化：环状数据

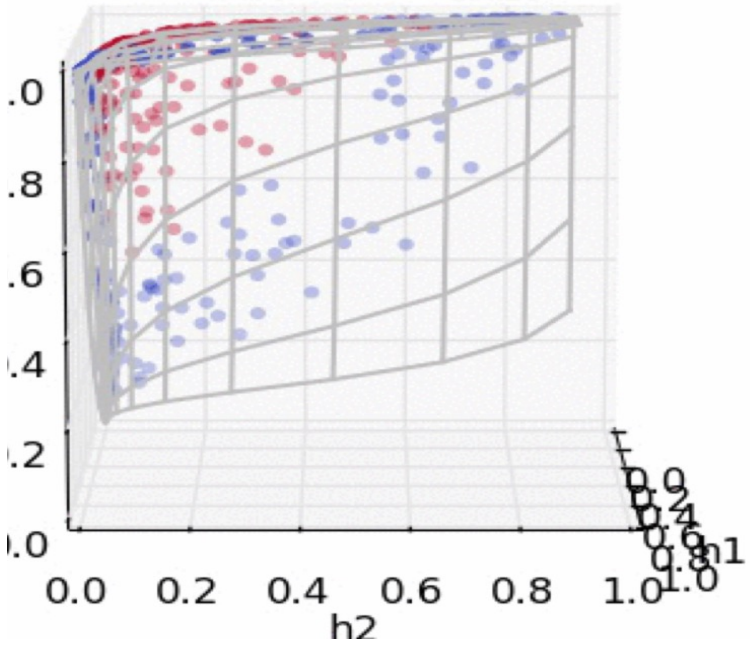




Playground 上的真实例子

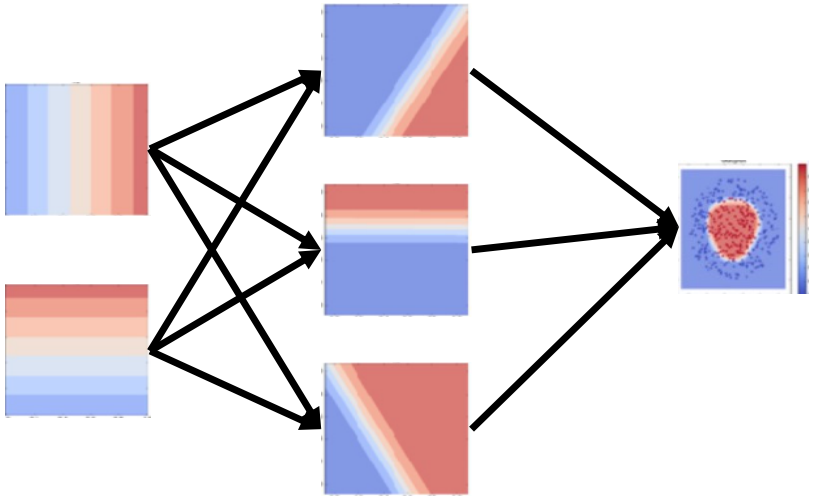
- 让我们来看看空间的变化：环状数据

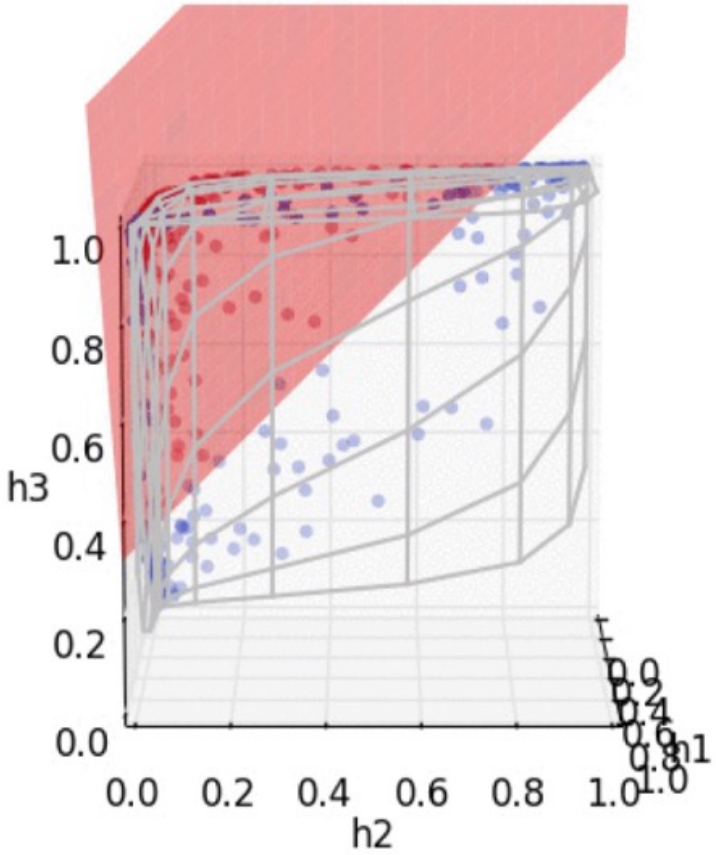




Playground 上的真实例子

- 让我们来看看空间的变化：环状数据





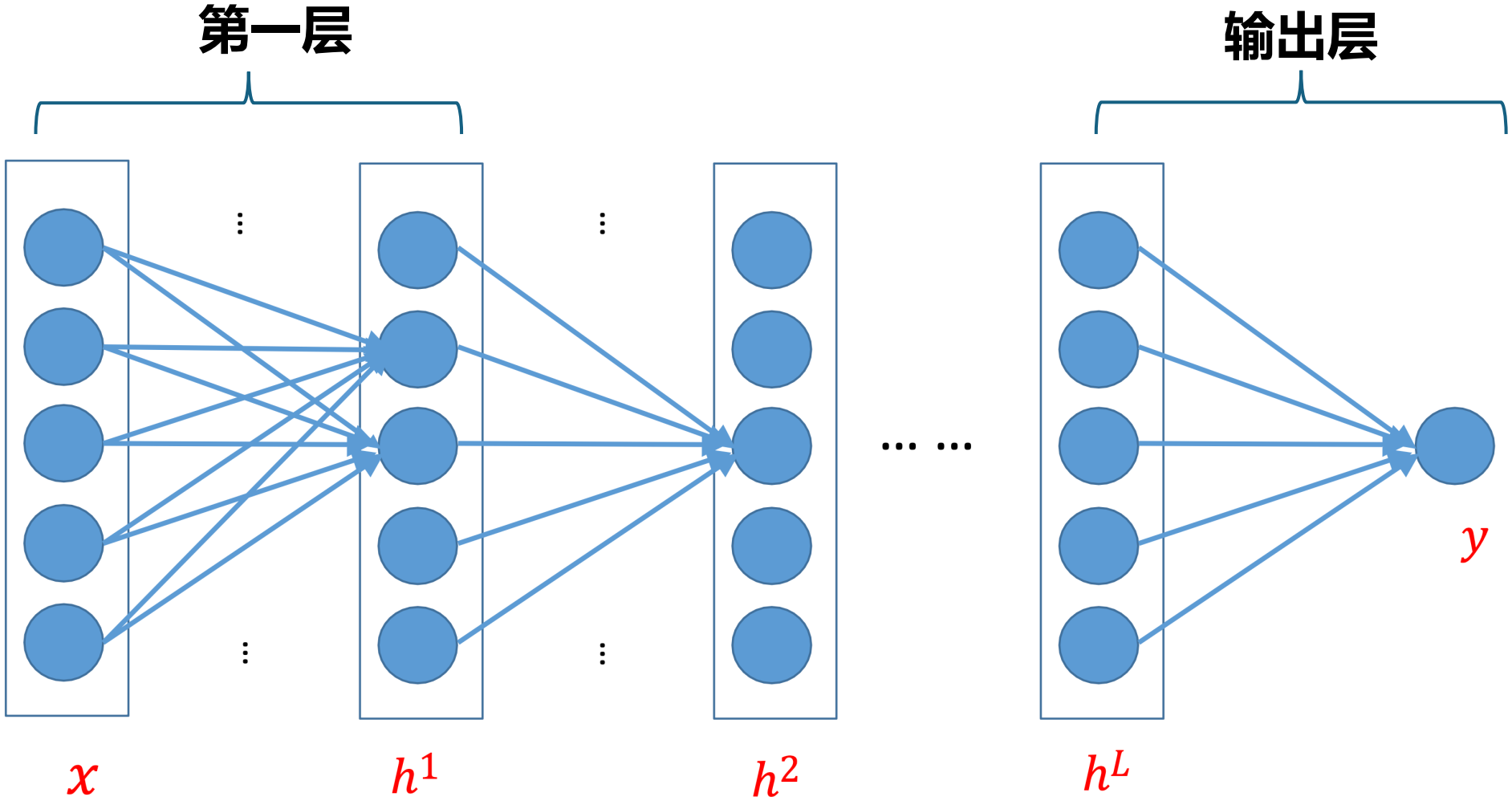
提纲

- 多分类问题
- 前馈神经网络
- **神经网络的组成部分**
  - 输入层
  - 输出层
  - 隐藏层
- 深度神经网络

组成部分

- **Representations（表示/表征）**
  - **Input（输入表示）**
    - 原始数据向量，作为模型学习表示的起点
  - **Hidden Variables（隐藏变量）**
    - 每一层的中间表示，由线性变换 + 非线性激活得到
    - 逐层抽象 → 从低级特征到高级语义
- **Layers / Weights（层与权重）：**
  - **Input（输入表示）**
    - 提取中间特征表示
  - **Output Layer（输出层）**
    - 将最后的隐藏表示映射到预测结果
      - 回归任务：线性输出；分类任务：Softmax 概率分布

组成部分

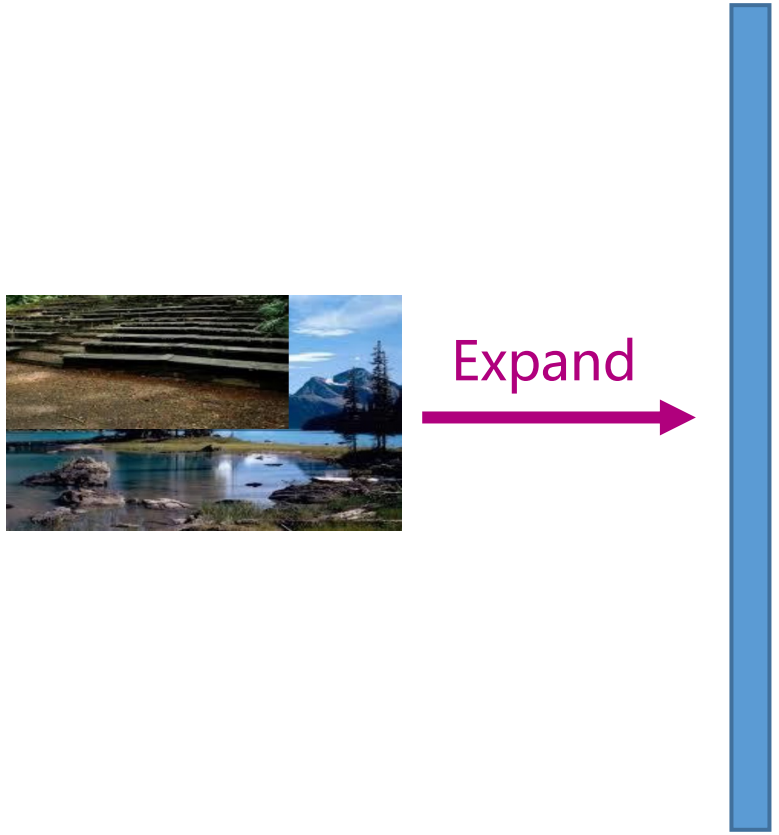


提纲

- 多分类问题
- 前馈神经网络
- **神经网络的组成部分**
  - 输入层
  - 输出层
  - 隐藏层
- 深度神经网络

输入

- 输入通常表示为一个向量
- 可能需要的预处理（Preprocessing）
  - 减去均值（Subtract mean）
    - 去除整体偏移，数据中心化。
    - 例如图像：对每个像素通道减去训练集均值。
- 归一化（Normalize）
  - 将特征缩放到指定范围，例如 [−1,1] 或标准正态分布。
  - 保证不同特征在相同量级，避免数值主导问题。

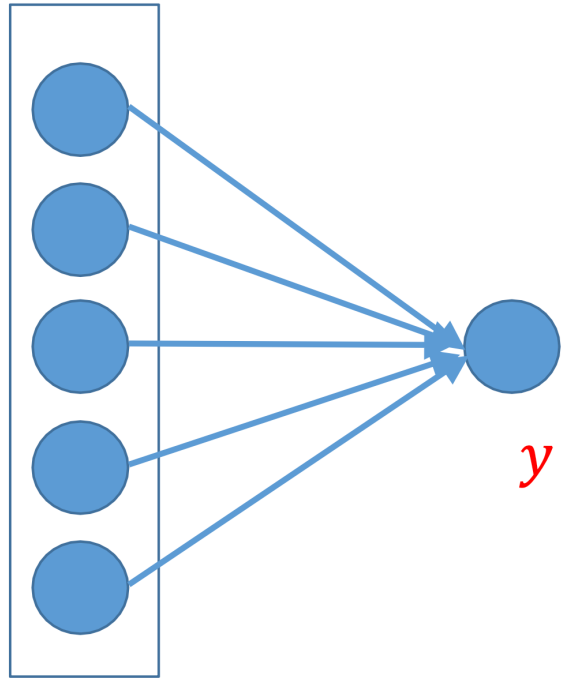


提纲

- 多分类问题
- 前馈神经网络
- **神经网络的组成部分**
  - 输入层
  - 输出层
  - 隐藏层
- 深度神经网络

输出层

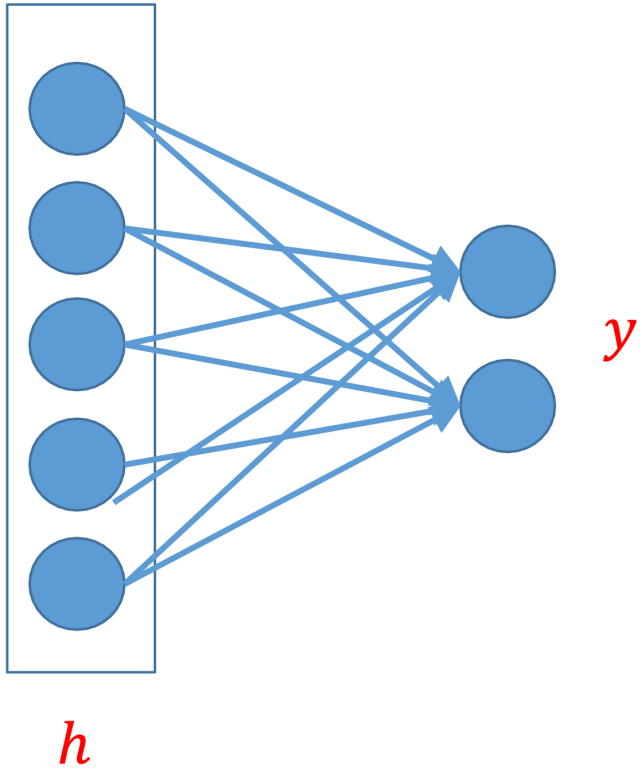
- 回归任务
  - 单维输出 $y = \mathrm{w}^{\mathsf{T}}\mathbf{h}$
  - 使用线性单元（linear unit），无非线性激活



输出层

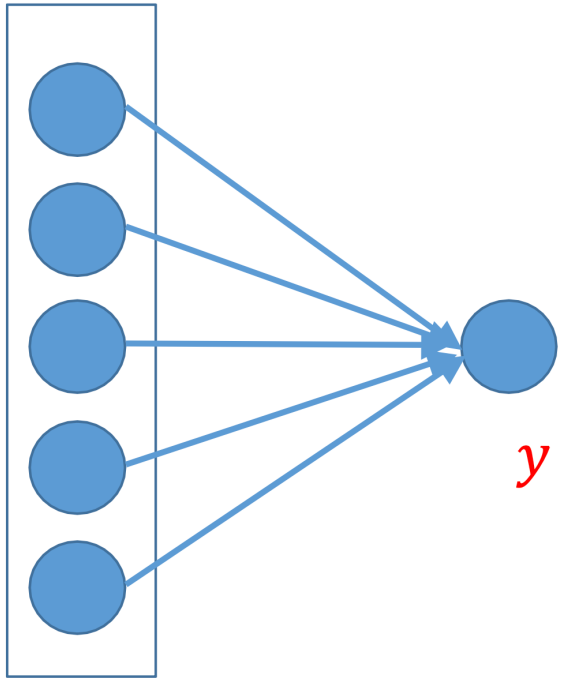
输出层

- 回归任务
  - 多维输出 $\boldsymbol{y} = \mathbf{W}^{\top}\boldsymbol{h}$
  - 使用线性单元（linear unit），无非线性激活



输出层

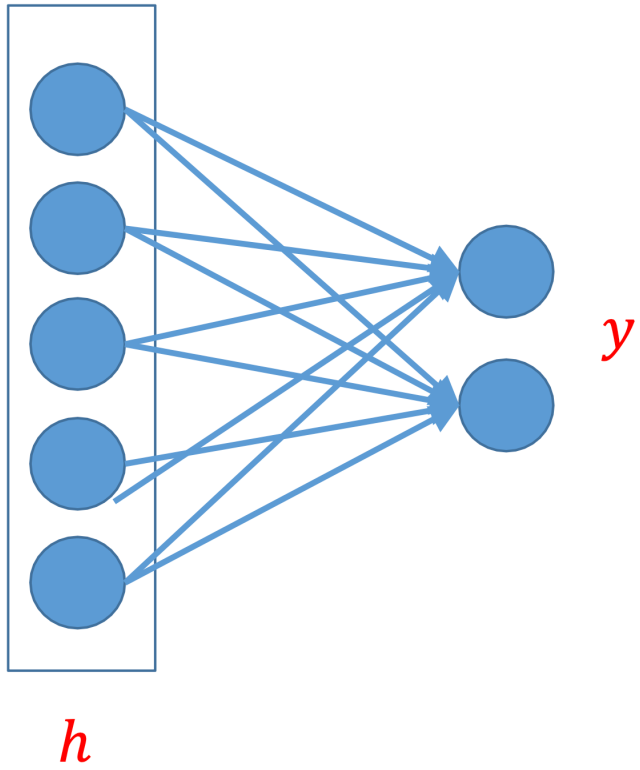
- 二分类任务
  - $y = \sigma(\mathbf{w}^{\top}\mathbf{h})$
  - 使用 Sigmoid 激活函数，相当于对隐藏表示 h 进行逻辑回归



输出层

输出层

- 多分类任务
  - $\boldsymbol{y} = softmax(\mathbf{z}),\ \mathbf{z} = \mathbf{W}^{\top}\mathbf{h}$
  - 使用 Softmax 激活函数，相当于对隐藏表示 h 进行多分类逻辑回归



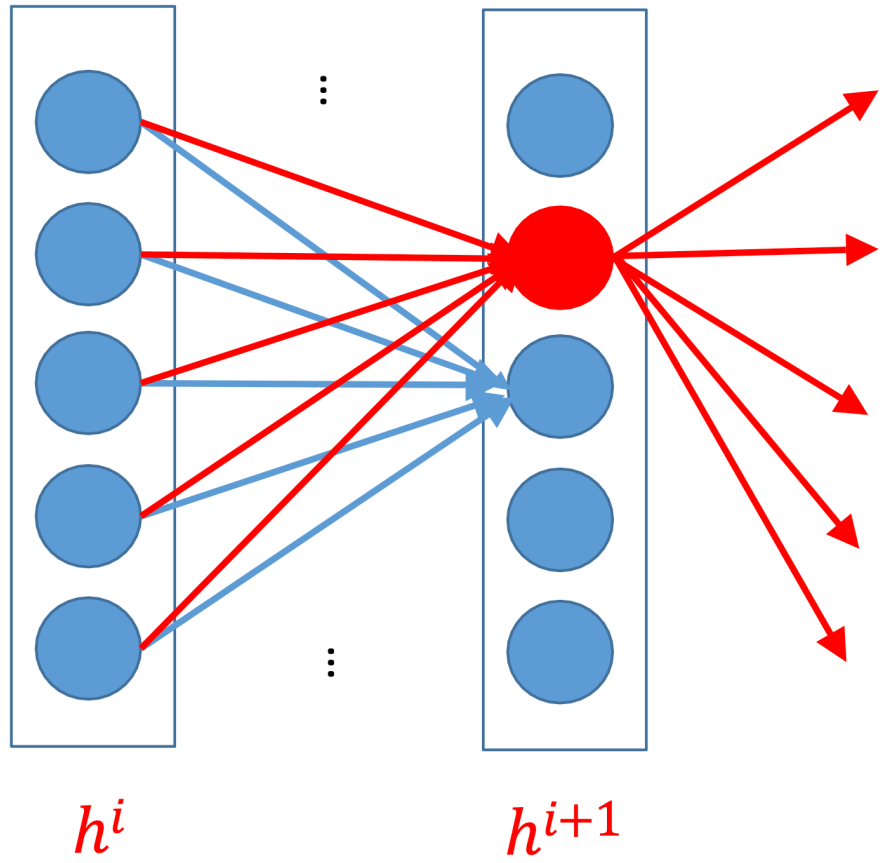
输出层

提纲

- 多分类问题
- 前馈神经网络
- **神经网络的组成部分**
  - 输入层
  - 输出层
  - 隐藏层
- 深度神经网络

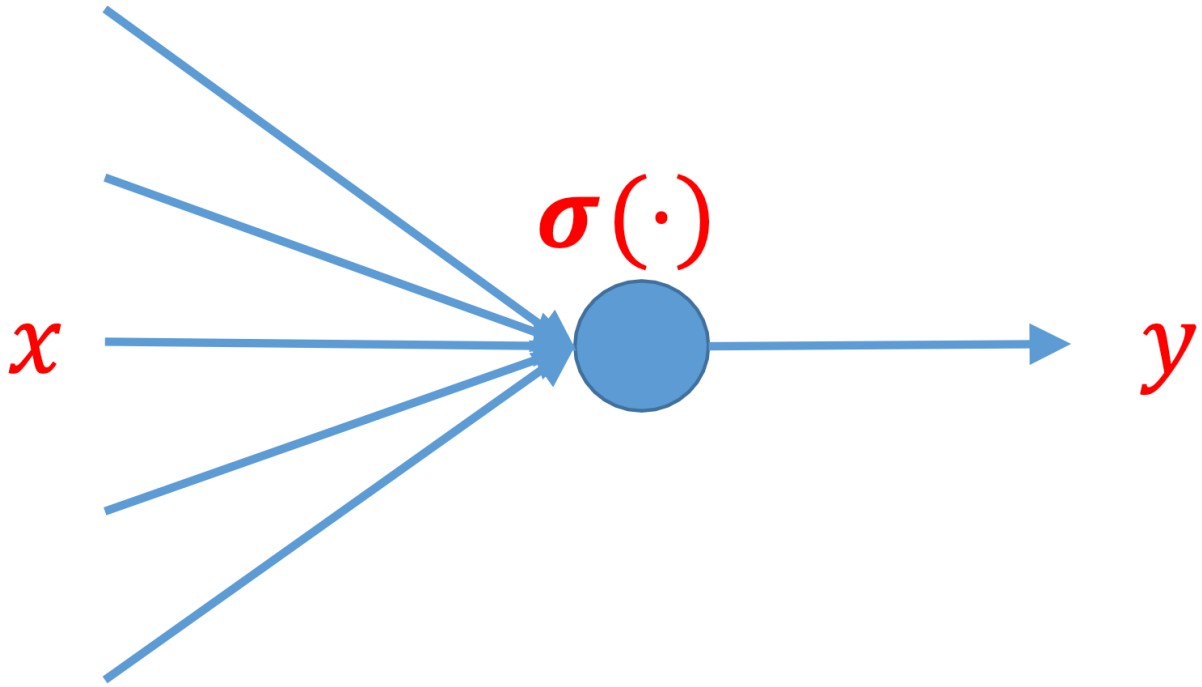
隐藏层

- 每个神经元从前一层输入中，取加权线性组合，再加上偏置
- 直观理解
  - 每个神经元输出一个数值，作为下一层的输入；
  - 隐藏层通过堆叠多个神经元逐步学习输入数据的抽象表示。

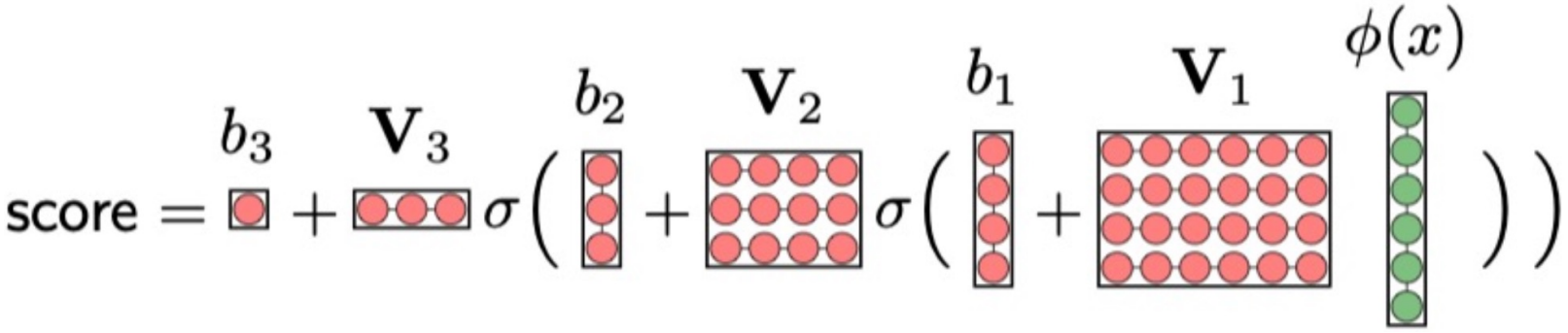


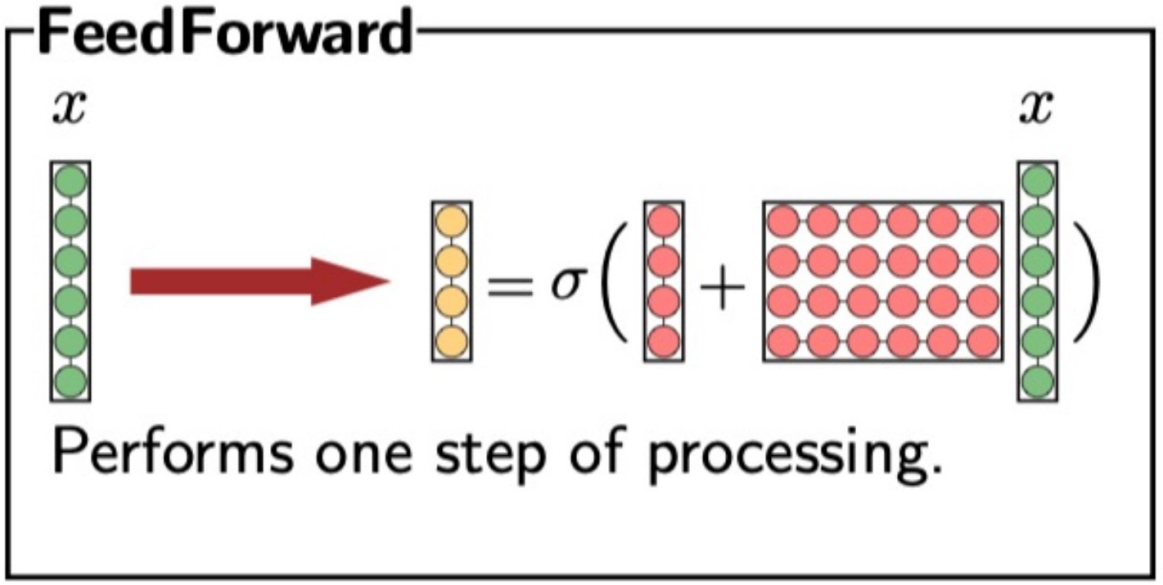
隐藏层

- $y = \sigma(\mathbf{w}^\top \mathbf{h})$
- 常见选择
  - Sigmoid, Tanh, ReLU, …



前馈神经网络



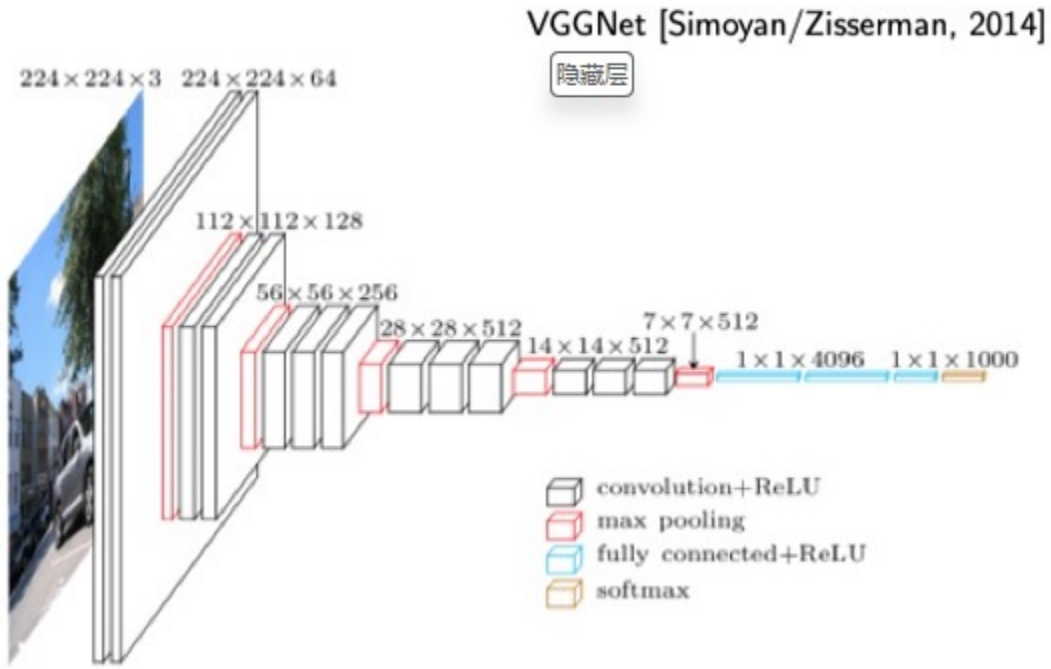


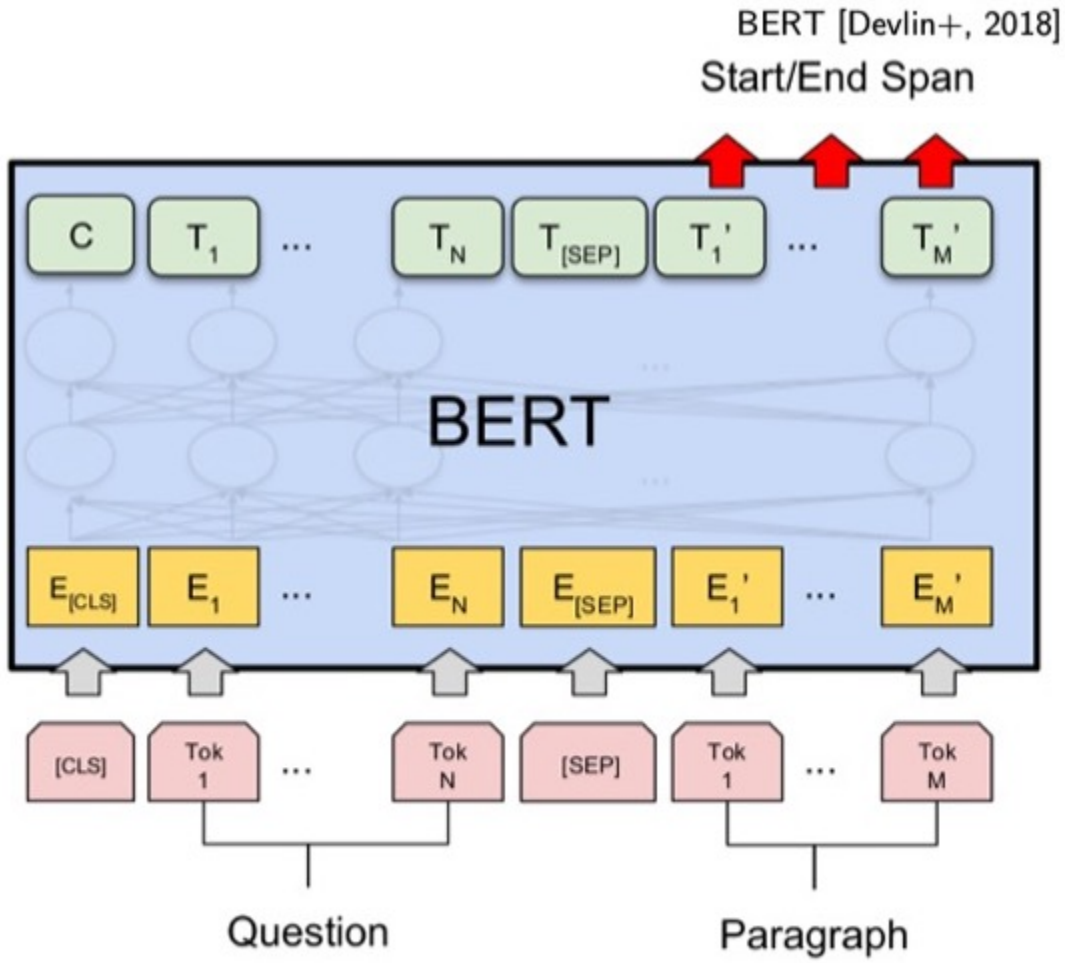
$$score = \mathbf{FeedForward}(\mathbf{FeedForward}(\mathbf{FeedForward}(\phi(x)))) = \mathbf{FeedForward}^3(\phi(x))$$

提纲

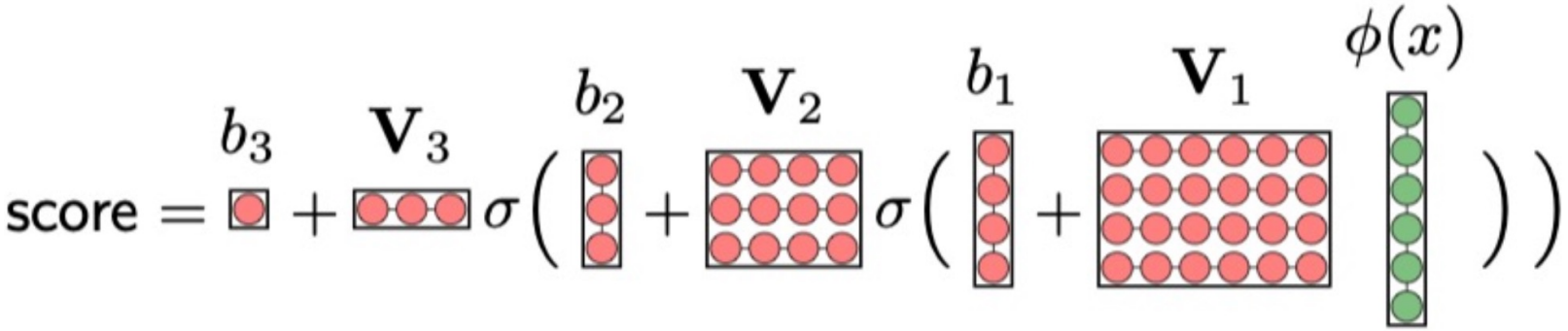
- 多分类问题
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  - 卷积神经网络
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  - Transformer

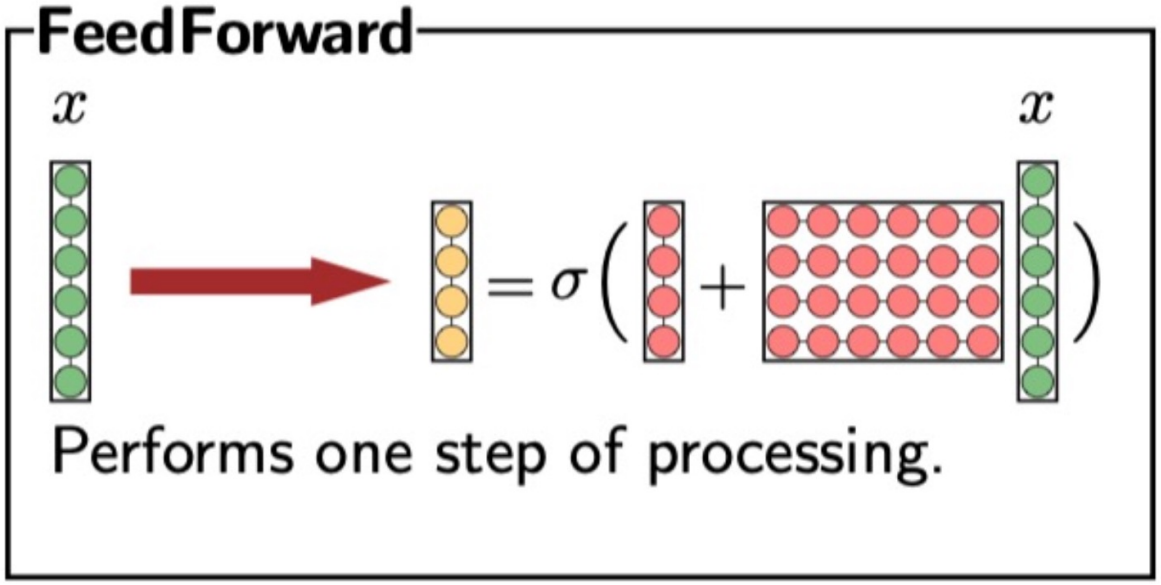
深度学习模型


VGGNet [Simoyan/Zisserman, 2014]
隐藏层
224×224×3
224×224×64
112×112×128
56×56×256
28×28×512
14×14×512
7×7×512
1×1×4096
1×1×1000
convolution+ReLU
max pooling
fully connected+ReLU
softmax


BERT [Devlin+, 2018]
Start/End Span
C
T1
...
TN
T[SEP]
T1'
...
TM'
BERT
E[CLS]
E1
...
EN
E[SEP]
E1'
...
EM'
[CLS]
Tok 1
...
Tok N
[SEP]
Tok 1
...
Tok M
Question
Paragraph

前馈神经网络



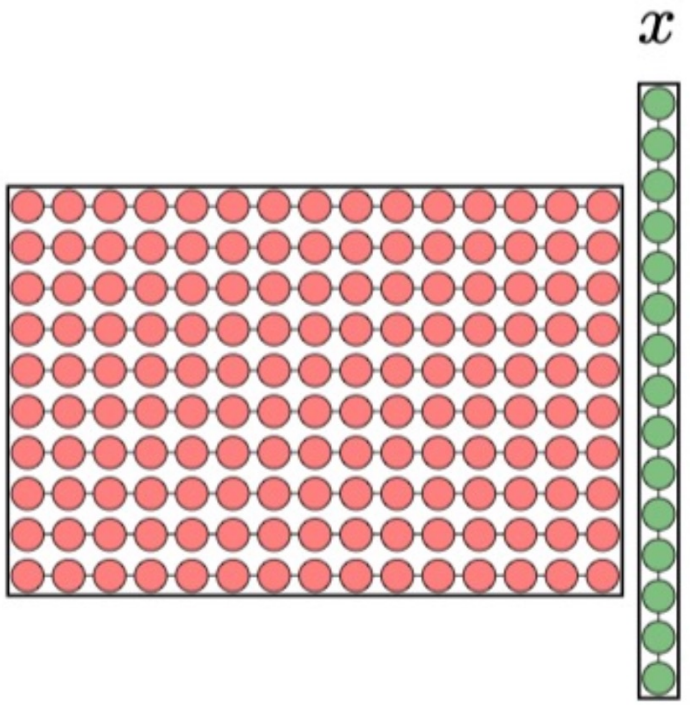


$$score = \mathbf{FeedForward}(\mathbf{FeedForward}(\mathbf{FeedForward}(\phi(x)))) = \mathbf{FeedForward}^3(\phi(x))$$

图片表征

- **问题**
  - 矩阵非常庞大（取决于图片的分辨率)
  - 无法捕捉图片的空间结构（局部性）

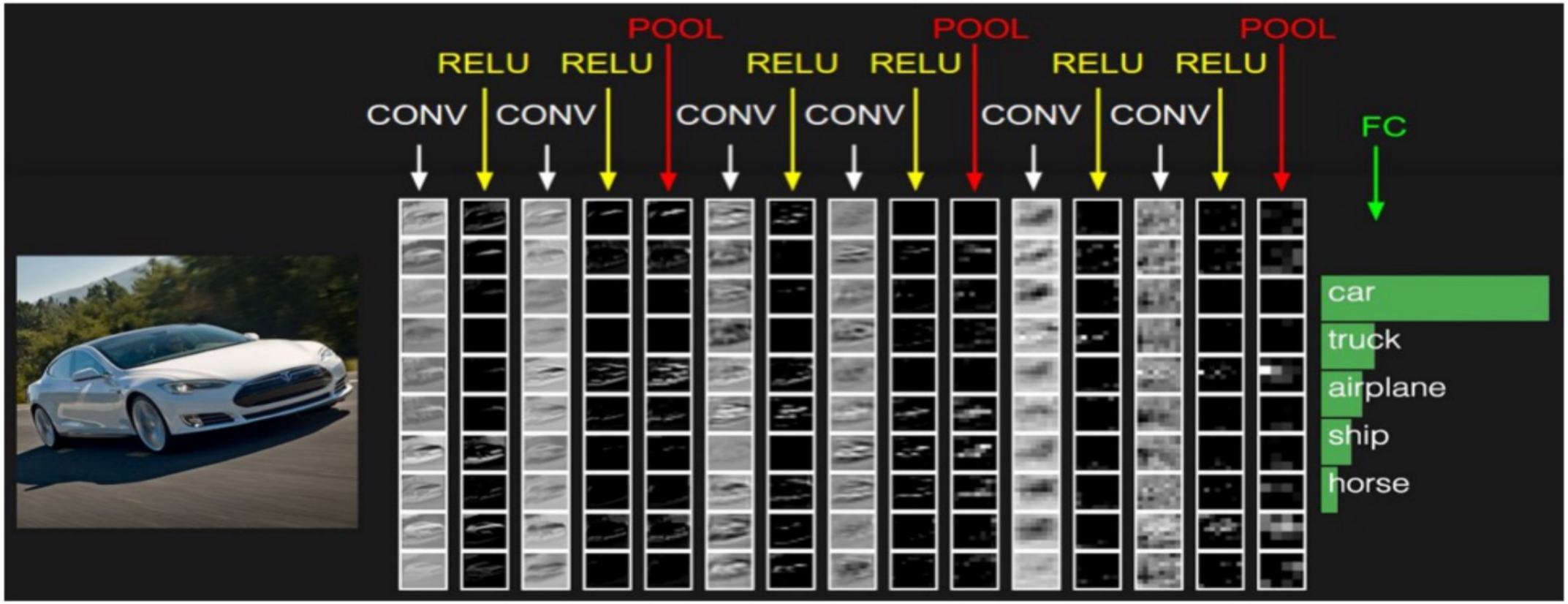




提纲

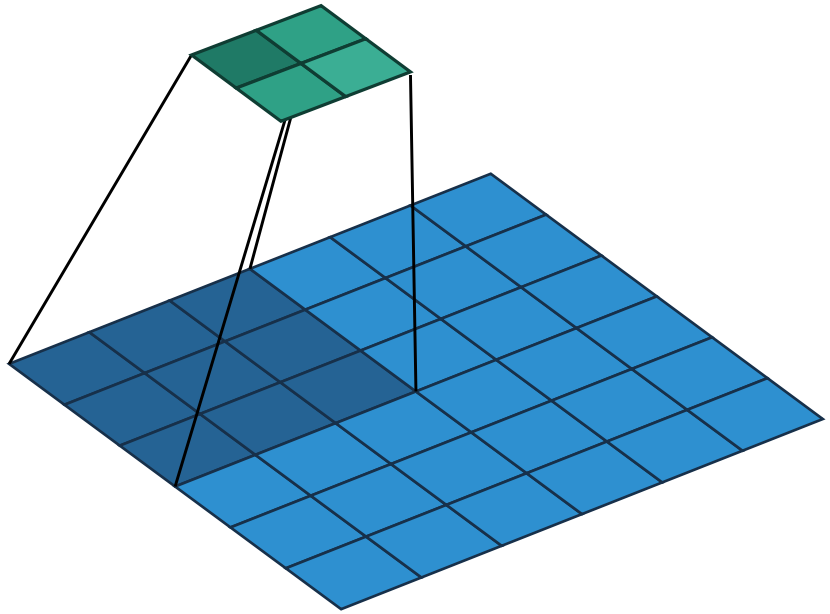
- 多分类问题
- 前馈神经网络
- 神经网络的组成部分
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  - 卷积神经网络
  - 循环神经网络
  - Transformer

卷积神经网络

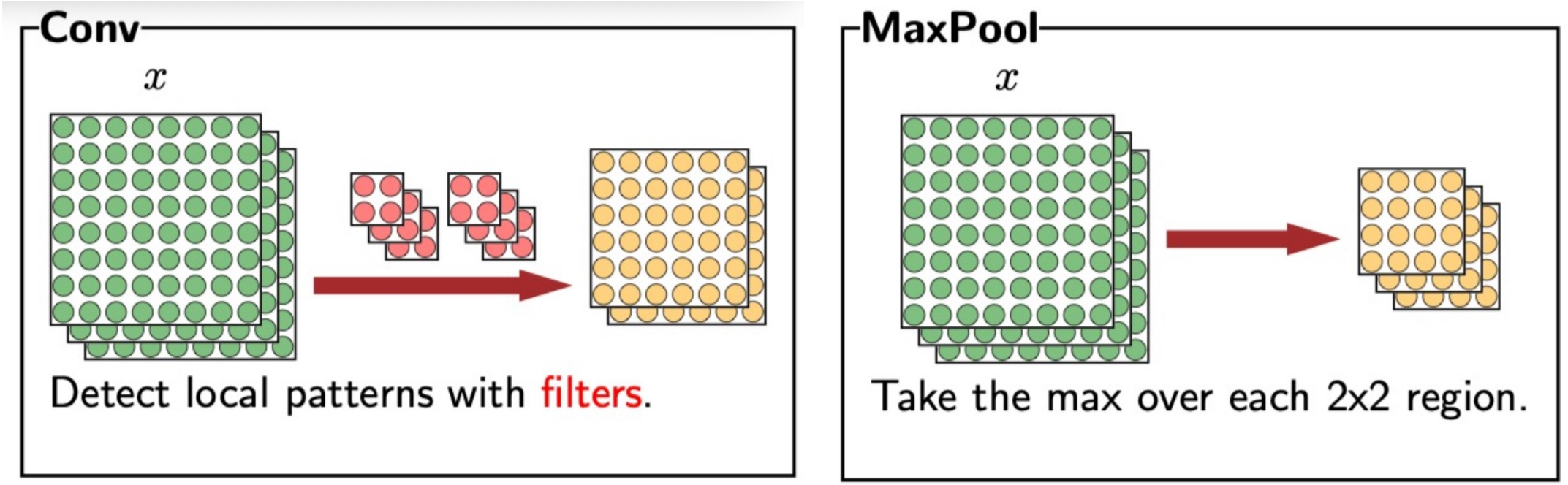


[Andrej Karpathy demo]

卷积神经网络



卷积神经网络

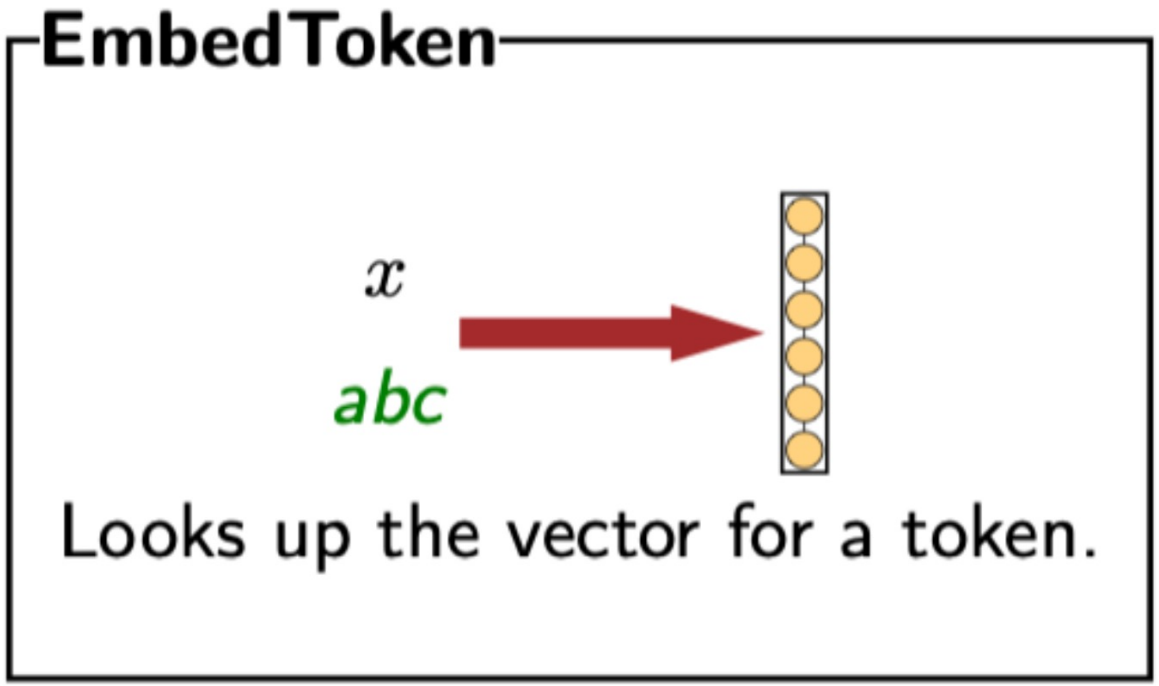


$$AlexNet(x) = \mathbf{FeedForward}^3(\mathbf{MaxPool}(\mathbf{Conv}^3(\mathbf{MaxPool}(\mathbf{Conv}(\mathbf{MaxPool}(\mathbf{Conv}(x)))))))$$

提纲

- 多分类问题
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词元嵌入

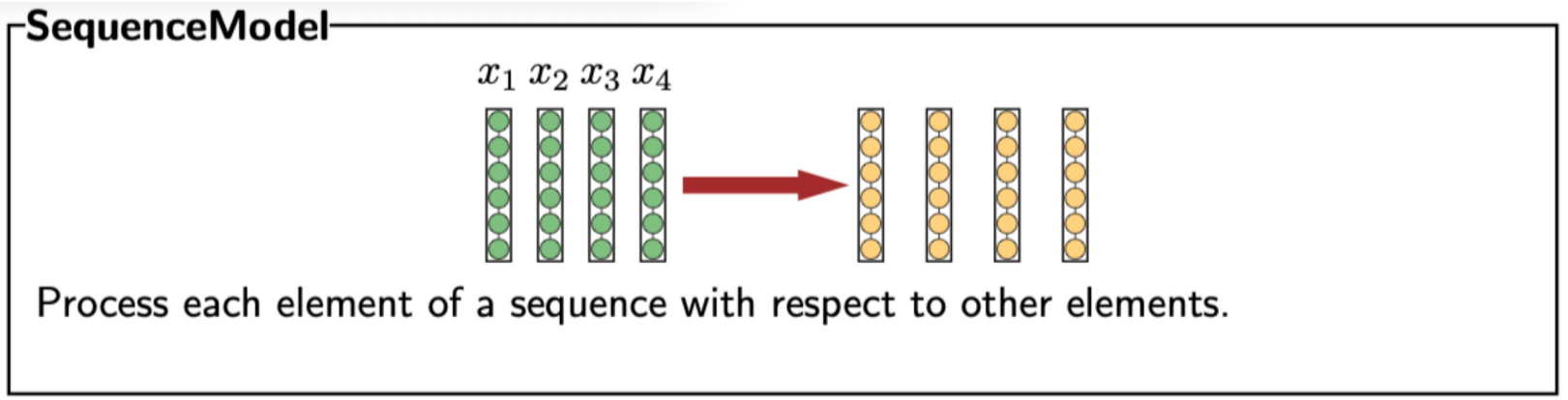


在气象学中，降水是大气水蒸气凝结后受重力下落的所有产物。

词 / token 的含义取决于上下文……

序列表征

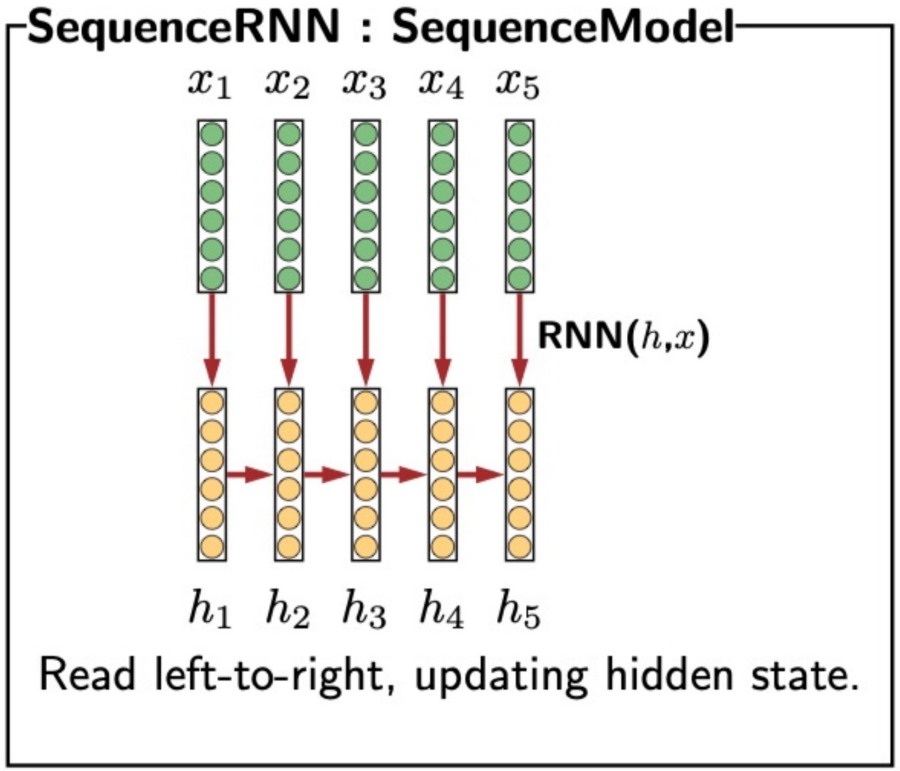
- **两种实现方式**
  - 循环神经网络
  - Transformers



根据序列中的其他元素处理每个元素

循环神经网络

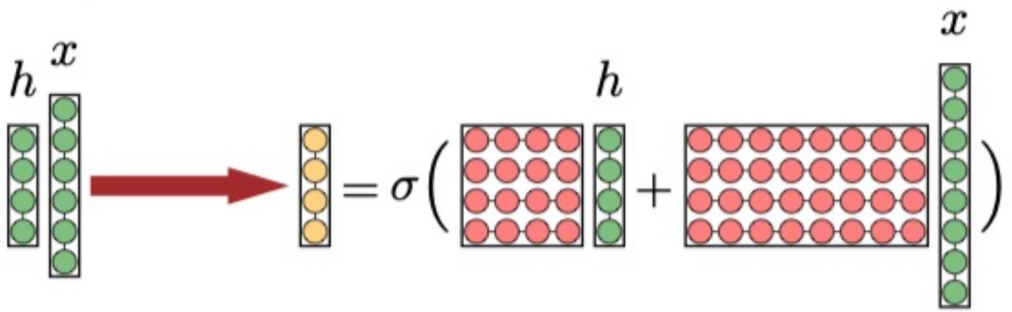
- 从左到右读取，更新隐状态。



在气象学中，降水是大气水蒸气凝结后受重力下落的所有产物。

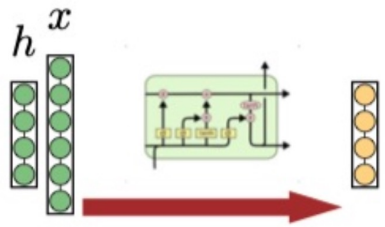
循环神经网络

SimpleRNN : RNN



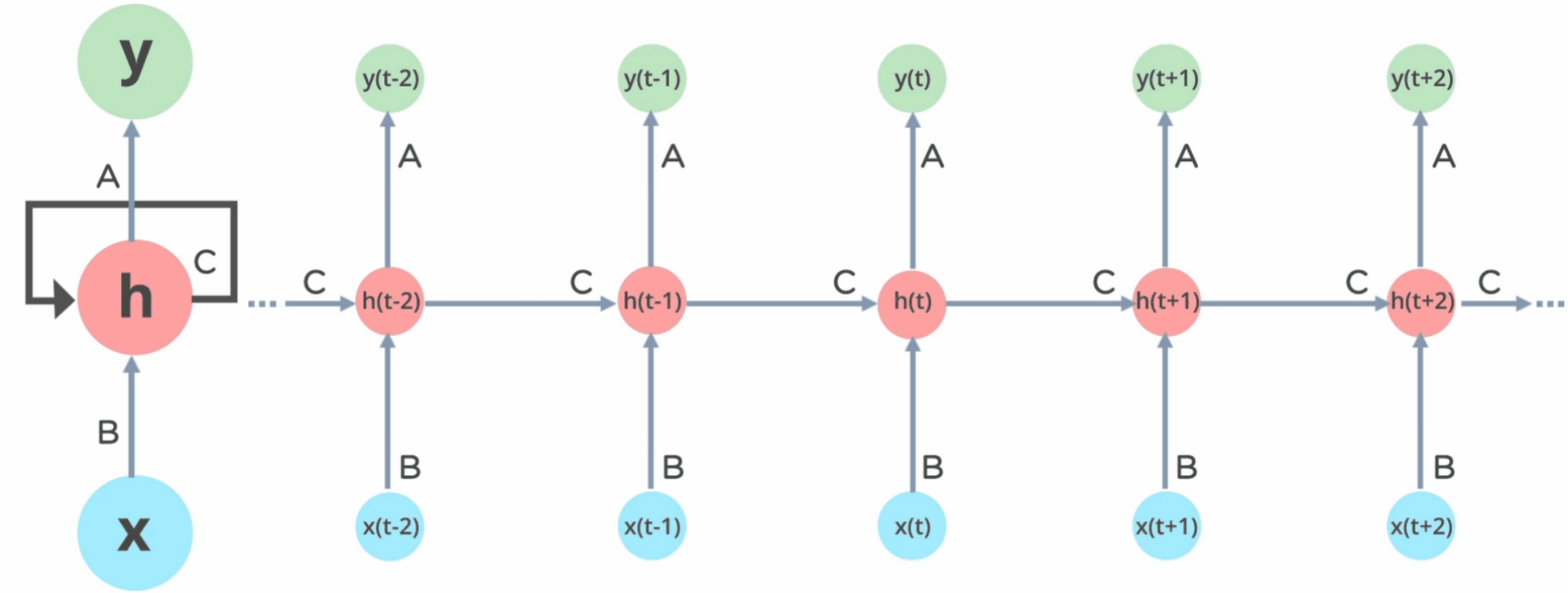
Update hidden state given a new input.

LSTM : RNN



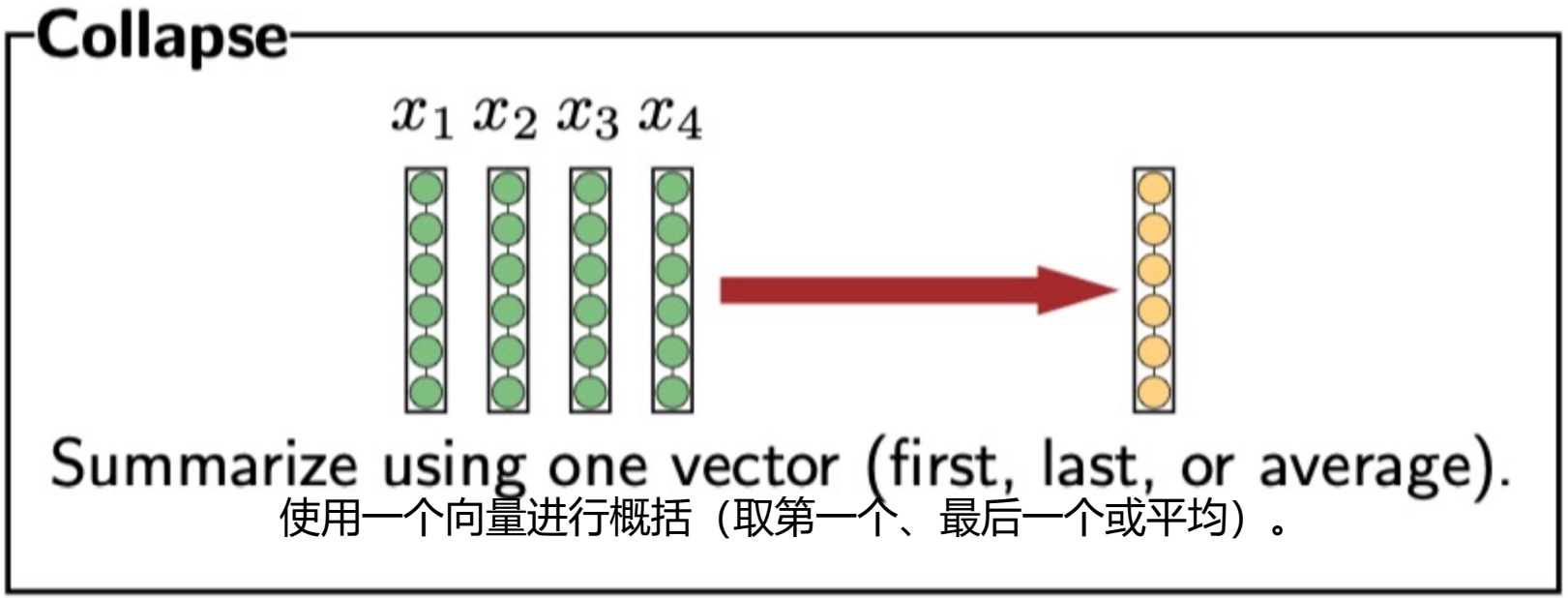
Update hidden state given a new input without forgetting the past.

循环神经网络



折叠成单一向量

- 文本分类模型示例

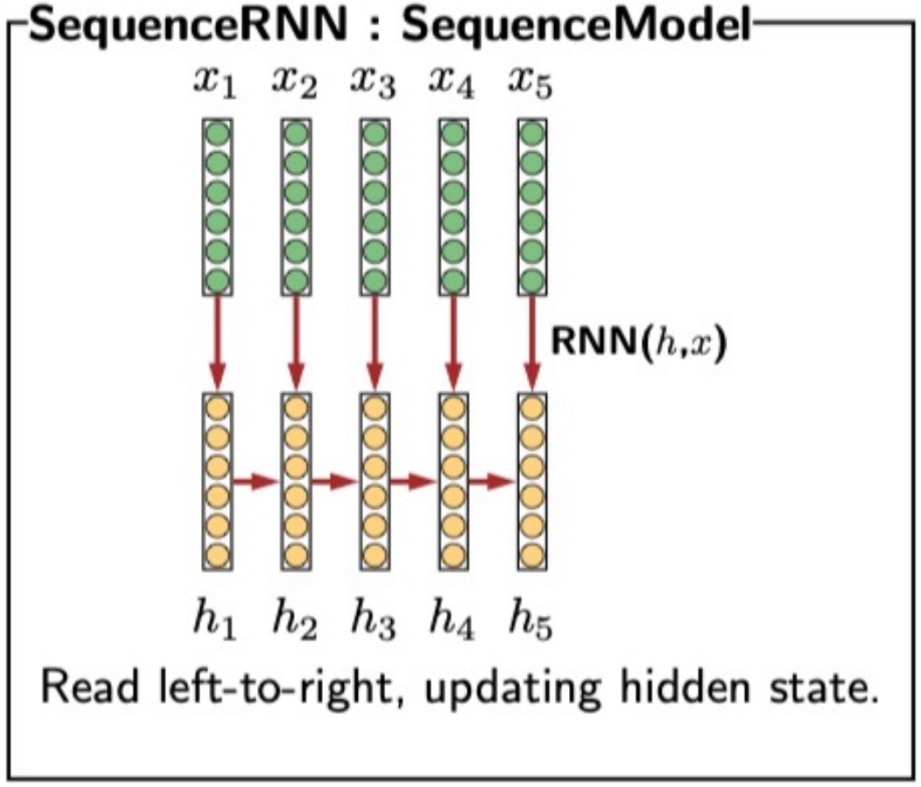


[CLS] 降水是由什么导致的？ [SEP] 在气象学中，降水是大气水蒸气凝结后受重力下落的所有产物。

$$score = \boldsymbol{w} \cdot \boldsymbol{Collapse}(\boldsymbol{SequenceModel}^{3}(\boldsymbol{EmbedToken}(x)))$$

长程依赖

- 问题
  - RNN（和 ConvNets）非常局部

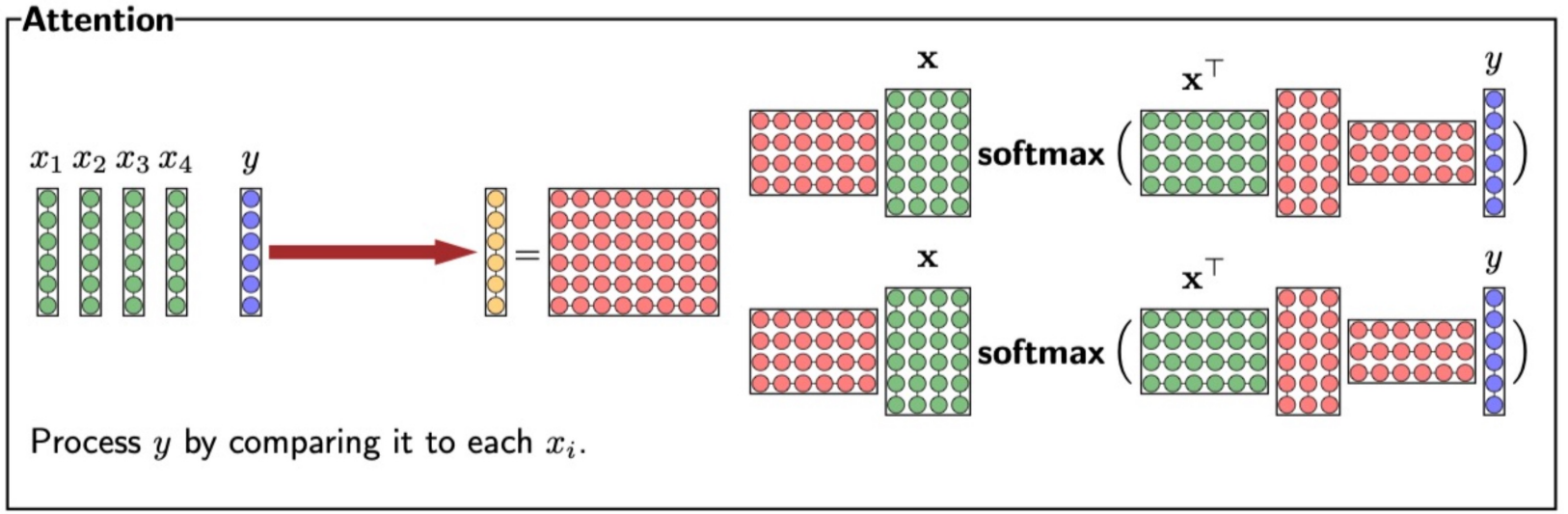


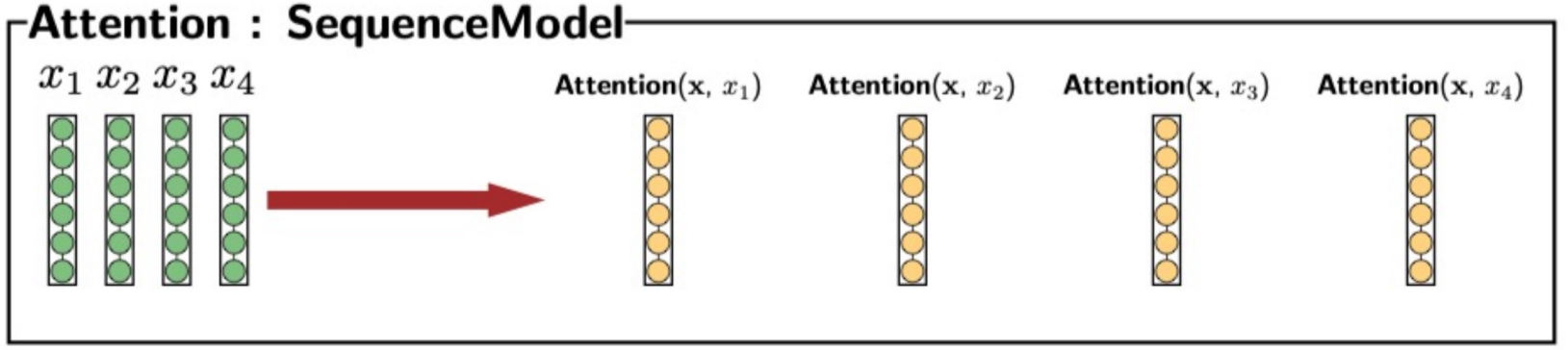
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提纲

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注意力机制

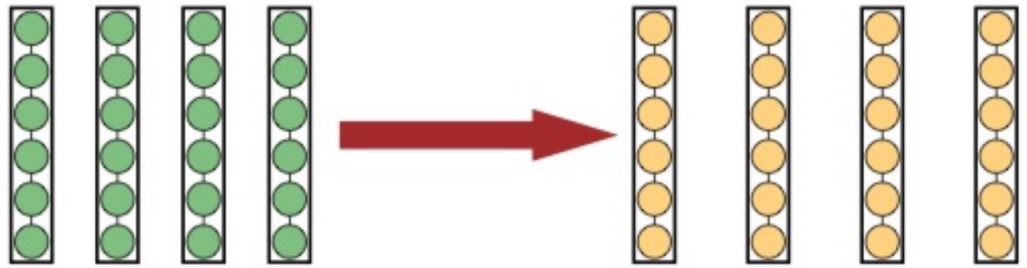

Attention
x_1 x_2 x_3 x_4
y
=
x
softmax
$\mathbf{x}^\top$
y
x
softmax
$\mathbf{x}^\top$
y
Process y by comparing it to each x_i.


Attention : SequenceModel
x_1 x_2 x_3 x_4
Attention(x, x_1)
Attention(x, x_2)
Attention(x, x_3)
Attention(x, x_4)

层归一化&残差连接

AddNorm(f) : SequenceModel

x_1 x_2 x_3 x_4

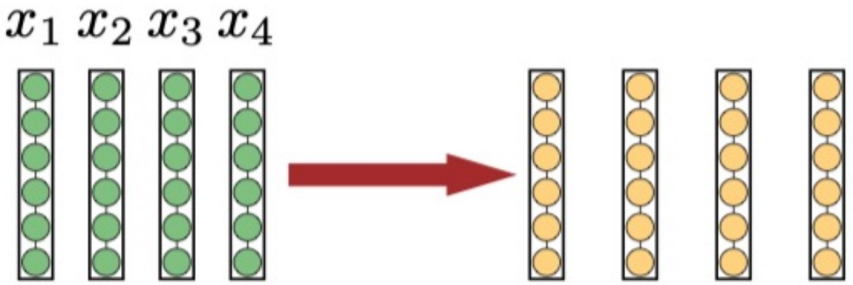


Applies f to $\mathbf{x}$ safely.

$$\textbf{AddNorm}(f, \mathbf{x}) = \textbf{LayerNorm}(\mathbf{x} + f(\mathbf{x}))$$

Transformer

TransformerBlock : SequenceModel

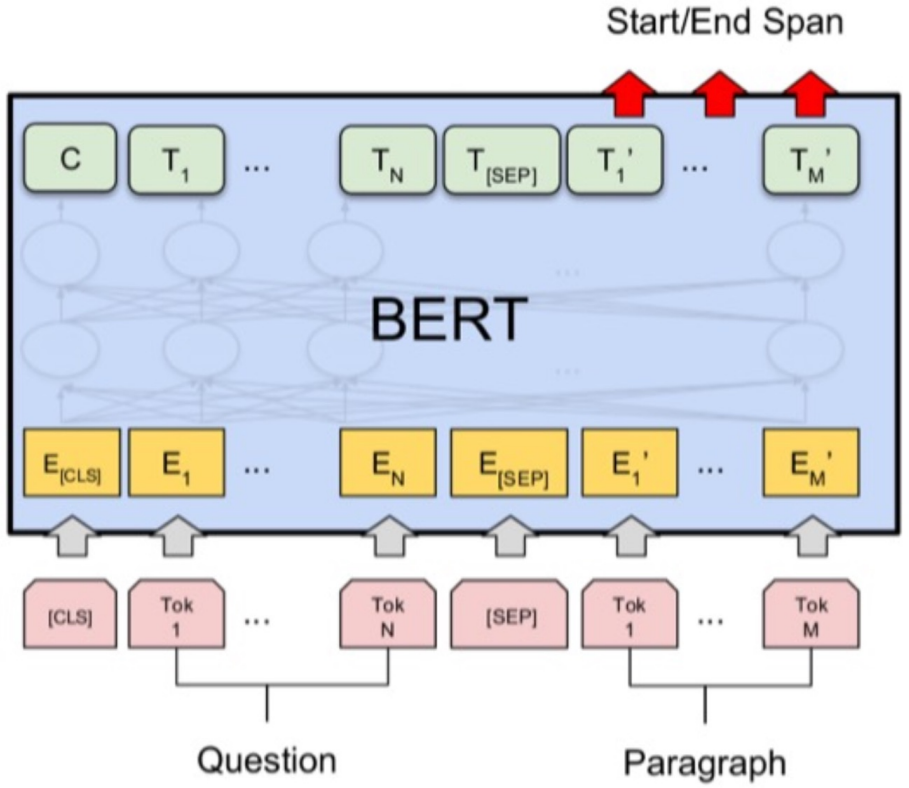


Processes each object x_i in context.

TransformerBlock$(\mathbf{x}) =$ **AddNorm**(**FeedForward**, **AddNorm**(**Attention**, $\mathbf{x}$))

BERT





[CLS] 降水是由什么导致的？[SEP] 在气象学中，降水是大气水蒸气凝结后受重力下落的所有产物。

$$BERT(\mathbf{x}) = \boldsymbol{TransformerBlock}^{\mathbf{24}}(\boldsymbol{EmbedToken}(\mathbf{x}))$$

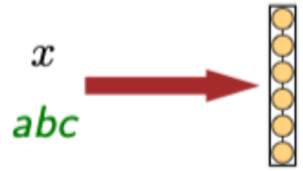
词元生成

GenerateToken



Generate token y based on x· **EmbedToken**(y).

EmbedToken



Looks up the vector for a token.

序列生成

LanguageModel

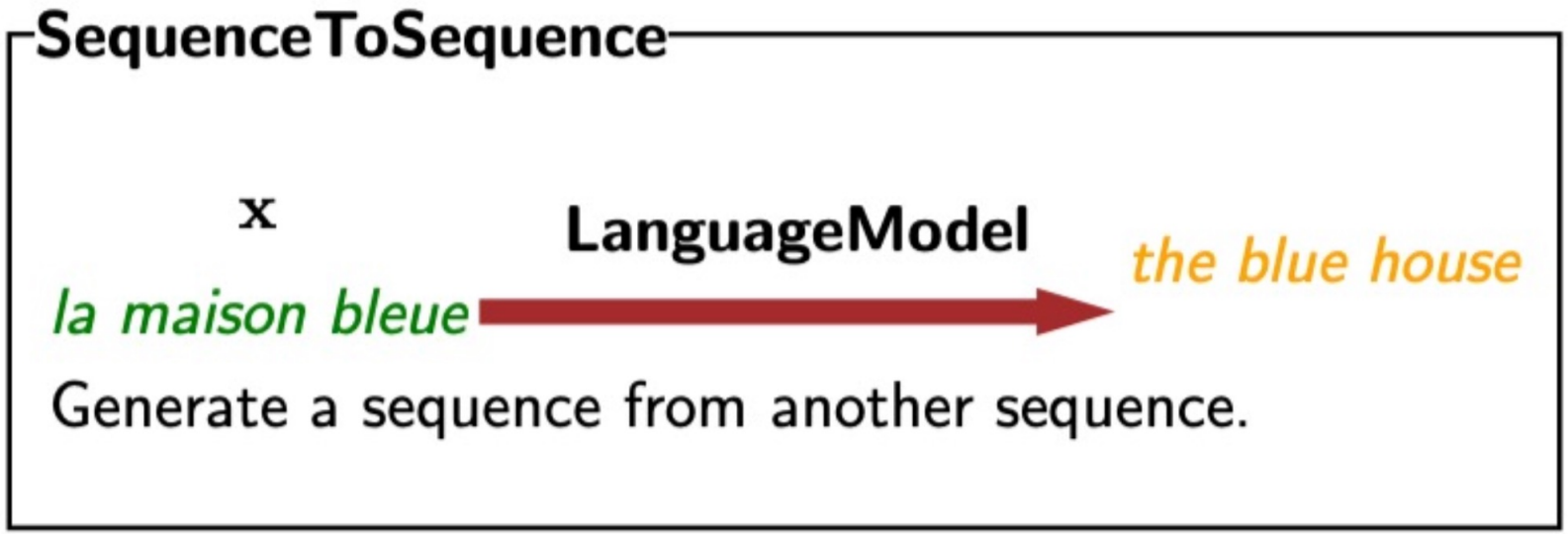


Generate next token in the sequence.

$$\textbf{LanguageModel}(\mathbf{x}) = \textbf{GenerateToken}(\textbf{Collapse}(\textbf{SequenceModel}(\textbf{EmbedToken}(\mathbf{x}))))$$

序列到序列模型

- **应用**
  - 机器翻译：句子 → 翻译
  - 文档摘要：文档 → 摘要
  - 语义解析：句子 → 代码



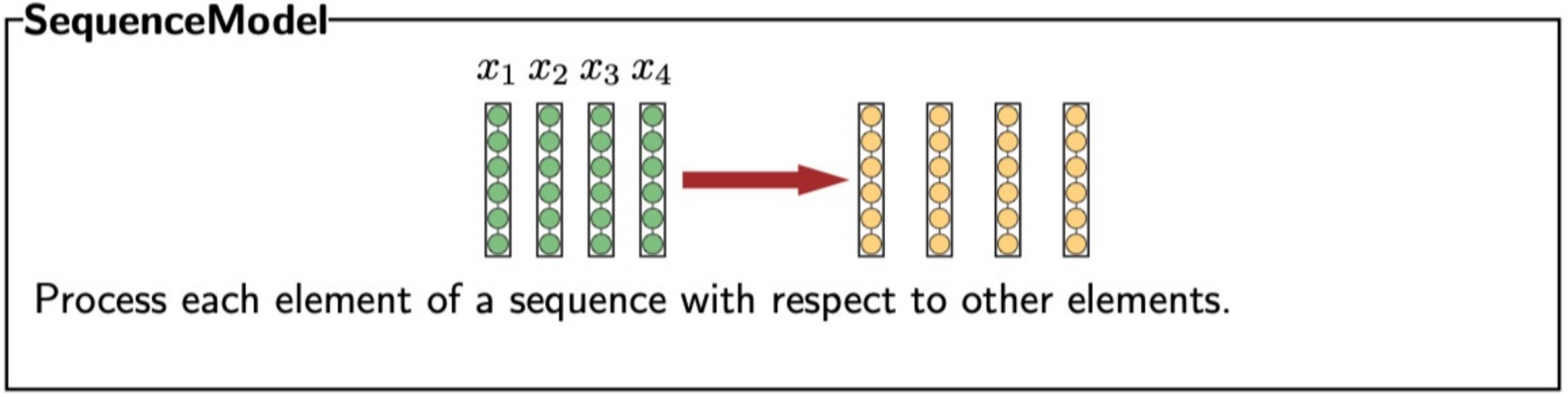
总结

FeedForward Conv MaxPool

EmbedToken SequenceRNN SimpleRNN LSTM

Attention AddNorm TransformerBlock BERT

Collapse GenerateToken LanguageModel SequenceToSequence



Questions



2026秋AI&ML
群号: 1109611031